American Urological Association

American Urological Association (AUA)/Canadian Urological Association (CUA)/
Society of Urodynamics, Female Pelvic Medicine & Urogenital Reconstruction (SUFU)

APPROVED BY THE AUA BOARD OF DIRECTORS AUGUST 2025

Authors' disclosure of potential conflicts of interest and author/staff contributions appear at the end of the article.

RECURRENT UNCOMPLICATED URINARY TRACT INFECTIONS IN WOMEN: AUA/CUA/SUFU GUIDELINE (2019, CONFIRMED 2022, AMENDED 2025)

This document is endorsed by the American Urogynecologic Society

Guideline Panel

Jennifer Anger, MD, MPH; Una Lee, MD; A. Lenore Ackerman, MD, PhD; Roger Chou, MD; Bilal Chughtai, MD; J. Quentin Clemens, MD; Duane Hickling, MD, MSCI; Anil Kapoor, MD; Kimberly S. Kenton, MD, MS; Melissa R. Kaufman, MD, PhD; Mary Ann Rondanina, Yahir A. Santiago-Lastra, MD; Ann Stapleton, MD; Lynn Stothers, MD; Toby C. Chai, MD

Amendment Panel

A. Lenore Ackerman, MD, PhD; Duane Hickling, MD, MSCI; Megan Bradley, MD

Amendment Staff and Consultants

Kristen E. D'Anci, PhD; Sennett K. Kim; Erin Kirkby, MS

SUMMARY

Purpose

Over the past few decades, our ability to diagnose, treat, and manage recurrent urinary tract infection (rUTI) long-term has evolved due to additional insights into the pathophysiology of rUTI, a new appreciation for the adverse effects of repetitive antimicrobial therapy ("collateral damage"),[1] rising rates of bacterial antimicrobial resistance, and better reporting of the natural history and clinical outcomes of acute cystitis and rUTI. For the purposes of this Guideline, the Panel considers only recurrent episodes of localized (restricted to the lower urinary tract) cystitis in women. This Guideline does not apply to individuals with complicating factors that place them at higher risk for development of a urinary tract infection (UTI) or for decreased efficacy of therapy as well as those with signs or symptoms of systemic bacteremia. In this document, the term UTI will refer to acute bacterial cystitis unless otherwise specified. This document seeks to establish guidance for the evaluation and management of patients with rUTIs to prevent inappropriate use of antibiotics, decrease the risk of antibiotic resistance, reduce adverse effects of antibiotic use, provide guidance on antibiotic and non-antibiotic strategies for prevention, and improve clinical outcomes and quality of life for women with rUTIs by reducing recurrence of UTI events.

Methodology

The systematic review utilized to inform this Guideline was conducted by a methodology team at the Pacific Northwest Evidence-based Practice Center (EPC). Scoping of the report and review of the final systematic review to inform Guideline statements was conducted in conjunction with the rUTI Panel. A research librarian conducted searches in Ovid MEDLINE (1946 to January [Week 1] 2018), Cochrane Central Register of Controlled Trials (through December 2017) and Embase (through January 16, 2018). Searches of electronic databases were supplemented by reviewing reference lists of relevant articles. An update literature search was conducted on September 20, 2018. In 2022, the EPC conducted an update review assessing abstracts from new studies published since the publication of the 2019 Guideline. The American Urological Association (AUA) asked the EPC to further assess a subset of studies included in the update report, to support potential changes to the 2019 Guideline. In 2025, the rUTI Guideline was updated through the AUA amendment process in which newly published literature were reviewed and integrated into previously published Guidelines. The methodologist searched for studies published on or after June 1, 2021 to November 1, 2024 using the same PubMed search strategies and strings from 2019 that were used for the original rUTI evidence report. New searches were constructed for new Key Questions addressing next generation sequencing and other non-culture testing, as well as for an existing Key Question to capture symptoms predictive of a UTI requiring treatment.

GUIDELINE STATEMENTS

Evaluation

1. Clinicians should obtain a complete patient history and perform a pelvic examination in women presenting with rUTIs. (*Clinical Principle*)

2. Clinicians should obtain urinalysis, urine culture and sensitivity with each symptomatic acute cystitis episode prior to initiating treatment in patients with rUTIs. (*Moderate Recommendation; Evidence Level: Grade C*)

3. To make a diagnosis of rUTI, clinicians should document evidence of inflammation (pyuria) and the presence of uropathogenic bacteria in association with symptomatic episodes. (*Clinical Principle*)

4. Clinicians should obtain repeat urine studies when an initial urine specimen is suspect for contamination, with consideration for obtaining a catheterized specimen. (*Clinical Principle*)

5. Cystoscopy and upper tract imaging should not be routinely obtained in the index patient presenting with rUTI. (*Expert Opinion*)

6. Clinicians may offer patient-initiated treatment (self-start treatment) to select rUTI patients with acute episodes while awaiting urine cultures. (*Conditional Recommendation; Evidence Level: Grade C*)

Asymptomatic Bacteriuria

7. Clinicians should omit surveillance urine testing, including urine culture, in asymptomatic patients with rUTIs. (*Moderate Recommendation; Evidence Level: Grade C*)

8. Clinicians should not treat asymptomatic bacteriuria (ASB) in patients. (*Strong Recommendation; Evidence Level: Grade B*)

Antibiotic Treatment

9. Clinicians should use first-line therapy (i.e., nitrofurantoin, trimethoprim-sulfamethoxazole [TMP-SMX], fosfomycin) dependent on the local antibiogram for the treatment of symptomatic UTIs in women. (*Strong Recommendation; Evidence Level: Grade B*)

10. Clinicians should treat rUTI patients experiencing acute cystitis episodes with as short a duration of antibiotics as reasonable, generally no longer than seven days. (*Moderate Recommendation; Evidence Level: Grade B*)

11. In patients with rUTIs experiencing acute cystitis episodes associated with urine cultures resistant to oral antibiotics, clinicians may treat with culture-directed parenteral antibiotics for as short a course as reasonable, generally no longer than seven days. (*Expert Opinion*)

Antibiotic Prophylaxis

12. Following discussion of the risks, benefits, and alternatives, clinicians may prescribe antibiotic prophylaxis to decrease the risk of future UTIs in women of all ages previously diagnosed with UTIs. (*Conditional Recommendation; Evidence Level: Grade B*)

Non-Antibiotic Prophylaxis

13. Clinicians should offer cranberry as an option for prophylaxis for women with rUTIs. (*Moderate Recommendation; Evidence Level: Grade B*)

14. Clinicians should inform patients with rUTIs that D-mannose alone for prophylaxis may not be effective in UTI prevention. (*Moderate Recommendation; Evidence Level: Grade B*)

15. Clinicians may offer methenamine hippurate for prophylaxis for women with rUTIs. (*Conditional Recommendation; Evidence Level: Grade C*)

16. When women with rUTIs have a water intake below 1.5 L/day (50 oz), clinicians may offer increased water intake for prophylaxis. (*Conditional Recommendation; Evidence Level: Grade C*)

Follow-up Evaluation

17. Clinicians should not perform a post-treatment test of cure urinalysis or urine culture in asymptomatic patients. (*Expert Opinion*)

18. Clinicians should repeat urine cultures to guide further management when UTI symptoms persist following antimicrobial therapy. (*Expert Opinion*)

19. For patients with persistent UTI symptoms after microbiological cure, clinicians should evaluate for alternative causes to patient symptoms. (*Expert Opinion*)

Estrogen

20. In perimenopausal and postmenopausal women with rUTIs, clinicians should recommend vaginal estrogen therapy to reduce the risk of future UTIs if there is no contraindication to vaginal estrogen therapy. (*Moderate Recommendation; Evidence Level: Grade B*)

INTRODUCTION

BACKGROUND

rUTI is a highly prevalent, costly, and burdensome condition affecting women of all ages, races, and ethnicities without regard for socioeconomic status, or educational level.[2] The incidence and prevalence of rUTI depend on the definition used. Approximately 60% of women will experience symptomatic acute bacterial cystitis in their lifetime.[3] An estimated 20% to 40% of women who have had one previous cystitis episode are likely to experience an additional episode, 25% to 50% of whom will experience multiple recurrent episodes.[4, 5] The exact numbers are unclear, as most epidemiologic studies utilize diagnosis codes that may overestimate true numbers due to overuse of UTI and rUTI codes in patients who have not yet undergone culture or evaluation.[3] Regardless of the definition, the evaluation and treatment of UTI have significant economic impact globally.[6]

Terminology and Definitions

For the purposes of this Guideline, the Panel considers only recurrent episodes in women of localized cystitis (restricted to the lower urinary tract); infections with suspected upper urinary tract or systemic involvement (i.e., systemic UTI) should be managed differently. This Guideline does not specifically consider patients in whom complicating factors may put them at higher risk for decreased treatment efficacy or for progression of a localized UTI to systemic infection. Such complicating factors may include an anatomic or functional abnormality of the urinary tract (e.g., stone disease, diverticulum, neurogenic bladder), an immunocompromised host, or urinary foreign bodies (e.g., indwelling urethral catheters, ureteral stents). In this Guideline, the term UTI will refer to acute bacterial cystitis. While most providers have confidence in making a diagnosis of acute cystitis, diagnostic criteria are imprecise and vary considerably. Strong evidence suggest that the diagnosis of acute bacterial cystitis should include the combination of acute-onset symptoms referable to the urinary tract, urinary inflammation on microscopic urinalysis (pyuria), and laboratory confirmation of significant bacteriuria.[7, 8] Without symptoms, bacteriuria of any magnitude is considered asymptomatic bacteriuria.

While there are multiple definitions for rUTI,[9] this Guideline supports the most commonly used definition of two episodes of acute bacterial cystitis within a six-month period sometime within the preceding year. This definition typically considers episodes to be separate infections with the resolution of symptoms between, and does not include those who require more than one treatment or multiple antibiotic courses for symptomatic resolution, as can occur with inappropriate initial or empiric treatment. Any patient experiencing episodes of symptomatic acute cystitis after previous resolution of similar symptoms meets the criteria for rUTI. However, it should be noted that those patients initially treated for localized bacterial cystitis who recur rapidly (i.e., within two weeks of initial treatment) after symptom resolution or display bacterial persistence without symptom resolution may require imaging, cystoscopy, or other further investigation for bacterial reservoirs. The definitions used in this Guideline can be found in **Table 1**.

TABLE 1: Guideline Definitions

Term	Definition
Acute bacterial cystitis	An infection of the urinary tract with: 1. Acute-onset symptoms such as dysuria in conjunction with variable degrees of increased urinary urgency and frequency, hematuria, and new or worsening incontinence 2. Urinary tract inflammation (pyuria ≥5 WBC/high power field (hpf) on microscopic urinalysis) 3. Detection of a bacterial uropathogen
Localized urinary tract infection (previously	An infection of the urinary tract in a healthy patient with an anatomically and functionally normal urinary tract, no signs or symptoms of upper

uncomplicated urinary tract infection)	urinary involvement or bacteremia, and no known complicating factors that would make the patient susceptible to progression to a systemic infection
Complicating Factors	Patient factors that place an individual at higher risk for development of a UTI and potentially decrease efficacy of therapy. Such factors include the following: • Anatomic or functional abnormality of the urinary tract (e.g., stone disease, diverticulum, neurogenic bladder) • Immunocompromised host • Indwelling urinary tract foreign body (e.g., indwelling urethral catheters, ureteral stents)
Systemic urinary tract infection	An infection of the urinary tract with signs and symptoms of systemic infection, with or without localized symptoms originating from any site in the urinary tract
rUTI	Two separate episodes of acute bacterial cystitis and associated symptoms over a six-month period within the preceding year
Asymptomatic bacteriuria	Presence of bacteria in the urine that causes no illness or symptoms
Pyuria	Presence of increased numbers of polymorphonuclear leukocytes (WBC) in the urine as evidence of an inflammatory response in the urinary tract[10, 11]

Index Patient

The index patient for this Guideline is an otherwise healthy adult female with localized rUTI in the absence of complicating factors. The infection should be associated with acute-onset symptoms attributable to the urinary tract as discussed below and occurs in the presence of both urinary inflammation (pyuria) and detection of a urinary bacterial uropathogen.[12] This Guideline does not apply to pregnant women, patients who are immunocompromised, those with anatomic or functional abnormalities of the urinary tract, women with rUTIs due to self-catheterization or indwelling catheters, or those exhibiting signs or symptoms of upper UTI or systemic bacteremia, such as fever and flank pain.[4] This Guideline also excludes those with neurological disease or illness relevant to the lower urinary tract, including those who exhibit genitourinary sequelae of peripheral neuropathy, diabetes, and spinal cord injury. Further, this Guideline does not discuss prevention of UTI in operative or procedural settings.

Symptoms

In UTI, acute-onset symptoms attributable to the urinary tract typically include dysuria in conjunction with variable degrees of increased urinary urgency and frequency, hematuria, and new or worsening incontinence. *Dysuria (urethral burning during voiding) is central in the diagnosis of UTI*; other symptoms of frequency, urgency, suprapubic pain, and hematuria are variably present. Acute-onset dysuria is a specific symptom, with more than 90% accuracy for UTI in young women in the absence of concomitant vaginal irritation or increased vaginal discharge.[13, 14] Dysuria can have other causes (e.g., atrophic vaginitis, pelvic floor myofascial pain, vulvar lichen sclerosus), however, the specificity of this symptom outside of this narrow population has not been defined.

In older adults, the symptoms of UTI may be less clear. Given the subjective nature of these symptoms, careful evaluation of their chronicity becomes an important consideration when the diagnosis of UTI is in doubt. Acute-onset dysuria, particularly when associated with new or worsening storage urinary symptoms, remains a reliable diagnostic criterion in older women living both in the community and in long-term care facilities.[15-17] Older women frequently have nonspecific symptoms that may be perceived as a UTI, such as chronic dysuria, cloudy urine, vaginal dryness, vaginal/perineal burning, bladder or pelvic discomfort, urinary frequency and urgency, or urinary incontinence, but these tend to be either chronic or fluctuating in nature. The lack of a correlation between symptoms and the presence of a uropathogen on urine culture was discussed in a systematic review of studies evaluating UTI in community-dwelling adults older than 65 years. Symptoms such as chronic nocturia, incontinence, and general sense of lack of well-being (e.g., fatigue, malaise, weakness), were common and not specific for UTI.[18] While this Guideline does not include women with chronic symptoms common in urology, such as overactive bladder (OAB), Guidelines from the American Geriatrics Society (AGS) and the Infectious Diseases Society of America (IDSA) agree that evaluation and treatment for suspected UTIs should be reserved for acute-onset (<1 week) dysuria or fever in association with other specific UTI-associated symptoms and signs, which primarily include gross hematuria, new or significantly worsening urinary urgency, frequency and/or incontinence, and suprapubic pain.[19-22]

DIAGNOSIS

Typically, for a diagnosis of cystitis, acute-onset symptoms should occur in conjunction with laboratory detection of a uropathogen from the urine, typically *Escherichia coli* (*E. coli*) (75% to 95%), but occasionally other pathogens such as other Enterobacteriaceae (e.g., *Proteus mirabilis* [*P. mirabilis*], *Klebsiella pneumoniae* [*K. pneumoniae*]) and *Staphylococcus saprophyticus (S. saprophyticus),* among other rare species.[23, 24]

While historically standard urine culture has been the mainstay of diagnosis of an episode of acute cystitis,[25] variations in specimen processing, the colony count thresholds designating positivity, and the infectious organism detected will result in wide variations in the accuracy of diagnosis.[26] Standard agar-based clinical culture has been used since the 19th century with few technical refinements; more recent studies demonstrate that a large proportion of urinary bacteria are not cultivatable using these standard conditions. The definition for clinically-significant bacteriuria of 10^5 colony-forming units (CFU)/mL was published more than 60 years ago and likely represents an arbitrary cut-off.[27-31] The origin of this cut-off derives from evidence that the use of this threshold in asymptomatic individuals is relevant to reducing the overdetection of contaminating organisms. More than 95% of subjects with $>10^5$ CFU/mL bacteria in a clean-catch specimen had definite bacteriuria on a catheterized specimen, while only a minority of patients with lower bacterial counts exhibited bacterial growth from a catheterized urine sample.[27] These data were obtained from asymptomatic women, however, and do not reflect the population in whom there is a suspicion of UTI. This threshold has been documented to provide a sensitivity of diagnosis of only 50% to 60%.[32]

In symptomatic women, however, several studies have identified subsets of women with pyuria and symptoms consistent with a UTI but colony counts $<10^5$ CFU/mL in voided urine.[33-40] One study of more than 200 premenopausal, non-pregnant women who presented with at least 2 symptoms of acute cystitis compared colony counts in a midstream, clean-catch urine sample to specimens obtained by urethral catheterization. Approximately 40% of the women who had *E. coli* grow from a catheterized specimen had colony counts $<10^5$ CFU/mL in the voided sample.[39] In multiple studies, a threshold of $\geq 10^2$ CFU/mL of *E. coli* from voided specimens had 88% to 93% positive predictive value for bladder bacteriuria **in patients with a high suspicion of UTI**.[35, 39] Lower midstream urine colony counts ($>10^2$ CFU/mL) have been associated with bladder bacteriuria on catheterization in symptomatic women with pyuria, suggesting that $\geq 10^2$ CFU/mL of a single uropathogen may be a more appropriate cut-off in selected patients in whom there is strong suspicion of infection.[41, 42]

Many laboratories, however, will not report colony counts $<10^3$ CFU/mL. In addition, it is likely that the strict use of a low threshold will lead to overdiagnosis. As such, clinical judgment in determining when a standard urine culture result represents clinically significant bacteriuria must factor in the clinical presentation of a patient, the urine collection method used, and the presence of other suggestive factors such as pyuria and the presence of urinary nitrites. Although a 10^5 CFU/mL threshold for bacterial growth on midstream voided urine may help distinguish bladder bacteriuria from contamination in **asymptomatic**, premenopausal women, a lower threshold may be appropriate in **symptomatic** individuals. Further, no specific threshold for urinary colony count has been demonstrated to identify those symptomatic patients at risk for progression to pyelonephritis or those who would benefit from more aggressive antimicrobial management.

The lack of clarity regarding the diagnostic threshold to define UTI may be influenced by both the population surveyed and the causative organism. Historically, standard urine culture has been highly reliable for the diagnosis of UTI caused by *E. coli* in acutely symptomatic women with 95% sensitivity and 85% specificity based on thresholds as low as 10^2 CFU/mL.[35] However, the ability of standard urine culture to diagnose UTI when the detected organism is not *E. coli* (e.g., gram-positive bacteria; *Enterococci* and Group B *Streptococci*) or when the patient has symptoms deviating from the classic pattern of acute-onset dysuria and urinary frequency without vaginal symptoms, has called into question the reliability of standard urine culture as a diagnostic tool.[14, 39] Hooton et al. reported that while standard urine culture had a positive predictive value of 93% for *E. coli* growth of at least 10^2 CFU/mL in women with acute dysuria, the positive predictive value was only 10% for *Enterococci* and 8% for Group B *Streptococci.*[39] In the same study, *Enterococci* and Group B *Streptococci* were frequently found in cultures from midstream urine, but not matched catheterized specimens, suggesting these positive cultures may represent false positives.

In addition to the lack of clarity about diagnostic thresholds, standard urine culture can be time consuming, taking between 1 to 4 days to receive a result and possibly longer to obtain corresponding antibiotic susceptibility profiles. The inability to obtain accurate information about the causative organisms and appropriate antimicrobial choice, delays effective treatment in cases where it is necessary to prevent unwanted complications, such as pyelonephritis. Fear of such complications encourages empiric antibiotic prescriptions that may increase antimicrobial overuse, individual patient antibiotic burden, and antimicrobial resistance.

In addition to the substantial potential for vulvovaginal contamination in all voided collections, standard urine culture is highly susceptible to contamination during specimen laboratory processing, leading to an average contamination rate of 15% reported for institutions across the nation.[43] Contamination is likely when urine culture results in low bacterial growth and/or growth of several bacterial species that are typically considered urethral and vaginal commensals, such as *Lactobacilli*, *Corynebacteria*, *Gardnerella*, *alpha-hemolytic Streptococci*, and aerobes. Contamination often occurs during the pre-analytic phase of urine culture (i.e., urine collection, storage, and transport), and quality control practices in many outpatient facilities are frequently lacking to ensure the adequacy of these steps.[44] False positive results created by contamination can promote suboptimal or unnecessary treatment, leading to poor patient outcomes and added healthcare costs contributing to the pervading problem of antimicrobial resistance.[45]

Human as habitat model

Some of the questions surrounding the accuracy of standard urine culture are also placed into new context by our evolving understanding of the microbiome, the microbial communities that live in concert with human hosts on body surfaces and within visceral organs.[46] Sensitive culture-dependent and -independent techniques have now revealed that the lower urinary tract, even in asymptomatic, healthy individuals, hosts a complex microbial community that is likely important in the maintenance of normal bladder function.[25, 47, 48] Previously, urine was widely considered to be sterile, and UTI had been defined as “microbial infiltration of the otherwise sterile urinary tract”.[49] This evolution in understanding, acknowledging that a urinary bacteria are the rule and not the exception, has been made possible by the development of a number of sensitive culture-dependent and -independent techniques. Thus, in the strictest definition, all individuals are likely “bacteriuric”. These commensal bacterial communities are generally symbiotic, supporting genitourinary immunity. In fact, these communities (constituting ASB) may protect patients with rUTI from additional symptomatic episodes. Of 673 young women with ASB, those treated with antibiotics experienced higher rates of subsequent, symptomatic UTI.[50] Intentional bladder colonization with a nonpathogenic *E. coli* (HU2117) has even been shown to safely reduce the risk of symptomatic UTI in patients with spinal cord injury.[51] Large population-based epidemiologic studies have documented the high prevalence (>75%) in the general population of chronic or fluctuating lower urinary tract symptoms that are not thought to be infectious in nature.[52] Given the presence of bacteria in a healthy urinary tract, our concept of when antibiotics are necessary to manage bacteriuria must evolve, even when urinary symptoms are present. In addition, symptomatic treatment and expectant management can be an appropriate treatment option for a subset of patients even when acute bacterial cystitis is confirmed. Thus, we must re-evaluate the dogma that relies heavily on the idea that all symptomatic bacteriuria should be managed with antibiotics.[53] Combined with a better understanding of the additional personal and societal harms of antibiotic overtreatment, the evaluation and management of rUTI should not be aimed at bacterial eradication, but at the amelioration of symptoms and prevention of complications.

Emerging Diagnostic Tools

A significant degree of uncertainty remains in diagnosing acute bacterial cystitis regardless of the method of testing. Each patient is unique; the specific clinical presentation, confounding symptoms, and equivocal testing can make it challenging to definitively identify the cause of uncomfortable urinary tract symptoms, even for the most experienced provider. The inability of standard urine culture to provide a definitive call about the presence or absence of infection is frustrating to providers and patients alike, which also contributes to antibiotic overtreatment.[54] The limitations in the clinical application of standard urine culture have raised interest in alternative, commercially available approaches to improve the clinical experience with UTI diagnosis; many patients are actively seeking these out due to their dissatisfaction with the current standards of UTI care.[55] New potential methodologies to diagnose UTI have evolved out of the scientific discoveries used to identify the genitourinary microbiome, including more sensitive culture-based methods and culture-independent techniques. As of yet, however, there is no laboratory test, including standard urine culture, that can provide reasonable diagnostic accuracy *without taking into consideration the totality of the patient presentation*.

Expanded quantitative urine culture (EQUC) utilizes more inoculum and a wider variety of growth media and atmospheric conditions for an extended culture period to dramatically enhance the number and varieties of bacteria that can be isolated.[25] While EQUC of catheterized specimens results in similar growth of *E. coli* as seen on standard urine culture, it can also detect a wide range of anaerobic, microaerophilic, and fastidious bacteria that do not grow in standard urine culture.[56] In a recent randomized controlled trial (RCT), women self-reported UTI symptoms and were randomized to treatment guided by either standard urine culture versus EQUC where symptom resolution was found to be similar between the two methodologies.[57] Further, no statistically significant difference was detected in symptomatic improvement based on treatment guided by standard urine culture versus EQUC in those with non-*E. coli* strains. An important caveat of this study is that specimens were obtained by catheterization, which is not typically performed in most clinical settings, such as outpatient laboratory facilities. Catheterization is likely necessary to avoid UTI overdetection, as analyzed voided urine specimens by EQUC in women with rUTI yielded high false positive results.[58] As EQUC requires microbial growth for identification, it does permit the determination of antimicrobial sensitivity, but remains vulnerable to the problems of poor specimen handling, susceptibility to contamination, and the delays needed to obtain final results. Thus, while EQUC can detect a greater range of urinary bacteria than standard urine culture, the lack of demonstrable advantages in current culture testing in terms of clinical outcomes means that the limited availability, increased labor and time to a result, and the requirement for catheterization limit the utility of EQUC in routine diagnosis of localized UTI.

In contrast, culture-independent approaches detect byproducts or components of microorganisms within a given sample, circumventing some of the limitations of conventional urine culture. Next-generation sequencing (NGS) typically refers to *amplicon sequencing* in which targeted primers are used to amplify a small, variable region of bacterial deoxyribonucleic acid (DNA), the sequence of which is then compared to a bacterial sequence database to allow identification of the taxa present.[59] *Shotgun sequencing* or *metagenomics* is a deep sequencing approach in which the entirety of DNA in a sample is fragmented, sequenced, then reassembled into whole genomes that represent the different organisms present, providing a more complete, but computationally expansive catalog of all DNA sources within the urine, including bacteria, viruses, fungi, as well as the patient.[60] *Multiplex polymerase chain reaction (PCR)* refers to the use of pathogen-specific primer probes to amplify a conserved region of the microbial genome to allow the identification of specific bacteria.[61] In contrast to NGS, PCR is only able to detect taxa targeted by pre-selected primers, while NGS provides a broader representation of all bacteria present. However, PCR can determine the quantitative amount of each pathogen present, while NGS cannot reliably provide a readout of overall bacterial levels. These techniques offer the promise of more rapid detection and a greater resistance to the contamination and handling issues that plague standard urine culture, but currently lack rigorous evidence that urine multiplex molecular tests improve diagnosis or clinical outcomes.

Preliminary evidence, however, suggests these alternative approaches may provide some benefits to patient care. In a small prospective randomized study of men and women with symptoms of acute cystitis (N=44), patients whose treatment was guided by NGS exhibited greater symptomatic improvement than those whose treatment was guided by standard urine culture.[62] In a large, retrospective analysis, PCR diagnostics were associated with reduced presentations to emergency departments and fewer subsequent hospitalizations.[63] Several randomized prospective trials of men and women with complicating factors of UTI have documented improved patient outcomes when treatment is guided by PCR-based diagnostics instead of standard urine culture, confirming shorter times to complete testing, reduced rates of empiric antibiotic prescribing, and fewer repeat visits.[64-66] While these results are reassuring, these studies examined patients with complicating factors of UTI, who may differ clinically from women without them. That population, however, tends to exhibit higher rates of bacteriuria, even in the absence of infection. Even higher rates of overdiagnosis would thus be expected in that population, particularly when using more sensitive detection methods. Overall, concerns for overdiagnoses may be ameliorated with thoughtful use of the technologies. It is also important to remember that these preliminary findings applied to participants with a high pre-test probability of UTI; no studies have demonstrated utility for these molecular diagnostic approaches in the evaluation of atypical presentations, such as for pelvic pain or chronic lower urinary tract symptoms.

Clinically, these techniques (both NGS and PCR) are often combined with the detection of antimicrobial resistance genes by PCR, which can provide guidance on antibiotic selection. PCR-based genotypic antimicrobial susceptibility testing (AST) detects the presence of genes or mutations that are associated with antimicrobial resistance. Genotypic AST is significantly faster than culture-based phenotypic susceptibilities as it is done directly from the source specimen, typically providing results within a few hours and can be performed without the delay of antecedent culturing. As with UTI diagnostics, PCR approaches can detect only a select subset of resistance markers for a set number of species. Genotypic AST methods predict resistance, not susceptibility, as the presence of a resistance gene does not necessarily equate to the expression of a resistance phenotype.[67] These tests cannot rule in antimicrobial therapy options unless there is a singular genetic mechanism known to confer resistance for a particular antibiotic in a specific bacterial species. In addition, genotypic AST may not detect novel resistance mechanisms or other non-enzymatic resistance mechanisms (e.g., porin loss or up-regulation of efflux pumps).[68-71] As the tests only detect the presence of resistance genes and/or mutations in antibiotic targets, but not their expression, genotypic AST may overdiagnose antibiotic resistance, leading to the use of more broad spectrum antibiotics than necessary. Regardless of these limitations, the rapidity of available genotypic susceptibility results may facilitate the more rapid administration of appropriate antibiotics for most multidrug-resistant bacteria, prevent inaccurate empiric prescribing, and help avert potential complications and reduce infection severity.[64, 72, 73]

While molecular diagnostic methods have transformed our understanding of the urinary microbiome and implicated more complex bacterial communities in the etiology of UTI symptoms,[59, 60, 74] their role in the diagnosis and management of UTI remains unclear. Without drastic changes in our diagnostic paradigm, more sensitive culture-based or molecular bacterial detection methods will likely be associated with increased diagnostic confusion and dilemmas, including overdiagnosis and associated overtreatment. Standard urine culture is not immune from overdiagnosis as well, which underscores the need for new approaches to diagnostic stewardship regardless of the bacterial detection method employed. There is some early evidence that the use of molecular diagnostic methods to rapidly identify uropathogens and identify an antibiotic susceptibility gene could help to avoid delayed or inappropriate antimicrobial treatment,[75] but the potential impact of such tests on global diagnostic and antibiotic stewardship is not clear. It is possible that more rapid detection, combined with appropriate clinical assessment and employment of urinalysis, could diminish antibiotic overuse by reducing the rampant empiric prescribing often employed while awaiting culture results. It is also possible that clinicians will apply the outdated principle that any bacterial presence in an otherwise sterile urinary tract constitutes infection, which could dramatically worsen an already dire global crisis of antibiotic resistance. While the current dependence of UTI diagnosis on standard urine culture relies on the unlikely principle that only those organisms detectable with agar-based culture are clinically concerning, the converse that all detectable organisms are pathogenic is also inaccurate. **Thus, despite a growing desire for more accurate diagnostics for UTI in patients with suggestive symptoms, the answer to improved antimicrobial stewardship lies not in better testing, but in better clinical judgement.**

Our novel understanding of a complex, generally beneficial microbiome requires a new diagnostic paradigm in which clinicians should no longer rely on the findings of standard urine culture or any other bacterial detection method as a substitute for clinical judgement. Whenever possible, patients should be evaluated at each symptomatic presentation, at least with documentation of specific signs and symptoms and laboratory assessment. Only with this data can clinicians decide whether the summation of information (i.e., the patient's clinical presentation, urinalysis findings, bacterial profiles, and current scientific evidence) suggests that antibiotic treatment is likely to be of benefit. Refocusing our diagnosis and management strategies from the current paradigm, which is centered on bacterial detection, to a measured approach by compiling the available clinical data for each patient and weighing the benefits and potential serious harms of antibiotics in each unique circumstance is not only warranted but imperative.

Antimicrobial Stewardship and the Consideration of Collateral Damage

In the past 20 years, antimicrobial resistance among uropathogens has increased dramatically. For example, increases in extended-spectrum β-lactamase (ESBL)-producing isolates has been described among patients with acute localized cystitis worldwide.[1, 76, 77] Localized UTI is one of the most common indications for antimicrobial exposure in otherwise healthy women. Fluoroquinolones have been linked to infection with methicillin-resistant *S. aureus* and increasing fluoroquinolone resistance in gram-negative bacilli, such as *Pseudomonas aeruginosa* (*P. aeruginosa*), while broad spectrum cephalosporins have been linked to subsequent infections with vancomycin-resistant *Enterococci*, ESBL–producing *K. pneumoniae*, β-lactam-resistant *Acinetobacter* species, and *Clostridioides difficile* (*C. difficile*).[1]

Adhering to a program of antimicrobial stewardship with attempts to reduce inappropriate treatment, decrease broad-spectrum antibiotic use, and appropriately tailor necessary treatment to the shortest effective duration, may significantly mitigate increasing fluoroquinolone and cephalosporin resistance.[78] Non-adherence to Guidelines for the treatment of acute cystitis, however, is more common in patients who have rUTIs than patients with an isolated episode of acute cystitis.[79] When patients present with acute cystitis and a history of rUTIs, many providers will employ strategies of lengthening the antimicrobial course, broadening antibiotic treatment, or increasing antibiotic doses for each episode, despite the absence of evidence to support such practices. Sometimes patients pressure providers to give non-guideline-based treatments with the hope that the number of recurrent episodes will be reduced or the time between acute cystitis episodes will be lengthened. These strategies have not been demonstrated to be efficacious and have the potential for harm to the individual and community, directly contradicting the principles of antibiotic stewardship.[80, 81] As antimicrobial resistance patterns vary regionally, the specific treatment recommendations for acute cystitis episodes and rUTI prophylaxis may not be appropriate in every community. Providers should combine knowledge of the local antibiogram with the selection of antimicrobial agents with the least impact on normal vaginal and fecal flora. An antibiogram provides a profile of the local results of antimicrobial sensitivity testing for specific microorganisms. Aggregate data from single hospital or healthcare systems are cumulatively summarized, usually annually, providing the percentage of a given organism sensitive to a particular antimicrobial.

In a study of more than 25 million emergency department visits during which a UTI was diagnosed, urinary symptoms were only identified in 32%. In the subset of older individuals (aged 65 to 84 years), this prevalence of symptoms fell to 24%.[82] The prevalence of antibiotic-resistant bacteria, risk of continued rUTIs as well as progression to later pyelonephritis is enhanced by unnecessary antibiotic treatment of ASB without any demonstrable benefit. These data demonstrate the important role of rUTI overtreatment in promoting antimicrobial resistance. While the Panel recognizes that there are financial and time costs associated with obtaining urinary cultures, such studies remain an important aspect of care, as culture-directed, not empiric, therapies are associated with fewer UTI-related hospitalizations and lower rates of intravenous antibiotic use.[83] The diligence of obtaining cultures for each symptomatic episode, which is associated with reduced rates of overtreatment and more appropriate antibiotic selection, is thought to be beneficial through minimizing collateral damage and the potential need for further treatment in the event of inappropriate empiric therapy.

Collateral damage describes ecological adverse effects of antimicrobial therapy, such as alterations of the normal gut microbiome that can help select drug-resistant organisms and promote colonization or infection with multidrug resistant (MDR) organisms.[1] The effects of specific antibiotics on the normal fecal flora promote drug resistance and increased pathogenicity. *E. coli* isolates continue to demonstrate high *in vitro* susceptibility to nitrofurantoin, fosfomycin, and mecillinam.[36, 84] These antimicrobials have minimal effects on the normal fecal microbiota.[85-87] In contrast, antimicrobials that alter the fecal flora more significantly, such as TMP-SMX and fluoroquinolones, promote increased rates of antimicrobial resistance.[87, 88]

Continued intermittent courses of antibiotics in rUTI patients are associated with significant adverse events, which may include allergic reactions, organ toxicities, future infection with resistant organisms, and *C. difficile* infections, particularly in older adults. Thus, substantial effort should be made to avoid unnecessary treatment unless there is a high suspicion of an acute cystitis episode.[89] Even with short courses of more targeted antibiotics, multiple treatments over time may in aggregate impact both the individual and community. Indeed, asymptomatic women with a history of rUTIs randomized to treatment for ASB in a placebo-controlled trial were more likely to have additional symptomatic cystitis episodes in a year of follow-up than those randomized to placebo.[50] In a longer study of over two years of follow up, women with rUTIs treated with the goal of eradicating residual bacteriuria demonstrated a higher prevalence of antibiotic resistance, a higher incidence of pyelonephritis, and a poorer quality of life in comparison to those in the non-treatment group.[90]

Education and Informed Decision-making

The prevalence of antibiotic-resistant bacteria is enhanced by the unnecessary antibiotic treatment of ASB.[90] Given the subjectivity of patient-reported symptoms and the lack of clear diagnostic criteria on laboratory testing, the diagnosis of UTI is highly imprecise. While no evidence exists to support the concept of withholding antimicrobials to patients with rUTIs, providers must bear in mind that continued intermittent courses of antibiotics are associated with significant adverse events, particularly in older patients. Substantial effort should be made to avoid unnecessary treatment unless there is a high suspicion of UTI.

For patients with episodes of acute cystitis without complicating factors, there is minimal risk of progression to tissue invasion or pyelonephritis. Additionally, urinary tract symptoms do not reliably indicate risk or presence of "bacteremic bacteriuria" ("urosepsis") or pyelonephritis. In a representative study of older patients with bacteremia who had the same bacterial species cultured from the urine, ascertainment of the patients' symptoms at the time of infection revealed that only 1 of 37 participants aged 75 years and older had symptoms consistent with UTI, such as dysuria.[91] Multiple randomized placebo-controlled trials have demonstrated that antibiotic treatment for acute cystitis offers little but mildly faster symptomatic improvement compared to placebo in patients with acute dysuria and significant bacteriuria.[92-95] The incidence of pyelonephritis in these patients is low and is not substantially different in individuals receiving antibiotics versus those treated with supportive care of analgesics and hydration.[96] The concern driving many antibiotic prescriptions has been that supportive care alone might be associated with a risk of progression to pyelonephritis, although this has not been conclusively proven even in prospective trials.[93] While fear of this complication has led to the pervasive treatment of suspected UTI with antibiotics, expectant management with analgesics and hydration while awaiting culture results is likely underutilized. Indeed, this evidence suggests that supportive care can be reasonably attempted with antibiotic treatment reserved for those patients in whom it would be anticipated to impact prognosis. For this reason, educating the patient regarding the safety of analgesics and hydration while awaiting test results and reassurance about the low risk of infection progression may also help assuage common patient fears.[54, 92-95] In weighing the rarity of pyelonephritis against the risks of empiric antibiotics, patients may choose supportive care if they perceive the treatment risks outweigh the personal benefits.

In a large clinical trial, a substantial proportion of women agreed to placebo randomization[97] without other treatments to ameliorate symptoms. This suggests that many women may be willing to attempt temporizing measures with symptomatic and non-antimicrobial management when the benefits and potential harms of intermittent antimicrobial treatment are adequately discussed. In fact, in focus groups, women with rUTI voiced concerns about the long-term adverse effects of repeated antibiotic courses. It is reasonable to consider an approach to the diagnosis and treatment of rUTI as one of shared decision-making, in which patients are educated about the inaccuracy of diagnostic testing, the benefits and potential risks of antimicrobial use, and the alternatives to standard antibiotic treatment.[54] We suggest documentation of the education provided and informed consent discussion. It is likely that far fewer patients will opt for more aggressive treatments when counseled appropriately. Many patients and providers do not know that localized cystitis typically is self-limited and rarely progresses to more severe disease.[18, 97, 98] If this were explained, the goals of care could be more clearly defined as the amelioration of symptoms, the prevention of long-term complications, and the more appropriate use of antibiotics to those situations in which it is likely to improve outcomes.[13]

The Panel also supports discussion with patients regarding certain modifiable behaviors, including changing mode of contraception and increasing water intake, that have been shown to reduce the risk of rUTI. Sexually active women may consider changing their mode of contraception if using either barrier contraceptives or spermicidal products.[99] The increased risk of UTI associated with spermicidal use is likely due to the deleterious effect on *Lactobacillus* colonization within the vaginal microbiome.[100] Increased water intake should be recommended to those consuming less than 1.5 L/day as increased water intake was associated with a lower likelihood of having at least 3 UTI episodes over 12 months (<10% versus 88%) and a greater interval between UTI episodes (143 versus 84.4 days; $p<0.001$).[101] Unfortunately, there are many commonly held myths surrounding rUTI lifestyle modifications. Case-control studies clearly demonstrate that changes in hygiene practices (e.g., front to back wiping), pre- and post-coital voiding, avoidance of hot tubs, tampon use, and douching do not play a role in rUTI prevention.[99, 102] This reframing of the discussion surrounding UTI is likely to benefit both individual patients and the health care system as a whole.

METHODOLOGY

The systematic review utilized to inform this Guideline was conducted by a methodology team at the Pacific Northwest EPC. Determination of the Guideline scope and review of the final systematic review to inform Guideline statements was conducted in conjunction with the rUTI Panel. In 2024, an independent methodologist conducted a search for studies published on or after June 1, 2021 to November 1, 2024 to capture new studies since the last update review.

Panel Formation

The rUTI Panel was created in 2017 by the American Urological Association Education and Research, Inc. (AUAER). This Guideline was developed in collaboration with the Canadian Urological Association (CUA) and the Society of Urodynamics, Female Pelvic Medicine & Urogenital Reconstruction (SUFU). The Practice Guidelines Committee (PGC) of the AUA selected the Panel Chairs who in turn appointed the additional panel members with specific expertise in this area in conjunction with CUA and SUFU. Additionally, the Panel included patient representation. Funding of the Panel was provided by the AUA with contributions from CUA and SUFU; panel members received no remuneration for their work.

In 2022, a small update panel was formed to review literature published since the original release of the Guideline in 2019.

The rUTI Amendment Panel was created in 2024 by the AUA to review new literature and provide updates herein. The Panel received no remuneration for their work.

Searches and Article Selection

A research librarian conducted searches in Ovid MEDLINE (1946 to January, Week 1, 2018), Cochrane Central Register of Controlled Trials (through December 2017) and Embase (through January 16, 2018). Searches of electronic databases were supplemented by reviewing reference lists of relevant articles. An update search was conducted for additional publications on September 20, 2018.

The methodology team developed criteria for inclusion and exclusion of studies based on the Key Questions and the populations, interventions, comparators, outcomes, timing, types of studies and settings (PICOTS) of interest. For populations, inclusion focused on women with rUTIs (defined as ≥3 UTIs in a 12-month period or ≥2 UTIs in a 6-month period; studies were also included in which rUTI was not defined, but the mean or median number of UTIs in a 12-month period was ≥3). Exclusions included pregnant women, women with rUTIs due to self-catheterization or indwelling catheters, and prevention of UTI in operative or procedural settings. Subgroups of interest were based on age, history of pelvic surgery, and the presence of diabetes mellitus. For interventions, evaluations included diagnostic tests for rUTI (urine dipstick, urinalysis with microscopy, urine culture, urine or serum biomarkers), antibiotics for treatment of acute UTI and prevention, cranberry, *Lactobacillus*, estrogen, and other preventive treatments. For studies on treatment and prevention of UTI, outcomes were UTI recurrence, UTI-related symptoms, recurrence rate, hospitalization, antimicrobial resistance, and adverse effects associated with interventions. The Panel included randomized and non-randomized clinical trials of treatments for acute UTI and preventive interventions in women with rUTIs, studies on the diagnostic accuracy of tests for rUTI, and prospective studies on the association between risk factors and progression to symptomatic UTI in women with ASB. For questions related to treatment of acute UTI, methodologists included systematic reviews, supplemented by primary studies published after the reviews.

Using the pre-specified criteria, two investigators independently reviewed titles and abstracts of all citations. The methodology team used a two-phase method for screening full-text articles identified during review of titles and abstracts. In the first phase, investigators reviewed full-text articles to identify systematic reviews for inclusion. In the second phase they reviewed full-text articles to address Key Questions not sufficiently answered by previously published systematic reviews, or recent publications to update previously published systematic reviews. Database searches resulted in 6,153 potentially relevant articles. After dual review of abstracts and titles, 214 systematic reviews and individual studies were selected for full-text dual review, and 65 studies in 67 publications were determined to meet inclusion criteria and were included in this review. An additional 10 publications were identified in the update literature search and were added to the review.

For the update review in 2022, the EPC team extracted Summary of Evidence tables from the 2019 review for the relevant Key Questions, added assessments of new studies to them, and combined results of old and new studies where appropriate. They updated or assessed the strength of evidence (SOE) for key comparisons and outcomes, using the approach described in the AHRQ EPC *Methods Guide for Comparative Effectiveness Reviews*.[103] The EPC reviewed abstracts from 19 studies in 21 publications. Full-text assessment was conducted on 11 of those studies for further review.[104-114]

In 2024, the rUTI Guideline was updated through the AUA amendment process in which newly published literature is reviewed and integrated into previously published Guidelines. An independent methodologist conducted literature searches of PubMed from June 1, 2021 to November 1, 2024, yielding 1,015 citations. Of those, 87 studies were selected for full-text review; 14 studies met inclusion criteria and were included in this review. Searches were run in PubMed and prior search strategies and strings were used from the 2019 rUTI evidence report. New searches were constructed to address NGS and other non-culture testing, and to capture symptoms predictive of a UTI requiring treatment.

Data Abstraction

Information was extracted on study design, year, setting (inpatient or outpatient), country, sample size, eligibility criteria, dose and duration of the intervention, population characteristics (age, race, UTI history, diabetes, prior genitourinary surgery, and other treatments), results, and source of funding for each study that met inclusion criteria. For included systematic reviews, an investigator abstracted study characteristics (number and design of included studies, definition of rUTI, study settings, study dates, treatment and follow up duration), population characteristics (age, diabetes history, surgical history, prior treatments), interventions, methods and ratings for the risk of bias, synthesis methods, and results. The relative risks (RRs) and 95% confidence intervals (CIs) were calculated if necessary for included outcomes, from data reported in the studies. All data abstractions were reviewed by a second investigator for accuracy. Discrepancies were resolved through discussion and consensus.

Risk of Bias Assessment

Risk of bias was independently assessed using predefined criteria. Disagreements were resolved by consensus. For clinical trials, we adapted criteria for assessing risk of bias from the U.S. Preventive Services Task Force (USPSTF).[115] Criteria included use of appropriate randomization and allocation concealment methods, clear specification of inclusion criteria, baseline comparability of groups, blinding, attrition, and use of intention-to-treat analysis. Methodologists assessed systematic reviews using AMSTAR 2 (Assessing the Methodological Quality of Systematic Reviews) criteria.[116] Studies were rated as “low risk of bias,” “medium risk of bias,” or “high risk of bias” based on the presence and seriousness of methodological shortcomings.

Studies rated “low risk of bias” are generally considered valid. “Low risk of bias” studies include clear descriptions of the population, setting, interventions, and comparison groups; a valid method for allocation of patients to treatment; low dropout rates and clear reporting of dropouts; blinding of patients, care providers, and outcome assessors; and appropriate analysis of outcomes.

Studies rated “medium risk of bias” are susceptible to some bias, though not necessarily enough to invalidate the results. These studies do not meet all the criteria for a rating of low risk of bias, but any flaw present is unlikely to cause major bias. Studies may be missing information, making it difficult to assess limitations and potential problems. The “medium risk of bias” category is broad, and studies with this rating vary in their strengths and weaknesses. Therefore, the results of some medium risk of bias studies are likely to be valid, while others may be only possibly valid.

Studies rated “high risk of bias” have significant flaws that may invalidate the results. They have a serious or “fatal” flaw in design, analysis, or reporting; large amounts of missing information; discrepancies in reporting; or serious problems in the delivery of the intervention. The results of high risk of bias studies could be as likely to reflect flaws in study design and conduct as true difference between compared interventions. Methodologists did not exclude studies rated high risk of bias *a priori*, but high risk of bias studies were considered to be less reliable than low or medium risk of bias studies, and methodologists performed sensitivity analyses without high risk of bias studies to determine how their inclusion impacted findings.

Data Synthesis

Evidence tables were constructed with study characteristics, results, and risk of bias ratings for all included studies, and summary tables to highlight the main findings.

For interventions to prevent rUTIs, meta-analysis was performed using the random effects DerSimonian and Laird model in RevMan 5.3.5 (Copenhagen, Denmark) when there were at least three studies that could be pooled.[117] The analyses of antibiotics were stratified by the specific antibiotic and stratified analyses of estrogen according to whether they were administered systemically or topically. Sensitivity analysis was performed by excluding high risk of bias trials. For antibiotic treatment of acute UTI, pooled estimates were reported from systematic reviews. Heterogeneity is reported via I^2 calculations. Meta-analyses were not updated from prior reviews with the results of new trials, but examined whether the findings of new trials were consistent with the reviews. For other Key Questions, there were too few studies to perform meta-analysis.

Determination of Evidence Strength

The Grading of Recommendations Assessment, Development, and Evaluation (GRADE)[118] system was used to determine the aggregate evidence quality for each outcome, or group of related outcomes, informing Key Questions. GRADE defines a body of evidence in relation to how confident Guideline developers can be that the estimate of effects as reported by that body of evidence, is correct. Evidence is categorized as high, moderate, low, and very low, and assessment is based on the aggregate risk of bias for the evidence base, plus limitations introduced as a consequence of inconsistency, indirectness, imprecision, and publication bias across the studies.[119] Additionally, certainty of evidence can be downgraded if confounding across the studies has resulted in the potential for the evidence base to overestimate the effect. Upgrading of evidence is possible if the body of evidence indicates a large effect or if confounding would suggest either spurious effects or would reduce the demonstrated effect.

The AUA employs a three-tiered strength of evidence system to underpin evidence-based Guideline statements. **Table 2** summarizes the GRADE categories, definitions, and how these categories translate to the AUA strength of evidence categories. In short, high certainty by GRADE translates to AUA A-category strength of evidence, moderate to B, and both low and very low to C.

The AUA categorizes body of evidence strength as Grade A, Grade B, or Grade C. By definition, Grade A evidence is evidence about which the Panel has a high level of certainty, Grade B evidence is evidence about which the Panel has a moderate level of certainty, and Grade C evidence is evidence about which the Panel has a low level of certainty.[120]

TABLE 2: Strength of Evidence Definitions

AUA Strength of Evidence Category	GRADE Certainty Rating	Definition
A	High	• Very confident that the true effect lies close to that of the estimate of the effect
B	Moderate	• Moderately confident in the effect estimate • The true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different
C	Low	• Confidence in the effect estimate is limited • The true effect may be substantially different from the estimate of the effect
	Very Low	• Very little confidence in the effect estimate • The true effect is likely to be substantially different from the estimate of effect

AUA Nomenclature: Linking Statement Type to Evidence Strength

The AUA nomenclature system explicitly links statement type to body of evidence strength, level of certainty, magnitude of benefit or risk/burdens, and the Panel's judgment regarding the balance between benefits and risks/burdens (**Table 3**). Strong Recommendations are directive statements that an action should (benefits outweigh risks/burdens) or should not (risks/burdens outweigh benefits) be undertaken because net benefit or net harm is substantial. Moderate Recommendations are directive statements that an action should (benefits outweigh risks/burdens) or should not (risks/burdens outweigh benefits) be undertaken because net benefit or net harm is moderate. Conditional Recommendations are non-directive statements used when the evidence indicates that there is no apparent net benefit or harm, or when the balance between benefits and risks/burden is unclear. All three statement types may be supported by any body of evidence strength grade. Body of evidence strength Grade A in support of a Strong or Moderate Recommendation indicates that the statement can be applied to most patients in most circumstances and future research is unlikely to change confidence. Body of evidence strength Grade B in support of a Strong or Moderate Recommendation indicates that the statement can be applied to most patients in most circumstances, but better evidence could change confidence. Body of evidence strength Grade C in support of a Strong or Moderate Recommendation indicates that the statement can be applied to most patients in most circumstances, but better evidence is likely to change confidence. Conditional Recommendations also can be supported by any evidence strength. When body of evidence strength is Grade A, the statement indicates that benefits and risks/burdens appear balanced, the best action depends on patient circumstances, and future research is unlikely to change confidence. When body of evidence strength Grade B is used, benefits and risks/burdens appear balanced, the best action also depends on individual patient circumstances and better evidence could change confidence. When body of evidence strength Grade C is used, there is uncertainty regarding the balance between benefits and risks/burdens, alternative strategies may be equally reasonable, and better evidence is likely to change confidence.

Where gaps in the evidence existed, the Panel provides guidance in the form of Clinical Principles or Expert Opinions with consensus achieved using a modified Delphi technique if differences in opinion emerged.[121] A Clinical Principle is a statement about a component of clinical care that is widely agreed upon by urologists or other clinicians for which there may or may not be evidence in the medical literature. Expert Opinion refers to a statement, achieved by consensus of the Panel, that is based on members' clinical training, experience, knowledge, and judgment.

TABLE 3: AUA Nomenclature Linking Statement Type to Level of Certainty, Magnitude of Benefit or Risk/Burden, and Body of Evidence Strength

Evidence Grade	Evidence Strength A (High Certainty)	Evidence Strength B (Moderate Certainty)	Evidence Strength C (Low Certainty)
Strong Recommendation (Net benefit or harm substantial)	-Benefits > Risks/Burdens (or vice versa) -Net benefit (or net harm) is substantial -Applies to most patients in most circumstances and future research is unlikely to change confidence	-Benefits > Risks/Burdens (or vice versa) -Net benefit (or net harm) is substantial -Applies to most patients in most circumstances but better evidence could change confidence	-Benefits > Risks/Burdens (or vice versa) -Net benefit (or net harm) appears substantial -Applies to most patients in most circumstances but better evidence is likely to change confidence (rarely used to support a Strong Recommendation)
Moderate Recommendation (Net benefit or harm moderate)	-Benefits > Risks/Burdens (or vice versa) -Net benefit (or net harm) is moderate -Applies to most patients in most circumstances and future research is unlikely to change confidence	-Benefits > Risks/Burdens (or vice versa) -Net benefit (or net harm) is moderate -Applies to most patients in most circumstances but better evidence could change confidence	-Benefits > Risks/Burdens (or vice versa) -Net benefit (or net harm) appears moderate -Applies to most patients in most circumstances but better evidence is likely to change confidence
Conditional Recommendation (Net benefit or harm comparable to other options)	-Benefits = Risks/Burdens -Best action depends on individual patient circumstances -Future Research is unlikely to change confidence	-Benefits = Risks/Burdens -Best action appears to depend on individual patient circumstances -Better evidence could change confidence	-Balance between Benefits & Risks/Burdens unclear -Net benefit (or net harm) comparable to other options -Alternative strategies may be equally reasonable -Better evidence likely to change confidence
Clinical Principle	a statement about a component of clinical care that is widely agreed upon by urologists or other clinicians for which there may or may not be evidence in the medical literature		
Expert Opinion	a statement, achieved by consensus of the Panel, that is based on members' clinical training, experience, knowledge, and judgment for which there may or may not be evidence in the medical literature		

Peer Review and Document Approval

An integral part of the Guideline development process at the AUA is external peer review. The AUA conducted a thorough peer review process to ensure that the document was reviewed by experts in the diagnosis and treatment of UTIs in women. In addition to reviewers from the AUA PGC, Science and Quality Council (SQC), and Board of Directors (BOD), the document was reviewed by representatives from CUA and SUFU as well as external content experts. Additionally, a call for reviewers was placed on the AUA website from November 19 to 30, 2018 to allow any additional interested parties to request a copy of the document for review. The Guideline was also sent to the Urology Care Foundation (UCF) to open the document further to the patient perspective. The draft Guideline document was distributed to 114 peer reviewers. All peer review comments were blinded and sent to the Panel for review. In total, 50 reviewers provided comments, including 38 external reviewers. At the end of the peer review process, a total of 622 comments were received. Following comment discussion, the Panel revised the draft as needed. Once finalized, the Guideline was submitted for approval to the AUA PGC, SQC, and BOD as well as the governing bodies of CUA and SUFU for final approval.

In 2025, as a part of the amendment process, the AUA conducted a thorough peer review process. A call for peer reviewers was posted on February 11th, 2025 and the draft Guideline document was distributed to 145 peer reviewers, 85 of which submitted comments. The Amendment Panel reviewed and discussed all submitted comments and revised the draft as needed. Once finalized, the Guideline was submitted for approval to the PGC and SQC as well as representatives from CUA and SUFU. It was then submitted to AUA BODs for final approval.

Guideline Statements

Evaluation

1. **Clinicians should obtain a complete patient history and perform a pelvic examination in women presenting with rUTIs. (*Clinical Principle*)**

Patients with rUTIs should have a complete history obtained, including LUTS such as dysuria, frequency, urgency, nocturia, incontinence, hematuria, pneumaturia, and fecaluria. Further information to obtain includes any history of bowel symptoms such as diarrhea, accidental bowel leakage, or constipation; recent use of antibiotics for any medical condition; prior antibiotic-related problems (e.g., *C. difficile* infection); antibiotic allergies and sensitivities; back or flank pain; catheter usage; vaginal discharge or irritation; menopausal status; post-coital UTI; contraceptive method; and use of spermicides or estrogen- or progesterone-containing products. Details of prior urinary tract or pelvic surgery should be obtained, and patients should be queried as to travel history and history of working or walking for long periods of time. Baseline genitourinary symptoms between infections may also be illuminative, including the number of voids per day, sensation of urge to void, straining to void, a sensation of incomplete emptying, pelvic pressure or heaviness, vaginal bulge, dysuria, dyspareunia, as well as the location, character, and severity of any baseline genitourinary or pelvic pain or discomfort. UTI history includes frequency of UTI, antimicrobial usage, and documentation of positive cultures and the type of cultured microorganisms. Complicating factors conferring an increased risk of progression to systemic UTI, as previously discussed, should also be elucidated.

Patient history should document the symptoms the patient considers indicative of a UTI, the relationship of acute episode to infectious triggers (e.g., sexual intercourse), antimicrobials used for each episode, responses to treatment for each episode, as well as the results of any prior diagnostic investigations. It is also important to note the relationship of infections to hormonal influences (e.g., menstruation, menopause, exogenous hormone use) as well as concomitant medication usage or behaviors that may alter infection susceptibility, including prior antimicrobial treatment, immunomodulatory medications, and topicals such as spermicides.

At the initial presentation for rUTI evaluation, a physical examination including an abdominal and detailed pelvic examination should be performed to look for any structural or functional abnormalities. Pelvic support for the bladder, urethra, vagina, and rectum should be documented, noting the compartment and stage of any clinically significant prolapse. The bladder and urethra should be palpated directly for evidence of urethritis, urethral diverticulum, Skene's gland cyst, or other enlarged vulvar or vaginal cysts, and a focused examination to document any other infectious and inflammatory conditions, such as vaginitis, vulvar dermatitis, and vaginal atrophy (genitourinary syndrome of menopause). The pelvic floor musculature should be examined for tone, tenderness, banding, and trigger/tender points,[122] as pelvic floor myofascial pain is associated with dysuria, urinary frequency, urgency, and pelvic pain in the absence of bacterial infection.[123] A focused neurological exam to rule out occult neurologic defects may also be considered. Evaluation for incomplete bladder emptying to rule out occult retention can be considered for all patients, but should be performed in any patient with suspicion of incomplete emptying, such as those with significant anterior vaginal wall prolapse, underlying neurologic disease, diabetes, or a subjective sensation of incomplete emptying.

The necessity of physical examination has been questioned, frequently citing data documenting that presumptive diagnosis of acute UTI made from symptomatic presentation was highly accurate with little additional diagnostic utility for physical exam.[14] A meta-analyses of 9 studies found that the combination of sudden-onset dysuria and frequency without vaginal discharge confers a 90% probability of UTI, defined in this report as >10^4 CFU/mL of a single uropathogen.[14] In the same study, approximately 50% of women with one or more UTI symptoms (e.g., dysuria, frequency, urgency, hematuria, or back pain), who lacked that very specific combination of symptoms did not have cystitis, suggesting misdiagnosis of acute UTI is common. While numerous studies have attempted to identify high yield signs and symptoms associated with UTI, few outside of acute dysuria provide diagnostic discrimination of UTI (for additional information, please see **Statement 8**).[124] In-person assessment significantly improves UTI diagnostic accuracy; culture positivity rates for those tested following virtual assessment (29%) are less than half that for patients seen in person (64%).[125] Given the number of conditions that share significant symptomatology with UTI and the common misdiagnosis of acute cystitis, patients with rUTI warrant evaluation with an examination at least at their initial presentation to rule out alternative explanations for the patient's relapsing symptoms and to identify any structural or functional abnormalities that may be contributing to infection recurrence. Additional examinations with subsequent infections are not required, particularly when presenting symptoms are consistent with prior episodes, but may be useful in situations where the patient's symptoms are atypical or fail to resolve with appropriate treatment.

2. **Clinicians should obtain urinalysis, urine culture and sensitivity with each symptomatic acute cystitis episode prior to initiating treatment in patients with rUTIs. (*Moderate Recommendation; Evidence Level: Grade C*)**

In women with a history of rUTIs with acute symptoms consistent with urinary infection, the Panel reviewed the literature related to obtaining urine culture or urinalysis versus not performing such urine tests to dictate treatment decisions. Although no randomized, prospective studies were identified that were specifically designed to document direct effects of procuring urinalysis and urine culture with antibiotic sensitivities prior to initiating treatment, the bulk of observational data supports the comprehensive laboratory diagnostic evaluation of each episode (e.g., urinalysis and standard urine culture). Despite older literature suggesting that it is reasonable to treat cystitis episodes in women with UTI presumptively, the epidemic of antibiotic resistance demands more careful management moving forward. In patients who present for rUTI management without any microbiological information regarding prior presumed episodes of localized cystitis, it is reasonable to proceed with the assumption of rUTI if their clinical history is consistent with that diagnosis (e.g., acute-onset dysuria, urinary frequency and urgency with resolution upon antimicrobial treatment) and institute appropriate preventative treatment. However, every effort should be made to obtain urinalysis, microbiological identification, and antimicrobial resistance patterns to confirm the diagnosis and follow clinical responses to management. The routine prescription of antibiotics based on patient symptoms without microbiological evaluation should be discouraged. While this practice is convenient for both the patient and provider, it is a common cause of rUTI overdiagnosis. Independent evaluation of each symptomatic episode not only allows for the modification of treatment plans as needed to address atypical or antimicrobial-resistant bacteria but will also help to identify those patients with one of the numerous alternative diagnoses who would benefit from a different approach to treatment.

Accumulating evidence suggests that urinalysis may have a larger role in diagnosing UTI than previously recognized. **While a positive urinalysis provides little increase in diagnostic accuracy,[25] a negative urinalysis (i.e., negative nitrites and <5 WBC/hpf) is useful in ruling out acute cystitis[126] and is associated with a very low risk of progression to bacteremia.**[127] In a study of more than 12,000 urinalyses, a negative urinalysis exhibited high predictive accuracy for the absence of clinically relevant UTI (specificity: 92.2%; positive predictive value [PPV]: 97.4%).[126] Implementation of stewardship programs in which initial urinalysis findings prompt a reflex urine culture only in the presence of a positive urinalysis are estimated to reduce the number of patients inappropriately treated with antibiotics by five-fold.[128]

However, a positive urinalysis has a low PPV for UTI; positive leukocyte esterase (≥1+), pyuria (≥10 WBC/hpf), nitrite (≥1+), and bacteriuria (≥50 cells/field) all have PPVs for UTI lower than 50%.[127] However, urinalysis findings may combine with urine culture results to prompt a need to retest the patient. Discordance between urinalysis and standard urine culture results suggests test inaccuracy. For example, high levels (>100,000 CFU/mL) of urinary *E. coli*, a nitrite producer, on urine culture with a corresponding urinalysis lacking urine nitrites may suggest poor technique in obtaining the urine sample or poor sample handling leading to false positives. As described in greater detail below, urinalysis can identify the presence of epithelial cells or mucus, which may suggest skin or vaginal contamination.[129] Such information from a urinalysis may indicate that obtaining a catheterized specimen is reasonable to evaluate the patient's culture results accurately.[130]

Thus, urinalysis can provide substantial evidence of informing the diagnosis for each symptomatic episode and should be considered in each unique presentation. Urinalysis is specifically helpful due to its high negative predictive value (NPV), however, positive findings should not be interpreted as evidence of a clinically significant UTI.

In patients whose clinical presentation and urinalysis findings are consistent with possible infection, examination for urinary bacteria (e.g., using standard urine culture) should be performed to inform appropriate antibiotic selection. Identifying the causative organism and antibiotic susceptibilities at each symptomatic episode not only lowers rates of treatment failures but improves longer-term outcomes. In a propensity-matched cohort study examining 48,283 women with localized UTIs, individuals having a urine culture >50% of the time experienced fewer UTI-related hospitalizations and lower rates of intravenous antibiotic use compared with not having cultures >50% of the time.[83] While obtaining a culture may not decrease health care utilization overall,[131] in rUTI specifically, tracking the development of antimicrobial resistance can help guide subsequent empiric treatment and inform prophylactic approaches. Additionally, the absence of urinary bacteria on laboratory assessment may prompt reevaluation for alternative diagnoses with similar symptoms.

Standard urine culture has historically been the gold standard for microbial identification but has significant limitations. Its accuracy varies widely, while highly predictive in select cases (e.g., acute dysuria in premenopausal women with *E. coli*), it performs poorly outside these criteria, with contamination, mishandling, and false negatives affecting up to 40% of clinically apparent UTIs.[39, 43, 44] Molecular diagnostics are more sensitive and faster but often detect bacteria in asymptomatic individuals, limiting their utility as a replacement.

No current test reliably distinguishes infection from colonization across all patients. Thus, microbial identification should guide care but must be interpreted in context—considering symptoms, urinalysis, test type, and specific findings. Culture or other methods can support or challenge a diagnosis, inform antibiotic selection, and help track resistance patterns over time. Conversely, negative or ambiguous results should prompt clinicians to consider alternative diagnoses.

Rather than endorse a specific test, organism list, or threshold, this Guideline emphasizes individualized interpretation and serial evaluation to improve outcomes, recognizing that all testing methods, including culture, are imperfect.

The Panel recognizes that, in select rUTI patients with symptoms of recurrence, presumptive treatment with antibiotics can be initiated prior to finalization of testing results based on prior speciation, susceptibilities, and local antibiogram. For reliable patients, the Panel recommends a process of shared decision-making with regards to deferring therapy prior to obtaining results from urine culture. Since progression of acute cystitis to pyelonephritis is uncommon, initiation of conservative non-antibiotic treatments, such as urinary analgesics, while awaiting urine culture results may be reasonable in select circumstances when the clinician deems that patient safety will not be compromised. The Panel does not advocate use of either point-of-care dipstick or home dipstick analysis to diagnose rUTI or guide treatment decisions due to the poor sensitivity and specificity of these modalities.

3. **To make a diagnosis of rUTI, clinicians should document evidence of inflammation (pyuria) and the presence of uropathogenic bacteria in association with symptomatic episodes. (*Clinical Principle*)**

While there are multiple definitions for rUTI, this Guideline stresses microbial confirmation of the underlying pathology. The Guideline defines rUTI as at least two acute episodes in six months of urinary tract-associated symptoms and signs. These symptoms and signs primarily include dysuria, urinary frequency and urgency, and new or worsening incontinence with or without gross hematuria without vaginal symptoms. These symptoms and signs should be in conjunction with evidence of a microbial infection at the time of symptom onset. Evidence of infection, which includes detection of significant urinary bacteria (e.g., standard urine culture) and urinary inflammation (i.e., pyuria with ≥5 WBC/hpf on microscopic urinalysis), are critical to establish a diagnosis of rUTI. While the episodes do not need to be in the immediate six months preceding the presentation, the duration and frequency of these recurrences should influence the suggested prophylactic intervention; expectant management may be less appropriate than active prophylaxis in an individual who has received more than three to four therapeutic courses of antibiotics in the past year.

The absence of pyuria rules out UTI in symptomatic women with confirmed bacteriuria, making pyuria a necessary but not sufficient factor that should be present during symptomatic episodes to consider a diagnosis of rUTI.[12] Documentation of bacterial uropathogens during each symptomatic period is also critical; if bacteriuria does not consistently accompany symptomatic periods or the bacterial organisms detected are inconsistent with UTI, it may be reasonable to consider alternative diagnoses. As discussed in Statement 2, while standard urine culture remains the historical standard for bacterial identification, its many limitations must be acknowledged. Clinicians should consider any additional available information, including alternative tests that detect bacteria, with critical deliberation when determining whether a patient meets criteria for rUTI.

Disorders such as interstitial cystitis/bladder pain syndrome, OAB, genitourinary syndrome of menopause, urinary calculi, bacterial or fungal vaginitis, vulvar dermatitis, non-infectious vulvovestibulitis, vulvo/vestibulodynia, pelvic floor myofascial pain, and less common lower urinary tract pathologies like carcinoma *in situ* of the bladder have significant symptom overlap with acute bacterial cystitis and may co-exist with isolated or rUTIs.[132] Given the large number of confounding diagnoses that present similarly to UTI and the high rates of bacteriuria, even in asymptomatic women, it is important to consider all clinical and laboratory data when deciding whether antibiotic treatment is needed. A lack of correlation between microbiological data and symptomatic episodes should prompt a diligent consideration of alternative or comorbid diagnoses, as may be the case in women with gross hematuria. Many such women lacking microbial confirmation may be incorrectly treated for UTI when they should be evaluated for bladder cancer.

In addition, patients with a long history of culture-proven symptomatic episodes of cystitis that occur at a lower frequency than that which is specified in the definition used in this document (two episodes within six months within the previous year) may also be appropriate to include under the umbrella of rUTI. Patients consistently presenting with one to two symptomatic infections per year for multiple years will likely benefit from a more proactive management strategy similar to that suggested herein for patients with rUTI.

4. **Clinicians should obtain repeat urine studies when an initial urine specimen is suspect for contamination, with consideration for obtaining a catheterized specimen. (*Clinical Principle*)**

It is important to establish the association of acute-onset urinary symptoms with laboratory evidence of infection, including both urinary inflammation (i.e., pyuria ≥5 WBC/hpf on microscopic urinalysis) and evidence of uropathogenic bacteria. Contamination of urine specimens with skin and vaginal bacteria can result in high rates of suboptimal or unnecessary treatment, resulting in poor patient outcomes and higher health care costs.[133] The potential for contamination on standard urine culture with midstream urine collection, necessitates careful evaluation of specimen quality and the cultured species reported. While variably defined, contamination should be suspected when the specimen exhibits growth of normal vaginal flora (e.g., *Lactobacillus*), mixed cultures containing more than one organism, or low quantities (i.e., <10^3 CFU/mL) of a pathogenic organism in an asymptomatic patient.[129, 134-137] Further, concomitant urinalysis can provide additional guidance; the presence of epithelial cells or mucus on microscopic urinalysis may also suggest contaminants.[129] Growth of organisms thought to be contaminants (e.g., *Lactobacilli*, Group B *Streptococci*, *Corynebacteria*, and non-saprophyticus coagulase-negative *Staphylococci*) generally do not require treatment. When there is high suspicion for contamination, clinicians can consider obtaining a catheterized specimen for further evaluation prior to treatment.[39, 138]

While a suprapubic aspirate provides the most accurate urinary sampling, it is not practical in most settings, and a mid-stream urine specimen is typically adequate to provide a sufficient quality specimen for analysis;[139-141] however, care must be taken to avoid contamination. Contamination of urinary samples varies considerably due to multiple factors associated with urine collection and storage. Under optimal conditions, a mid-stream voided specimen may provide contamination rates of less than 1%, with specificity and sensitivity for UTI greater than 98% and 95%,[142-144] respectively. Poor collection, storage, and processing techniques, however, can produce contamination rates of 30% to 40%.[43, 138]

The largest contribution to this variability results from post-collection processing, particularly with regards to specimen storage.[43, 145] As urine can be easily seeded with commensal flora, low numbers of contaminant bacteria can continue to proliferate when stored at room temperature, leading to increased numbers of false-positive cultures or uninterpretable results. IDSA and the American Society for Microbiology (ASM) agree that urine should not sit at room temperature for more than 30 minutes to facilitate accurate laboratory diagnosis of UTI.[10, 130, 146] Several observational studies describe significant increases in colony counts after storage at room temperature for more than a few hours,[147-149] while delayed cultures on urine specimens kept refrigerated or preserved in urine transport solutions, such as boric acid or other preservatives, demonstrate high agreement with the results of immediate culture.[149-152] Thus, samples should either be transported to the lab in urine transport media in vacuum-filled tubes or refrigerated (2°C to 10°C) immediately to reduce artifactual bacterial proliferation. Further, the clinician should discourage patients from bringing samples from home due to the high potential for inadequate storage and erroneous results. Clinicians should also have a suspicion for such contamination particularly when high levels of bacteriuria are seen in the context of a urinalysis without detectable pyuria.[126]

While there is no definitive evidence that urethral cleansing improves specimen quality or reduces contamination,[129, 134-136] clinical laboratories and expert opinions still support preparation of the urethral meatus and surrounding vaginal epithelium with a cleaning or antiseptic solution prior to providing a voided specimen.[130] Care should also be taken to avoid contact of the collection cup with the skin or vaginal epithelium. Labial spreading is highly effective at reducing contamination, halving the contamination rates seen without attention to this detail.[153] The initial urinary stream should be discarded, and the subsequent midstream sample sent to the laboratory for analysis.[154] Oral instruction provided to patients may not be sufficient; written instructions for sample collection may be more effective at reducing contamination rates for voided specimens.[43] Such instructions can even be placed on the wall of the clinic bathroom.

The vaginal and skin microbiota in asymptomatic women can contain many bacterial species thought of as pathogens, including *S. aureus*, *S. viridans*, *Enterococci*, Group B *Streptococci*, low-temperature–tolerant *Neisseriae*, and members of the family *Enterobacteriaceae*, including *E. coli*.[138] It is important to note that several conditions can present with dysuria unrelated to acute cystitis, such as atrophic vaginitis, and are also associated with increases in vaginal bacteria and/or other disturbances in the vaginal microbiota that increase the likelihood of UTI misdiagnosis. In these circumstances and in patients who may have a difficult time performing a high-quality clean-catch specimen (e.g., morbidly obese or wheelchair-bound patients), it is reasonable to consider straight or “in-and-out” catheterization after sterile preparation of the urethra to reduce specimen contamination.[130]

The lack of clear-cut rules for the distinction of contamination from clinically significant bacteriuria stresses the importance of provider judgment in the interpretation of urine culture results. The diagnosis of a cystitis episode in patients with or without a history of rUTI should be based on the combination of thorough clinical assessment with urine testing, with careful consideration of the specimen quality, bacterial identity and quantity, and possible comorbid microbial disturbances.

5. **Cystoscopy and upper tract imaging should not be routinely obtained in the index patient presenting with rUTI. (*Expert Opinion*)**

Cystoscopy and upper tract imaging are not routinely necessary in patients with localized rUTI due to low yield of anatomical abnormalities. However, if a patient does not respond appropriately to treatment of localized UTI (i.e., poor symptomatic or microbiological response to initial treatment or rapid recurrence of infection, if particularly with the same organism repeatedly), clinicians may suspect that the patient exhibits complicating factors which necessitate further evaluation of the urinary tract via cystoscopy and upper tract imaging. Cystoscopy may be useful in these circumstances to assess for anatomical or structural abnormalities (e.g., bladder diverticuli, ectopic ureteral orifices, ureteral duplication, presence of foreign bodies). In patients with previous pelvic surgery, cystoscopy can be helpful to assess for anatomic abnormalities from the previous surgery, including urethral stricture or obstruction, foreign body such as mesh, bladder stones, fistula, or urethral/bladder diverticulum.

In a single-institutional cohort study of 163 women who had abdominopelvic imaging available, cystoscopy identified only 9 cases with significant clinical findings. Of those, only five cases were uniquely identified on cystoscopy and missed on imaging modalities.[155] Further, a meta-analysis reviewing the utility of cystoscopy, imaging, and urodynamics found that cystoscopy was not warranted, and imaging was unlikely to be of value in the absence of symptoms of upper tract disease or other gynecological problems in women presenting with rUTI.[156] In patients with gross hematuria in the presence of a positive urine culture and no risk factors for urothelial malignancy (e.g., age under 40, non-smoker, no environmental risk), cystoscopy is not necessary. If any risk factors are present, cystoscopy should be performed. Additionally, further evaluation for bladder cancer should be performed in the presence of gross hematuria without documented infection.

Upper tract imaging is not routinely necessary in the evaluation of localized rUTI, due to low yield for clinically significant findings. In a prospective observational study of the diagnostic yield of intravenous urography (IVU) with respect to referral source and presenting features, 91.7% of patients presenting with rUTI had normal IVU.[157] Further, Fair et al. reported that only 5.5% of IVUs were considered to have positive findings in a population of 164 female patients with a history of rUTI; however, none of the findings affected management approach.[158] Higher yield may be found in "high-risk" patients, such as those presenting with gross hematuria, persistent microscopic hematuria between infections, pyelonephritis, or other instances of atypical presentation.[159] For any patient with suspicion for pyelonephritis, or history of hematuria or renal calculi, upper tract imaging is recommended.

6. **Clinicians may offer patient-initiated treatment (self-start treatment) to select rUTI patients with acute episodes while awaiting urine cultures. (*Conditional Recommendation; Evidence Level: Grade C*)**

In select circumstances, employing a shared decision-making process with informed patients, initiation of a short treatment course of antibiotic therapy at the discretion of the patient (self-start) therapy may be offered for acute symptomatic episodes in patients with diagnosis of rUTI. (**Table 4**) Two trials emerged in the literature analysis that compared intermittent versus daily dosing for self-start treatment.[160, 161] These 2 medium risk of bias trials found no difference between intermittent dosing versus daily dosing in risk of ≥1 UTI over 12 months (2 studies; RR: 1.15, 95% CI: 0.88 to 1.50, I^2=0%). One of the trials that examined self-start therapy in the context of prophylaxis after exposures to different possible UTI-predisposing conditions (e.g., sexual intercourse, traveling, working or walking for a long time, diarrhea or constipation) found that a single dose of antibiotics was no different than a short course of daily antibiotics (RR: 1.15; 95% CI: 0.87 to 1.51).[161] Initial and subsequent antibiotics varied (e.g., nitrofurantoin, TMP-SMX, norfloxacin, ciprofloxacin, amoxicillin, cefaclor, ceruroxime); selection of antibiotics was based on susceptibility testing and prior use. One medium risk of bias crossover trial (N=38) found that intermittent self-administered TMP-SMX for treatment of acute symptoms was associated with increased risk of ≥1 UTI versus daily prophylactic TMP-SMX (68% versus 6.1%; RR: 11.16; 95% CI: 2.86 to 43.63).[162] Intermittent dosing was also associated with increased UTI frequency (2.2 versus 0.2 microbiologically confirmed episodes per patient-year; p<0.001). Most UTI episodes in women on intermittent dosing resolved with single-dose TMP-SMX treatment, and the rest responded to 10- to 20-day courses of antibiotics. There was no difference in risk of any adverse event (8.8% versus 15.2%; RR: 0.58; 95% CI: 0.15 to 2.24).

Although the original concept behind self-start therapy allowed for women to treat their UTI without obtaining a culture, given more recent goals to reduce overuse of antibiotics and the development of antibacterial resistance, the Panel recommends obtaining culture data for symptomatic recurrences when feasible. However, the Panel appreciates that, in certain situations, procurement of a urine culture will not be possible and empiric therapy may be allowed in select circumstances when the clinician deems such patients reliable with communication and self-assessment of symptoms. Patients must also understand the need to limit frequent or extended courses of antimicrobial therapy. Self-start therapies should utilize the choices of antibiotics that would be prescribed for acute symptoms (**Table 4**), accounting for the patient's prior culture and sensitivities as well as local antibiograms. Antibiograms provide the clinician with critical data regarding choice of agents, particularly when selecting empiric antibiotics pending urine culture and sensitivity results. Documentation by the clinician of the frequency of such self-initiated treatment episodes and course of symptom resolution will assist in defining an individualized strategy for therapy and determining necessity for alterations in strategy.

Asymptomatic Bacteriuria

7. **Clinicians should omit surveillance urine testing, including urine culture, in asymptomatic patients with rUTIs. (*Moderate Recommendation; Evidence Level: Grade C*)**

Without appropriate symptoms, bacteriuria of any magnitude is considered ASB. An expanding body of literature is examining the factors that define asymptomatic status in individuals with ASB. These patients may often report urinary odor, cloudy urine, or other lower urinary tract symptoms (e.g., chronic urgency and frequency).[163] At present, there is no single, definitive symptom that clearly distinguish between ASB and a UTI, but the presence of an acute change in symptoms is likely predictive. While the IDSA notes that pregnant women and patients scheduled to undergo invasive urinary tract procedures where a breach of the urothelial mucosa is anticipated to benefit from treatment for ASB, substantial evidence supports that other populations, including women with diabetes mellitus and long-term care facility residents, do not require or benefit from additional evaluation or antimicrobial treatment.[164] Therefore, clinicians should, in general, omit testing for malodorous and/or cloudy urine when not accompanied by an acute change in other lower urinary tract symptoms. This diagnostic stewardship intervention can reduce overtreatment of ASB.

In women with rUTIs, there is no evidence that identification of ASB between UTI episodes provides useful prognostic information. Prospective observational studies have found no differences in rates of hypertension, chronic kidney disease, renal dysfunction, abnormal renal imaging, or mortality in women with or without bacteriuria.[10] Additionally, evidence exists to suggest a lack of effectiveness of treatment for ASB, which serves as indirect evidence that identification of ASB by surveillance testing would not result in improved clinical outcomes, unless an alternative effective treatment exists.[50]

8. **Clinicians should not treat ASB in patients. (*Strong Recommendation; Evidence Level: Grade B*)**

Evaluation and treatment of rUTIs should be performed only when acute cystitis symptoms are present. There is no evidence that treatment of ASB results in improved clinical outcomes, and there is clear evidence that these practices can cause harm (e.g., antibiotic side effects, development of opportunistic infections [e.g., *C. difficile*], antibiotic resistance). Thus, in those with rUTI, it is paramount to identify and treat only acute cystitis episodes.[165] One randomized trial of women (N=673; median 40 years of age) with a history of rUTIs and ASB found that antibiotic treatment versus no antibiotics was associated with an increased risk of symptomatic recurrence (47% versus 13%; RR: 3.17; 95% CI: 2.55 to 3.90) and development of antibiotic-resistant organisms.[50] These findings suggest that ASB may actually prevent the development of symptomatic UTIs. In addition, a recent systematic review concluded that antimicrobial treatment of ASB does not appear to improve microbiologic outcomes, morbidity, or mortality.[166] Current evidence also indicates that screening/treatment of ASB does not reduce UTI rates, morbidity, or mortality in patients with complicating factors (i.e., elderly, immunosuppressed, renal transplant patients, diabetics).[10, 167] A recent large cohort study sought to determine predictors of bacteremia from a presumed urinary source in hospitalized patients with ASB and no systemic signs of infections. The study reported a mixed population but when looking at just female patients, those without vital sign abnormalities or leukocytosis, had a predicted probability of bacteremia of 2% or less.[168] This evidence continues to strengthen the arguments for non-treatment of ASB. The only clearly recognized indications for screening/treatment of ASB are 1) pregnant women, and 2) patients undergoing elective urologic surgery.[10, 169] At this time, it is clear that symptoms such as malodorous and/or cloudy urine, although associated with bacteriuria, do not warrant treatment.

ASB and Struvite Stones

Certain bacteria (most commonly *P. mirabilis*) produce urease and are associated with the development of infection (struvite) stones in the urinary tract. When infection stones are present, complete removal of the stones is required in order to eradicate the associated UTI. However, there is no clear evidence that identification and treatment of ASB caused by urease-producing organisms prevents struvite stone formation. Furthermore, this practice exposes patients to the inherent risks associated with recurrent antibiotic therapy. For these reasons, the Panel does not recommend the routine treatment of urease-producing bacteriuria (including *P. mirabilis*) in the absence of UTI symptoms or documented urinary tract stones. However, in certain patients with recurrent struvite stones, screening for and treating urease-producing bacteriuria may be indicated if other measures have not been able to prevent stone formation. This is an area where more research is required.

Antibiotic Treatment

9. **Clinicians should use first-line therapy (i.e., nitrofurantoin, TMP-SMX, fosfomycin) dependent on the local antibiogram for the treatment of symptomatic UTIs in women. (*Strong Recommendation; Evidence Level: Grade B*)**

There is limited but older data from a Cochrane review of studies published from 1977 to 2003 that compares antibiotics for uncomplicated UTIs. This systematic review included 21 RCTs (N=6,016) which compared one antibiotic versus another for treatment of uncomplicated UTI.[170] The systematic review found no differences between fluoroquinolones, β-lactams (e.g., penicillins and its derivatives, cephalosporins), or nitrofurantoin versus TMP-SMX in the likelihood of short-term (within two weeks of treatment) or long-term (up to 8 weeks) symptomatic or bacteriological cure; RR estimates were close to 1.0 for all comparisons and outcomes. Results were similar when trials of fluoroquinolones or β-lactams were stratified according to whether the duration of treatment was 3 days or 7 to 10 days, or when trials of fluoroquinolones were stratified according to the specific medication (i.e., ciprofloxacin, ofloxacin, or norfloxacin). Fluoroquinolones (2 trials; pooled RR: 0.08; 95% CI: 0.01 to 0.43; I^2=0%) and nitrofurantoin (3 trials; pooled RR: 0.17; 95% CI: 0.04 to 0.76; I^2=0%) were each associated with lower likelihood of rash than TMP-SMX. There was no difference in risk of discontinuation due to adverse events, though estimates were imprecise and favored fluoroquinolones (3 trials; RR: 0.37; 95% CI: 0.12 to 1.14; I^2=39%) and nitrofurantoin (3 trials; pooled RR: 0.69; 95% CI: 0.34 to 1.41; I^2=0%). There was no difference between fluoroquinolones or nitrofurantoin with respect to risk of resistance or other adverse events (e.g., pyelonephritis, diarrhea), though some estimates were imprecise and not all harms were reported for all comparisons. There was no difference between β-lactams and TMP-SMX in rates of rash or other harms.

The systematic review also found no differences between nitrofurantoin or fluoroquinolones versus β-lactams in short- or long-term symptomatic or bacteriological cure.[170] Fluoroquinolones were associated with decreased risk of rash compared with β-lactams (2 trials; RR: 0.10; 95% CI: 0.02 to 0.56; I^2=0%); there were no other statistically significant differences between fluoroquinolones or nitrofurantoin versus β-lactams in likelihood of short- or long-term symptomatic or bacteriological cure, though some estimates were imprecise. Data on risk of resistance was very sparse and imprecise.

A systematic review that evaluated the comparative effectiveness of different antibiotics for localized UTIs[171] included 12 RCTs (N=5,514), 11 of which were published from 2002 to 2009. Antibiotics assessed in the studies reviewed were amoxicillin-clavulanate, gatifloxacin, ciprofloxacin, norfloxacin, TMP-SMX, nitrofurantoin, fosfomycin, and pivmecillinam. A network meta-analysis was performed with results reported using ciprofloxacin as the reference treatment. The network meta-analysis found amoxicillin-clavulanate to be inferior to ciprofloxacin for likelihood of short-term (5 days to 2 weeks) clinical cure (OR: 0.07; 95% CI: 0.02 to 0.24), long-term (29 to 49 days) clinical cure (OR: 0.31; 95% CI: 0.19 to 0.53), and short-term bacteriological cure (OR: 0.17; 95% CI: 0.08 to 0.35). However, there was only a single trial of amoxicillin-clavulanate. There were no statistically significant differences between other antibiotics versus placebo in the likelihood of short- or long-term clinical or bacteriological cure. In a randomized trial of women with localized UTI, five-day nitrofurantoin compared with single-dose fosfomycin resulted in a significantly greater likelihood of clinical and microbiological resolution at four weeks after therapy.[172]

Gatifloxacin, which is not currently available in the United States or Canada at the time of this publication, generally performed similarly to ciprofloxacin, with other antibiotics trending towards inferior results. Therefore, the review concluded that ciprofloxacin and gatifloxacin appear to be the most effective treatments for UTI, and amoxicillin-clavulanate the least effective. However, all analyses were based on small numbers of trials; no antibiotic other than ciprofloxacin was evaluated in more than three trials. There were no statistically significant differences between other antibiotics versus ciprofloxacin in risk of adverse events, though estimates were imprecise. In addition to the small number of trials available for each comparison within the network, other shortcomings of this analysis include failure to report direct and indirect estimates separately, the consistency between direct and indirect estimates, and uncertainty in treatment rankings.[173]

This systematic review highlights a key concept discussed in the IDSA 2011 Guideline for treatment of acute cystitis. Specifically, if antimicrobial therapies for UTI are compared based upon efficacy in achieving clinical and/or bacteriological cure, there is relatively little to distinguish one agent from another. However, the IDSA Guideline introduced the concepts of *in vitro* resistance prevalence and ecological adverse effects of antimicrobial therapy, or collateral damage, as key considerations in choosing UTI treatments.[146] Although the aforementioned studies suggest the efficacy of fluoroquinolones, they are not among the suggested first-line agents in the United States due to increasing resistance and potential adverse events. The three first-line agents available in the United States (i.e., nitrofurantoin, TMP-SMX, fosfomycin) are effective in treating UTI and not associated with significant risk of treatment-associated toxicity or complications.[146] Nitrofurantoin specifically shows exceptional durability against emergence of resistance.[174] TMP-SMX is not recommended for empiric use in areas where local resistance rates exceed 20%.[146] **Table 4** shows first-line agents recommended by the IDSA Guideline.[146] Second-line or alternate therapies include β-lactam agents or fluoroquinolones and are generally chosen because of resistance patterns and/or allergy considerations. With the exception of fosfomycin, single-dose antibiotics should not be used in the treatment of patients with rUTI.[146] As noted, fluoroquinolone agents have potentially adverse side effect profiles, including QTc prolongation and tendon rupture, and increased risk of aortic rupture leading to U.S. Food and Drug Administration (FDA) black box warnings in recent years.[175]

The recent FDA approval of pivmecillinam, an oral prodrug of the amidinopenicillin antibiotic mecillinam, for uncomplicated UTI presents a valuable opportunity to address the need for new treatments for acute bacterial cystitis. A recent systematic review of the safety of pivmecillinam revealed a benign profile.[176] Together with the evidence for efficacy of and minimal resistance to pivmecillinam, the findings of this review support the position of pivmecillinam as an available treatment for uncomplicated UTI when multi-drug resistance is a concern. The FDA approved sulopenem etzadroxil and probenecid oral tablets for the treatment of uncomplicated UTI caused by certain bacteria (i.e., *E. coli, K. pneumoniae, or P. mirabilis*) in adult women who have limited or no alternative oral antibacterial treatment options (https://www.fda.gov/drugs/news-events-human-drugs/fda-approves-new-treatment-uncomplicated-urinary-tract-infections-adult-women-who-have-limited-or-no).

Finally, gepotidacin (FDA-approved on March 2025), is the first new antibiotic in almost 30 years for uncomplicated UTI in female adults and pediatric patients 12 years of age and older. Gepotidacin is the first of a new class of oral triazaacenaphthylene antibiotics that inhibits bacterial DNA replication by blocking two different Type II topoisomerase enzymes. This provides activity against most target uropathogens (i.e., ***E. coli*** and ***S. saprophyticus)***, and ***Neisseria gonorrhoeae (N. gonorrhoeae)***, including isolates resistant to current antibiotics. Gepotidacin was compared to nitrofurantoin and was shown to have similar therapeutic success (nitrofurantoin: 47% therapeutic success; gepotidacin: 51% therapeutic success).[177, 178]

Although these recommendations are from the uncomplicated UTI literature, the Panel supports management of each individual UTI episode, even among those with rUTI, per the IDSA guidelines for cystitis.[146]

TABLE 4: First-line therapy for the treatment of localized symptomatic UTI

Treatment effects	Nitrofurantoin (monohydrate/macrocrystals)	TMP-SMX	Fosfomycin
Cure rate	88% to 93%	90% to 100%	83% to 91%
Antimicrobial spectrum	narrow: *E. coli, S. saprophyticus*	typical uropathogens	Covers VRE, ESBL GNRs
Collateral damage*	No	Minimal	No
Resistance	Low, stable X 50y	Increasing	Currently low
Dose & duration	100 mg BID X 5d	One DS BID X 3d	3 g single dose

*Collateral damage is defined as ecological adverse effects of antimicrobial therapy

10. Clinicians should treat rUTI patients experiencing acute cystitis episodes with as short a duration of antibiotics as reasonable, generally no longer than seven days. (*Moderate Recommendation; Evidence Level: Grade B*)

There is limited high quality up-to-date evidence of comparative trials on the length of antibiotic therapies for complete resolution of UTI symptoms. Generally, all antibiotics have risks; as such, stewardship should be exercised to balance symptom resolution with reduction in risk of recurrence.

Two systematic reviews compared shorter versus longer courses of antibiotics for UTI.[179-181] One review included 15 trials (N=1,644) which found that single-dose antibiotics were associated with increased risk of short-term (<2 weeks after treatment) bacteriological persistence versus short-course (3 to 6 days; 5 studies; RR: 2.01; 95% CI: 1.05 to 3.84; I^2=36%) or long-course (7 to 14 days; 6 studies; RR: 1.93; 95% CI: 1.01 to 3.70; I^2=31%) antibiotic therapy.[181] There were no differences in risk of longer-term (>2 weeks) bacteriological persistence, short-term symptomatic persistence, risk of reinfection, any adverse event, or discontinuation due to adverse events.

Another systematic review included 32 trials (N=9,605) and it found that three-day courses of antibiotics, irrespective of class, were associated with increased risk of long-term (4 to 10 weeks from end of treatment) bacteriological failure (18 studies; RR: 1.31; 95% CI: 1.08 to 1.60; I^2=30%) versus more prolonged (5 to 10 day) therapy, but there were no differences in risk of short-term (2 to 15 days from end of treatment) bacteriological failure (31 studies; RR: 1.19; 95% CI: 0.98 to 1.44; I^2=0%) or short- or long-term symptomatic failure (24 studies; RR: 1.06; 95% CI: 0.88 to 1.28; I^2=15% and 10 studies; RR: 1.09; 95% CI: 0.94 to 1.27, respectively).[179, 180] Short-course therapy (3 day) was associated with increased risk of short- and long-term bacteriological failure (18 studies; RR: 1.37; 95% CI: 1.07 to 1.74; I^2=0% and RR: 1.43; 95% CI: 1.19 to 1.73; I^2=0%, respectively), but effects on short- or long-term bacteriological failure was not statistically significant. A 3-day course of antibiotics was associated with decreased risk of adverse effects (29 studies; RR: 0.83, 95% CI: 0.74 to 0.93; I^2=14%), discontinuation due to adverse events (24 studies; RR: 0.28 to 0.91; I^2=42%), and gastrointestinal adverse events (24 studies; RR: 0.81; 95% CI: 0.67 to 0.97; I^2=11%) compared with longer duration therapy.[179, 180] As such, clinicians should treat rUTI patients with as short a duration of antibiotics as reasonable, generally no longer than seven days.

11. In patients with rUTIs experiencing acute cystitis episodes associated with urine cultures resistant to oral antibiotics, clinicians may treat with culture-directed parenteral antibiotics for as short a course as reasonable, generally no longer than seven days. (*Expert Opinion*)

Many such infections will be caused by organisms producing ESBLs. Generally, such organisms are susceptible only to carbapenems. However, before considering that these infections require intravenous antimicrobials, clinicians should order fosfomycin susceptibility testing, as many MDR uropathogens, including ESBL-producing bacteria, retain susceptibility to fosfomycin and/or nitrofurantoin. Consultation with an infectious diseases specialist may be appropriate for assistance in the management of such infections.

Antibiotic Prophylaxis

12. Following discussion of the risks, benefits, and alternatives, clinicians may prescribe antibiotic prophylaxis to decrease the risk of future UTIs in women of all ages previously diagnosed with UTIs. (*Conditional Recommendation; Evidence Level: Grade B*)

For evidence-based treatment of rUTIs, a large body of evidence exists in support of antibiotic prophylaxis. The systematic review for this Guideline identified 28 trials evaluating antibiotics for prevention of rUTI.[160, 161, 182-208] Most were rated as medium to high risk of bias, predominately for non-reporting of factors used in the assessment of bias (e.g., unclear randomization, allocation concealment, or blinding methods; high or unclear attrition; failure to report intention-to-treat analysis). Sample sizes ranged from 26 to 308 (total N=2,758). Ten trials demonstrated that antibiotics perform better than placebo,[182, 192, 195, 196, 199, 203-205, 207, 208] and the results were consistent across antibiotics. Ten trials evaluated nitrofurantoin,[182, 186, 187, 189, 190, 194, 198, 200, 202, 207] five trials TMP-SMX,[183, 184, 201, 207, 208] four trials TMP,[187, 188, 197, 206] one trial cephalexin,[192] one trial fosfomycin,[191, 198] and one trial tested various antibiotics in intermittent versus daily regimens.[161] In addition, some older studies used antibiotics that are no longer used routinely in practice (e.g., norfloxacin,[188, 199, 200, 203] perfloxacin,[193] prulifloxacin,[191] cinoxacin,[195, 196, 204-206] and cefaclor[187]). The duration of preventive treatment ranged from 6 to 12 months. In 8 trials, the mean age of enrollees was ≥50 years;[161, 184, 191, 194, 197, 198, 202, 207] in the other trials the mean age was in the 30's or low 40's, so both perimenopausal and postmenopausal women and younger premenopausal women have been studied in these trials. The number of UTIs in the 12 months prior to initiating prophylaxis ranged from 2 to 7 in trials that reported this information.

While the studies reviewed are relevant to the issue of UTI prevention, it must be noted that most of the relevant RCT studies on antibiotic prophylaxis were published prior to 1995. While the quality of these studies is acceptable, results from them may be less applicable (i.e., prophylaxis may be less effective) given the changing antibiotic resistance patterns that have occurred over time. As such, results should be interpreted in light of current resistance patterns.

Prophylactic Antibiotics Associated with Decreased Likelihood of UTI Recurrence Compared with Placebo

When comparing prophylactic antibiotic use to placebo or no antibiotic, antibiotics were associated with a decreased likelihood of experiencing ≥1 UTI recurrence versus placebo or no antibiotics (11 studies; RR: 0.26, 95% CI: 0.18 to 0.37, I^2=14%; absolute risk difference [ARD]: -46%, 95% CI: -56% to -37%).[182, 192, 194-196, 199, 203-205, 207, 208] All of the trials evaluated daily dosing of antibiotics except for one,[208] which evaluated intermittent dosing of TMP-

SMX with sexual intercourse. This trial also found antibiotics to be more effective than placebo for preventing rUTI (RR: 0.11; 95% CI: 0.02 to 0.75). All of the trials compared antibiotics versus placebo except for one study of nitrofurantoin versus no antibiotics that reported similar effects on risk of rUTI (RR: 0.34; 95% CI: 0.22 to 0.51).[194]

With antibiotic use, there is an increased risk of adverse events, including pulmonary and hepatic side effects. Antibiotics were associated with increased risk of any adverse event (6 studies; RR: 1.73, 95% CI: 1.08 to 2.79, I^2=0%; ARD: 12%, 95% CI: 1% to 22%)[192, 195, 196, 199, 205, 208] and vaginitis (3 studies; RR: 3.01, 95% CI: 1.27 to 7.15, I^2=0%; ARD: 18%, 95% CI: 0.05 to 0.32).[195, 199, 208] There was no interaction between the antibiotic used and risk of adverse events. There were no differences in risk of withdrawal due to adverse events (4 studies; RR: 2.76, 95% CI: 0.64 to 11.84, I^2=0%)[192, 195, 199, 204] or gastrointestinal adverse events specifically (2 studies; RR: 2.52, 95% CI: 0.28 to 22.87, I^2=0%),[199, 208] but data were sparse and estimates imprecise.

Overall, antibiotic prophylaxis reduced the number of clinical recurrences when compared to placebo in premenopausal and postmenopausal women with rUTIs. The results of the trials on prophylactic antibiotics consistently demonstrate the positive effect of this preventive treatment, while acknowledging the increase in mild, moderate, and severe adverse events associated with antibiotic use. The effect of the antibiotic prophylaxis lasted during the active intake time period. Once the antibiotics were stopped, UTIs recurred and equaled the placebo arm outcomes.

Comparison of Prophylactic Antibiotics

Among eight trials of one antibiotic versus another for prevention of rUTI,[187, 189-191, 198, 200, 206, 207] six evaluated comparisons involving nitrofurantoin.[187, 189, 190, 198, 200, 207] Nitrofurantoin was compared against fosfomycin (one trial),[198] TMP (one trial),[190] TMP-SMX (one trial),[207] norfloxacin (two trials),[189, 200] and cefaclor (one trial).[187]

There was no difference between nitrofurantoin versus other antibiotics in risk of experiencing ≥1 UTI (6 studies; RR: 0.81, 95% CI: 0.63 to 1.03, I^2=0%). When stratified according to the specific antibiotic to which nitrofurantoin was compared, findings were also generally consistent in showing no differences in risk of UTI recurrence, with no differences versus fosfomycin, TMP-SMX, norfloxacin, and cefaclor (p for interaction is 0.79). However, nitrofurantoin was associated with a decreased risk of rUTI compared to TMP in one trial (RR: 0.58, 95% CI: 0.36 to 0.94; ARD: -28%, 95% CI: -50% to -5%).[190] One trial of nitrofurantoin versus fosfomycin was published in 2007 (RR for ≥1 UTI: 0.87; 95% CI: 0.44 to 1.71);[198] all of the other trials were published in or before 1995.

While quinolones have been studied as prophylaxis, the use of fluoroquinolones, such as ciprofloxacin, for prophylactic antibiotic use is not recommended in current clinical practice. In 2008, the FDA issued a black box warning on the increased risk of tendinitis and tendon rupture associated with ciprofloxacin.[209] These serious side effects associated with fluoroquinolone use, which also include QT interval prolongation, seizures, and *C. difficile* infection, generally outweigh the benefits of its use for localized UTI.

There is little evidence on the benefits of rotating antibiotics used for prophylaxis. In a different population of inpatient hospital treatment of infection, informed switching strategies have been used that take the frequency of antibiotic resistance mutations into account.[210, 211] They used local antibiogram-guided therapy, which can potentially serve as a valuable strategy to curb resistance. However, there is not enough evidence in the existing published literature to reach reliable conclusions regarding the efficacy of cycling antibiotics as a means of controlling antibiotic resistance rates.

Adverse Events Associated with Prophylactic Antibiotics

There was no difference in risk of any adverse event (4 studies; RR: 1.59; 95% CI: 0.58 to 4.42; I^2=89%), but estimates were inconsistent,[189, 190, 200, 207] and nitrofurantoin was associated with increased risk of study withdrawal compared to other antibiotics (i.e., norfloxacin, TMP, and TMP-SMX) (4 studies; RR: 2.42, 95% CI: 1.14 to 5.13, I^2=5%; ARD: 7%, 95% CI: 1% to 13%).[187, 189, 190, 200] All trials except for one[200] found nitrofurantoin associated with increased risk of any adverse event (RR estimates ranged from 2.00 to 2.40). There were no differences between nitrofurantoin and other antibiotics in risk of gastrointestinal adverse events (3 studies; RR: 1.78; 95% CI: 0.57 to 5.50; I^2=0%),[189, 198, 207] or vaginitis (2 studies; RR: 0.45; 95% CI: 0.13 to 1.54; I^2=0%),[189, 207] but estimates were imprecise. Other side effects included vaginal and oral candidiasis, skin rash, and nausea.

While nitrofurantoin remains a first-line choice for treatment of acute UTI as recommended by IDSA[146] and has been shown to be effective as a prophylactic antibiotic for UTI prevention, all antibiotics including nitrofurantoin have potential risks. These risks should be discussed with patients prior to prescribing for short-, medium-, or long-term prophylaxis. Nitrofurantoin is commonly prescribed in women of all ages and has rare but potentially serious risks of pulmonary and hepatic toxicity.[212-215] The rate of possible serious pulmonary or hepatic adverse events has been reported to be 0.001% and 0.0003%, respectively.[216] One 2015 systematic review[217] observed no pulmonary or hepatotoxic events related to nitrofurantoin among 4,807 patients from 27 controlled trials. A 2018 retrospective chart audit[218] of an urban academic medical center found 0.7% of patients experienced possible serious pulmonary or hepatic adverse effects, and 0.15% (5/3,400 patients) were highly suspicious for having a serious lung or liver reaction. These patients were more likely to have long-term exposure to nitrofurantoin, highlighting the need for caution when prescribing long-term and avoiding nitrofurantoin in patients with chronic lung disease.

Nitrofurantoin use in older adults has been controversial. Nitrofurantoin is listed as a potentially inappropriate medication for older adults by the AGS Beers Criteria,[219] with the strength of recommendation by the Panel as strong and a quality of evidence of low. The 2015 Beers update (maintained in the 2023 update) has been modified to recommend avoidance of nitrofurantoin when creatinine clearance is below 30 mL/min. The rationale for avoiding nitrofurantoin included pulmonary toxicity, hepatotoxicity, and peripheral neuropathy, with concern about long-term use if other alternatives are available for use. Nitrofurantoin-induced lung injury[220-225] can occur in the acute, subacute or chronic setting, most commonly presenting with a dry cough and dyspnea.[226] The mechanism underlying pulmonary toxicity is related to the direct effects of nitrofurantoin metabolites on lung tissue.[227] Acute pulmonary reactions appear after a mean of nine days from starting nitrofurantoin therapy, while symptoms of subacute and chronic pulmonary reactions develop between one and six months of treatment, respectively.[222] In a 1980 analysis of 921 reported cases by Holmberg et al.,[212] 47% of cases of chronic respiratory disease occurred after more than 12 months of nitrofurantoin therapy. Risk assessment, shared decision-making, and clinical monitoring is important to avoid the potential adverse events associated with nitrofurantoin.

Potential adverse effects of gastrointestinal disturbances and skin rashes are commonly associated with antibiotics, including TMP, TMP-SMX, cephalexin, and fosfomycin.[228, 229] Gastrointestinal disturbances and skin eruptions are the most common adverse reactions associated with TMP and TMP-SMX.[230] TMP-SMX has been uncommonly associated with other adverse effects.[231] These adverse effects include neurologic effects (e.g., aseptic meningitis, tremor, delirium, gait disturbances), decreased oxygen carrying capacity (e.g., methemoglobinemia, blood dyscrasia), toxic epidermal necrolysis (e.g., drug hypersensitivity, fixed drug eruption), reproductive toxicity (e.g., structural malformations including neural tube, small for gestational age, hyperbilirubinemia), interactions with other drugs (e.g., inhibition of the P450 system), hypoglycemia, hyperkalemia and nephrotoxicity. Long-term administration of TMP-SMX appears to be safe, though hematologic and laboratory monitoring may be indicated.

Consideration of Antibiotic Resistance

In general, there is sparse reporting of antibiotic resistances, with little data specifically on the impact of long-term antibiotic therapy on antibiotic resistance. However, there are data on the effects of antibiotic prescribing on antimicrobial resistance in individual patients.[232] A 2010 systematic review and meta-analysis demonstrated that individuals prescribed with an antibiotic for UTI develop bacterial resistance to that antibiotic. In 5 studies of urinary tract bacteria (14,348 participants), the pooled odds ratio for resistance was 2.5 (95% CI: 2.1 to 2.9) within 2 months of antibiotic treatment and 1.33 (95% CI: 1.2 to 1.5) within 12 months. The effect is greatest in the month immediately after treatment but may persist for up to 12 months.[232] Antibiotic resistance is related to an individual's bacterial gene pool since resistance is carried on plasmids and integrons and can be transferred between commensal organisms and potential pathogens. As such, even transient use of antibiotics can affect the carriage of resistant organisms and impact the endemic level of resistance in the population. The potential harms related to acquiring an antibiotic-resistant infection should be factored into the decision for antibiotic treatment and for prophylaxis for UTI.

Dosing and Duration of Prophylactic Antibiotics

The most tested schedule of antibiotic prophylaxis (TMP, TMP-SMX, nitrofurantoin, cephalexin) was daily dosing. However, fosfomycin used prophylactically is dosed every 10 days. Four trials compared different antibiotic dosing strategies.[160-162, 193] Three trials[160-162] compared intermittent versus daily dosing, and one trial[193] compared once weekly versus once monthly dosing.

As previously reviewed under the discussion of self-start therapy, two medium risk of bias trials found no difference between intermittent dosing versus daily dosing in risk of ≥1 UTI (2 studies; RR: 1.15, 95% CI: 0.88 to 1.50, I^2=0%).[160, 161] One of the trials compared a single dose of antibiotics for exposures to different UTI-predisposing conditions (e.g., sexual intercourse, travelling, working or walking for a long time, diarrhea or constipation) versus daily antibiotics (RR: 1.15; 95% CI: 0.87 to 1.51).[161] The other intermittent dosing trial compared a single dose of ciprofloxacin after sexual intercourse with daily dosing (RR: 1.24; 95% CI: 0.29 to 5.32).[160]

The duration of antibiotic prophylaxis in the literature ranged from 6 to 12 months, and after stopping, the frequency of UTI has been shown to resume to the prior state of rUTI frequency. In clinical practice, the duration of prophylaxis can be variable, from three to six months to one year, with periodic assessment and monitoring. Some women carry on with continuous or post-coital prophylaxis for years to maintain the benefit without adverse events, but it should be noted that continuing prophylaxis for years is not evidence-based.

Continuous antimicrobial prophylaxis regimens for women with rUTIs have been recommended by several trials.[187, 189, 190, 206-208] The dosing options for continuous prophylaxis include the following:

- TMP 100 mg once daily
- TMP-SMX 40 mg/200 mg once daily
- TMP-SMX 40 mg/200 mg thrice weekly
- Nitrofurantoin monohydrate/macrocrystals 50 mg daily
- Nitrofurantoin monohydrate/macrocrystals 100 mg daily
- Cephalexin 125 mg once daily
- Cephalexin 250 mg once daily
- Fosfomycin 3 g every 10 days

Antibiotic Prophylaxis in Women Who Experience Post-Coital UTIs

In women who experience UTIs temporally related to sexual activity, antibiotic prophylaxis taken before or after sexual intercourse has been shown to be effective and safe. This use of antibiotics is associated with a significant reduction in recurrence rates. Additionally, intermittent dosing is associated with decreased risk of adverse events including gastrointestinal symptoms and vaginitis.

In a 1990 randomized double-blind placebo-controlled trial of 27 sexually active women with a median age of 23, post-coital antibiotics were shown to be more effective than placebo in reducing UTI recurrences.[208] Other older studies of post-coital antibiotic prophylaxis published between 1975 and 1989[233-236] were non-randomized but had similar results supporting post-coital dosing. In one study of 135 women, post-coital dosing was as effective as daily dosing.[160] Antibiotic prophylaxis should be offered to women with sexual activity-related rUTIs which is to be taken before or after sexual intercourse. The antibiotic prophylaxis approach targets the preventive therapy to the time frame when these women are most vulnerable to UTIs, thus minimizing use of antibiotics, decreasing risk of adverse events, and potentially reducing direct and indirect costs of rUTIs.

Recommended instructions for antibiotic prophylaxis related to sexual intercourse include taking a single dose of an antibiotic immediately before or after sexual intercourse. Dosing options for prophylaxis include the following:

- TMP-SMX 40 mg/200 mg
- TMP-SMX 80 mg/400 mg
- Nitrofurantoin 50 to 100 mg
- Cephalexin 250 mg

Non-Antibiotic Prophylaxis

13. Clinicians should offer cranberry as an option for prophylaxis for women with rUTIs. (*Moderate Recommendation; Evidence Level: Grade B*)

There has been a growing concern regarding antibiotic resistance in the setting of rUTI. In 2015, the World Health Organization (WHO) increased awareness of the issue of the growing world-wide phenomenon of antimicrobial resistance through its publication of the Global Action Plan on Antimicrobial Resistance (AMR).[237] AMR is one factor that has led to an increasing interest in the scientific community to study non-antibiotic modalities in the prevention of rUTI, including the use of probiotics and the consumption of cranberry products.

Cranberries have been studied as a preventative measure for UTI for decades, but recently cranberry has been subject to an increasing number of randomized clinical trials. These studies have used cranberry in a variety of formulations, including juice, cocktail, and tablets. The proposed mechanism of action is thought to be related to proanthocyanidins (PACs) present in cranberries and their ability to prevent the adhesion of bacteria to the urothelium. Most healthcare providers would agree that a cranberry supplement standardized to at least 36 mg of bioavailable PAC is a good option for rUTI prophylaxis. A recent systematic review and meta-analysis supports this, suggesting that cranberry supplements standardized to at least 36 mg PACs are more effective in reducing UTI recurrence than those which contain less than 36 mg.[238] It must be noted however, that PACs are found in varying concentrations depending on formulation used, and many of the cranberry products used in the studies noted below were explicitly formulated for research purposes. The availability of such products to the public is a severe limitation to the use of cranberries for rUTI prophylaxis outside the research setting and must be discussed with patients. Juice studies have used a variety of juices and cocktails in varying volumes of daily consumption and have included cranberry of varying concentrations within the overall volume of product ingested. Likewise, cranberry tablets include variability in dosing and are not subject to the same regulatory environment as antimicrobial drugs. Many studies do not include validation of PAC dosage. Clinical studies have also not routinely reported side effects, but the available evidence suggests the potential harms from cranberry use are minimal.

The 2019 systematic review identified 8 randomized trials including cranberry versus placebo/no cranberry (6 RCTs; one with a *Lactobacillus* arm)[239-244] and cranberry versus antibiotics (2 RCTs).[183, 197] Four RCTs studied cranberry in a beverage form, and five studied cranberry tablets/capsules. Risk of bias was variable across the studies. Cranberry was associated with decreased risk of experiencing at least 1 UTI recurrence than placebo or no cranberry (5 trials; RR: 0.67, 95% CI: 0.54 to 0.83; ARD: -11%, 95% CI: -16% to 5%).[239-243] Kontiokari et al. found a 20% reduction in UTIs (versus control) with 50 mL of daily cranberry-lingonberry juice concentrate over 6 months.[239] Maki et al. used one 240 mL serving of cranberry beverage daily versus placebo and found the antibiotic use-adjusted incidence rate ratio (IRR) to be 0.61 (95% CI: 0.41 to 0.91; $P=0.016$).[240] Stothers et al. found that both cranberry juice and cranberry tablets significantly decreased the number of patients experiencing at least 1 symptomatic UTI per year (to 20% and 18%, respectively) compared with placebo (to 32%; $p<0.05$).[241] Takahashi et al. randomized women to 125 mL of daily cranberry juice (UR65) or placebo over 24 weeks. In a subgroup analysis of women aged 50 years or more, relapse of UTI was observed in 16 of 55 patients (29.1%) in the cranberry group versus 31 of 63 (49.2%) in the placebo group.[242] Cranberry fruit powder was also found to reduce UTIs significantly (10.8% versus 25.8%; $p=0.04$) in women who received 500 mg daily for 6 months.[243] This study noted that the cranberry fruit powder, which includes the pulp, seeds, and peel, had a PAC content of 0.56%.

There was no statistically significant difference between daily cranberry versus antibiotics in risk of experiencing ≥1 UTI after 6 or 12 months, but the pooled estimate was based on only 2 trials, favored antibiotics (not statistically significant), and was imprecise (RR: 1.30; 95% CI: 0.79 to 2.14; $I^2=68\%$).[183, 197] Beerepoot et al.[183] compared cranberry versus TMP-SMX (RR: 1.09; 95% CI: 0.92 to 1.28). The study found cranberry was associated with a decreased number of clinical UTI recurrences (mean 4.0 versus 1.9; $p=0.02$) and shorter time to first recurrence (median 4 versus 8 months; $p=0.03$); however, effects were no longer present in the three months following discontinuation of treatment. Researchers noted cranberry was associated with a lower risk of resistance in *E. coli* isolates than TMP-SMX in patients with symptomatic recurrence (resistance to TMP or TMP-SMX ~15% versus ~90% and resistance to amoxicillin ~25% versus ~80%). McMurdo et al.[197] compared cranberry to TMP (RR: 1.76; 95% CI: 1.00 to 3.09) and found no difference in time to recurrence (median: 84 versus 91 days; p=0.48). Additionally, it was found that 31.6% of microbial isolates in symptomatic UTI recurrences were TMP-susceptible in the TMP and cranberry groups combined.

Not all studies have included a methodology to examine a hypothesized mechanism of action in humans, which have included both inhibition of adherence mechanisms and urinary content changes that make the urine generally less habitable to uropathogens. Clinical studies have also not routinely reported side effects. Cranberry, in a formulation that is available and tolerable to the patient, may be offered as prophylaxis including oral juice and tablet formulations as there is not sufficient evidence to support one formulation over another when considering this food-based supplement. In addition, there is little risk to cranberry supplements, further increasing their appeal to patients. However, it must be noted that fruit juices can be high in sugar content, which is a consideration that may limit use in diabetic patients.

For the 2022 update report, four additional studies addressed cranberry prophylaxis, three of these were in combination with different non-antibiotic agents, and one comparing high and low doses of proanthocyanidins. Thus, the new studies could not be combined with those assessed in 2019 or with each other. The study comparing doses of cranberry (proanthocyanidins) did not show a difference in UTI recurrences or in adverse events between doses,[106] nor did a study of cranberry, propolis (produced by bees), and zinc show differences compared with placebo.[107] A combination of cranberry, D-mannose, and *Lactobacillus* also did not show differences in outcomes compared with no treatment in one trial.[105] There were some data discrepancies between the abstract, tables, and figures in this study, and recalculated p-values based on reported numbers of participants; the study had reported differences in UTI rates as statistically significant, but calculated p-values were not significant. Strength of evidence for findings in all three of these studies was very low. The fourth study[104] compared high-dose cranberry with *Lactobacillus* and vitamin A to placebo, and provided low-strength evidence that fewer patients had UTI recurrences with treatment (9.1% versus 33.3%; p=0.0053). Rates of adverse events were low and did not differ between groups in any new study (very low SOE). This study did not impact the current Guideline statement because it combined cranberry with two other substances.

Two recent RCTs (N=218) investigated use of cranberry prophylaxis[245, 246] which support the data included in the 2019 rUTI Guideline.[247] Rondanelli et al.[245] randomized 46 women older than 70 years of age with diabetes taking an SGLT-2 inhibitor and with a history of at least one UTI in the past year to either a highly standardized cranberry extract formulated in phospholipids (e.g., Anthocran™ Phytosome™ 120 mg capsule with 6% to 9% PAC content) or placebo every day for 6 months. The UTI rate in the intervention group was a third of that documented in the placebo group (3% versus 9%). No adverse events or safety concerns were reported in either group. Tsiakoulias et al.[246] randomized 172 females with a history of rUTI to Cysticlean™, a standardized, high dose PAC (240 mg/day) cranberry supplement, or placebo. The number of UTIs was significantly lower in the intervention arm compared to the placebo arm (IRR: 0.49; p<0.001) and quality of life was slightly improved.

A 2024 systematic review and network meta-analysis compared cranberry juice, cranberry tablets, and increased fluid intake for preventing and managing UTIs in over 3,000 participants across 20 trials.[248] The study found that cranberry juice consumption led to a 54% lower rate of UTIs compared to no treatment and a 27% lower rate than placebo liquid. Cranberry juice also reduced antibiotic use by up to 50% compared to no treatment and significantly lessened UTI symptoms. While both increased fluid intake and cranberry tablets offered some benefits, cranberry juice provided the most substantial reduction in UTI incidence and antibiotic reliance, supporting its use as an effective non-antimicrobial intervention.

14. **Clinicians should inform patients with rUTIs that D-mannose alone for prophylaxis may not be effective in UTI prevention. (*Moderate Recommendation; Evidence Level: Grade B*)**

At the time of the last rUTI Guideline update (2022),[249] evidence was insufficient to recommend or refute the use of D-Mannose for the prevention of rUTI. Contemporary evidence from a high-quality large RCT[250] (N=598) showed no difference in UTI recurrence rate between use of D-mannose (2 g/day) and placebo (RR: 0.92; 95% CI: 0.80 to 1.05). Moderate quality evidence from this RCT shows no difference in microbiological recurrence (RR: 0.90; 95% CI: 0.60 to 1.33), number of prescribed antibiotics (adjusted IRR: 0.88; 95% CI: 0.69 to 1.12), UTI-related symptoms (adjusted IRR: 0.88; 95% CI: 0.72 to 1.08) or hospital admissions (RR: 1.47; 95% CI: 0.47 to 4.61) between the two groups. The incidence of harm were low in both groups. In addition, very low-quality evidence from one RCT[251] (N=44) showed no difference in UTI recurrence rate in participants using topical vaginal estrogen and D-mannose (2 g/day) compared to participants using vaginal estrogen alone. Very low-quality evidence suggests that treatment-related adverse events were mild and mostly gastrointestinal-related (flatulence). Thus, while D-mannose administration is unlikely to do harm, there is insufficient evidence to support its clinical benefit in preventing UTI episodes, and therefore cannot be recommended.

15. **Clinicians may offer methenamine hippurate for prophylaxis for women with rUTIs. (*Conditional Recommendation; Evidence Level: Grade C*)**

Previous studies suggested that methenamine salts provide comparable protection against UTI recurrence compared to antibiotics.[252] However, the strength of this evidence was considered very poor and deemed insufficient to recommend methenamine in previous Guideline updates. Harding et al.'s non-inferiority RCT[253] adds to this growing body of supporting evidence and the Panel now believes this provides sufficient evidence supporting methenamine as an effective alternative to antibiotics in the prevention of UTI. In this study, 240 women with rUTI were randomized to receive methenamine hippurate 1 g twice daily or a daily antibiotic (nitrofurantoin 50/100 mg, cephalexin 250 mg, or trimethotrim 100 mg) for 12 months. During treatment, the incidence rate of symptomatic, antibiotic-treated UTIs decreased substantially in both arms to 1.38 episodes per person-year (95% CI: 1.05 to 1.72 episodes per person-year) for methenamine hippurate and 0.89 episodes per person-year (95% CI: 0.65 to 1.12 episodes per person-year) for antibiotics (absolute difference: 0.49; 90% CI: 0.15 to 0.84). This absolute difference did not exceed the predefined, strict, non-inferiority limit of one UTI per person-year. There was no difference in microbiologically confirmed UTIs or harms during the study period. The limitation of the study was that it was unblinded and the prophylactic antibiotics used were heterogeneous. Moreover, long term safety data of methenamine hippurate is not available. Urine acidifiers, such as ascorbic acid (vitamin C), have been previously recommended to promote the conversion of methenamine into its bacteriostatic components, formaldehyde, and ammonia. However, urinary acidifiers have not been found to significantly lower urinary pH[254] and, in RCTs, do not enhance the effects of methenamine.[252, 255] Therefore, concominant use of vitamin C and methenamine is not recommended.

16. **When women with rUTIs have a water intake below 1.5 L/day (50 oz), clinicians may offer increased water intake for prophylaxis. (*Conditional Recommendation; Evidence Level: Grade C*)**

One medium risk of bias trial of women with rUTIs who reported <1.5 L/day of fluid intake at baseline (N=140; mean age 36 years) found increased water intake above 1.5 L was associated with fewer UTI recurrences compared with no additional fluids (mean: 1.7 versus 3.2 UTI episodes over 12 months; p<0.001).[101] Increased water intake was also associated with lower likelihood of having at least 3 UTI episodes over 12 months (<10% versus 88%) and greater interval between UTI episodes (143 versus 84.4 days; p<0.001). The increased fluid intake intervention was based on provision of three 500 mL bottles of water to be consumed daily. Daily fluid intake increased from 0.9 L/day to 2.2 L/day in the increased water intake group compared with no change in the no additional fluids group. Particularly in patients with lower fluid consumption, increasing water intake may be a low-risk, effective method to reduce the risk of rUTI.

Other Preventive Methods

Several trials were identified evaluating various other prophylactic agents, including *Lactobacillus*, herbs/supplements, intravesical hyaluronic acid/chondroitin, biofeedback, and immunoactive therapy, for prevention of rUTI.[186, 188, 194, 201, 256-261] However, the Panel cannot recommend these agents as it was not possible to draw reliable conclusions regarding their effectiveness due to the small number of trials for each treatment, imprecise estimates, and methodological shortcomings.

Lactobacillus. While *Lactobacillus* probiotics have been studied with greater interest in recent years given growing concerns for antibiotic resistance, the Panel is unable to recommend the use of *Lactobacillus* as a prophylactic agent for rUTI given the current lack of data indicating benefit in comparison to other available agents. Lack of access to the specific *Lactobacillus* strains used in the clinical studies outlined below remains a significant barrier for most healthcare providers.

A systematic review identified five trials evaluating *Lactobacillus* for prevention of rUTI.[185, 262-265] Sample sizes ranged from 30 to 238 (total N=464) and the duration of treatment ranged from 5 days to 12 months. Three trials compared *Lactobacillus* to placebo,[262, 263, 265] one trial compared *Lactobacillus* to an antibiotic,[185] and one trial compared *Lactobacillus* to a skim milk-based *Lactobacillus* growth factor.[264] All of the trials evaluated *Lactobacillus* via vaginal suppository, except for one trial[185] of oral *Lactobacillus* versus an antibiotic. *Lactobacillus* species were *rhamnosus, reuteri*, and *crispatus*.

There was no difference between *Lactobacillus* vaginal suppositories versus placebo in risk of experiencing ≥1 UTI in 3 trials of younger (mean age between 20 to 30 years of age) women (RR: 1.01; 95% CI: 0.45 to 2.26; I^2=55%).[262, 263, 265] The *Lactobacillus* species and dosing schedules varied across trials (twice weekly, daily for 5 days, or daily for 5 days then weekly for 10 weeks).

One trial (N=138; mean age of 64 years) found no differences between daily oral *Lactobacillus* (*rhamnosus* GR-1 and *reuteri* RC-14) versus TMP-SMX at 12 months in mean number of clinical UTI recurrences (3.3 versus 2.9; mean difference: 0.4; 95% CI: -0.4 to 1.5) or likelihood of experiencing ≥1 UTI (79% versus 69%; RR: 1.15; 95% CI: 0.98 to 1.34), although *Lactobacillus* was associated with shorter time to first recurrence (median: 3 versus 6 months; p=0.02).[185]

A double-blind, placebo-controlled study involving 174 premenopausal women with rUTI evalutated the effectiveness of prophylactic oral and/or vaginal probiotic supplementation over four months, with follow-up over one year. Participants were randomized into four groups: placebo, oral probiotics, vaginal probiotics, and a combination of both oral and vaginal probiotics. Results showed that vaginal probiotics, either alone or combined with oral probiotics, significantly reduced the incidence and recurrence of symptomatic UTIs compared to placebo or oral probiotics alone. At four months, UTI incidence was 70.4% in the placebo group, 61.3% with oral probiotics, 40.9% with vaginal probiotics, and 31.8% with combination vaginal and oral probiotics. The time to first UTI recurrence was significantly longer in groups receiving vaginal probiotics. No serious adverse events were reported.[266]

Herbal Therapies. Two trials evaluated herbal therapies for prevention of rUTI.[256] One medium risk of bias trial (N=174; mean age 54 years) found no difference between herbal therapy (i.e., nasturtium and horseradish) versus placebo in the mean number of UTIs after 3 months of treatment (mean UTIs: 0.43 versus 0.37; p=0.28) or 3 months following the end of treatment (mean UTIs: 0.74 versus 0.63; p=0.26).[256] A high risk of bias (open-label) trial found no differences between treatment with 3 different herbal therapies (berberine/arbutin/birch, berberine/arbutin/birch/forskolin, or PAC) for 12 weeks in risk of ≥1 UTI at 24 weeks.[257]

Intravesical Hyaluronic Acid/Chondroitin. Two small, medium risk of bias trials evaluated intravesical hyaluronic acid plus chondroitin for prevention of rUTI.[258, 259] One trial (N=54; mean age 35 years) found intravesical hyaluronic acid plus chondroitin (weekly for 4 weeks, then monthly for 5 months) was associated with decreased risk of experiencing ≥1 UTI at 12 months (52% versus 100%; RR: 0.52; 95% CI: 0.36 to 0.75), mean number of UTIs (0.67 versus 4.19; p<0.001), and longer time to UTI recurrence (185.2 versus 52.7 days; p<0.001) than intravesical saline.[258] Intravesical hyaluronic acid was also associated with better scores on the SF-36 (78.6 versus 53.1; p<0.001). Harms were not reported. Another trial (N=26; mean age 60 years) found intravesical hyaluronic acid plus chondroitin (weekly for 4 weeks, then biweekly for 4 weeks) was associated with fewer UTI episodes (1 versus 2.3; p<0.01), lower Visual Analogue Scale (VAS) pain score (1.6 versus 7.8; p<0.001), less pelvic pain and urgency/frequency symptoms (Pelvic Pain and Urgency/Frequency [PUF] scale: 11.2 versus 19.6; p<0.001), better sexual function (sexual function questionnaire: 2.4 versus 6.3; p=0.001), and better quality of life (King's Health Questionnaire: 18.4 versus 47.3; p<0.001) versus once weekly oral SMX 200 mg and TMP

40 mg 12 months after the end of treatment, though there were no differences on any of these outcomes 2 months after the end of treatment.[259] No harms were reported. While these studies show promise, further studies are needed to assess generalizability, long-term outcomes, and overall feasibility.

Immunotherapy and Biofeedback. A medium risk of bias trial (N=451; mean age of 44 years) found no differences between an immunoactive therapy (oral OM-89S, a lyophilized lysate of 18 *E. coli* strains) versus placebo or nitrofurantoin in mean number of UTIs, UTI incidence, likelihood of experiencing at least 1 UTI, or time to next UTI.[260] In a high risk of bias trial (N=86; mean age of 23 years), 12 months of uroflowmetry biofeedback reduced the prevalence of UTI to 25% while biofeedback training of the pelvic floor muscles reduced the UTI prevalence to 24%. Administration of both concurrently decreased the prevalence of UTI to only 20%. In contrast, 90% of participants who received no treatment continued to experience UTI.[261] The trial was rated as high risk of bias due to open-label design, high attrition, and failure to conduct intention-to-treat analysis.

Follow-up Evaluation

17. Clinicians should not perform a post-treatment test of cure urinalysis or urine culture in asymptomatic patients. (*Expert Opinion*)

There are no studies that address whether or not screening urinalysis or urine culture following clinical cure of a documented UTI is beneficial in those with a history of rUTI. Extrapolating from the ASB literature, the Panel does not endorse microbiological reassessment (i.e., repeat urine culture) after successful UTI treatment as this may lead to overtreatment. The Panel does recognize, however, that certain clinical scenarios, such as planned surgical intervention in which mucosal bleeding is anticipated, may prompt screening. It should again be emphasized that symptom clearance is sufficient. In patients with rapid recurrence (particularly with the same organism), clinicians may consider evaluation on and off therapy to help identify those patients who warrant further urologic evaluation. Additionally, repeated infection with bacteria associated with struvite stone formation (e.g., *P. mirabilis*) may prompt consideration of imaging to rule out calculus.

Where possible, and particularly in cases where diagnostic data may have some uncertainty, a clinical assessment ensuring symptomatic resolution after treatment may facilitate improved management. Documentation of resolved symptoms after antibiotic treatment in cases where initial symptom and lab diagnostics were atypical may facilitate future decision-making. If the symptoms do not resolve, re-evaluation may be indicated. Should the initial infectious organism persist, the patient may warrant additional work up (e.g., imaging or cystoscopy), while the persistence of symptoms despite bacterial clearance may warrant consideration of alternative diagnoses.

18. Clinicians should repeat urine cultures to guide further management when UTI symptoms persist following antimicrobial therapy. (*Expert Opinion*)

After initiating antimicrobial therapy for UTI, clinical cure (i.e., UTI symptom resolution) is expected within three to seven days. Although there is no evidence, the Panel felt it reasonable to repeat microbiological assessment (e.g., standard urine culture) if UTI symptoms persist beyond seven days. Although a second antibiotic can be given empirically, this should only be done after a urine sample is obtained. This will minimize unnecessary treatment of patients with persistent UTI/pain symptoms whose laboratory data is inconsistent with persistent infection.

19. For patients with persistent UTI symptoms after microbiological cure, clinicians should evaluate for alternative causes to patient symptoms. (*Expert Opinion*)

Clinicians should explore alternative diagnoses for patients who continue to experience persistent lower urinary tract symptoms (LUTS) after achieving microbiological cure of a UTI. While it is common to attribute ongoing symptoms solely to residual inflammatory effects of infection, symptoms may stem from other underlying conditions such as overactive bladder, bladder pain syndrome, pelvic floor disorders or even undiagnosed genitourinary malignancy. For example, acquired myofascial tension of the pelvic floor musculature, characterized by pelvic pain, urinary frequency/urgency and difficulties voiding, can mimic UTI symptoms but requires a different management approach. Additionally, the presence of psychological factors such as anxiety or depression may also contribute to LUTS. A comprehensive evaluation, including a detailed history, physical examination including pelvic exam, and possibly further diagnostic testing (such as cystoscopy and/or urodynamics), can help identify these alternative causes. By adopting a broader diagnostic perspective, clinicians can ensure that patients receive appropriate treatment that addresses the actual underlying issue, ultimately improving their symptoms and health-related quality of life.

Estrogen

20. In perimenopausal and postmenopausal women with rUTIs, clinicians should recommend vaginal estrogen therapy to reduce the risk of future UTIs if there is no contraindication to vaginal estrogen therapy. (*Moderate Recommendation; Evidence Level: Grade B*)

Clinicians should recommend vaginal estrogen therapy to all perimenopausal and postmenopausal women with rUTI to reduce the risk of rUTI. This is in contrast to oral or other formulations of systemic estrogen therapy, which have not been shown to reduce UTI and are associated with different risks and benefits. Patients who present with rUTI and are already on systemic estrogen therapy can and should still be placed on vaginal estrogen therapy. There is no substantially increased risk of adverse events. However, systemic estrogen therapy should not be recommended for treatment of rUTI. Multiple randomized trials using a variety of formulations of vaginally applied estrogen therapy demonstrated a decreased incidence and time to recurrence of UTI in hypoestrogenic women. **Table 5** shows the formulations and dosing of several commonly used types of vaginal estrogen therapy. A systematic review of vaginal estrogen therapy for genitourinary syndrome of menopause concluded there was insufficient evidence to favor one formulation of vaginal estrogen over another.[267] However, a Cochrane Review suggested that vaginal cream may be more effective than the estrogen ring in preventing UTI.[268] Given the lack of clear superiority of one type of vaginal estrogen, clinicians should recommend the formulation of vaginal estrogen that is preferred by the patient.

The systematic review identified four trials (mean age ≥65; N=313) comparing estrogen versus placebo or no estrogen, and found estrogen to be associated with a reduced risk of experiencing ≥1 UTI versus placebo or no estrogen that was nearly statistically significant (4 trials; RR: 0.59; 95% CI: 0.35 to 1.01; I^2=76%).[269-272] There were no statistically significant differences in risk of rUTI when trials were stratified according to use of oral (2 trials; RR: 0.95, 95% CI: 0.63 to 1.44, I^2=2%)[269, 270] or topical estrogen (2 trials; RR: 0.42, 95% CI: 0.16 to 1.06, I^2=85%).[271, 272]

One trial evaluating estriol vaginal cream (0.5 mg nightly for 2 weeks, then twice weekly)[272] found topical estrogen associated with a decreased risk of experiencing ≥1 UTI (RR: 0.25; 95% CI: 0.13 to 0.50), decreased annualized UTI incidence (median: 0.5 versus 5.9 episodes; $p<0.001$), and fewer days of antibiotic use after 8 months (6.9 versus 32.0; $p<0.001$) than placebo.

In the systematic review for the update report, one new study with high risk of bias compared estrogen therapy to placebo in 35 women.[114] This study showed that women treated with estrogen had fewer UTI recurrences, similar to the findings of four studies on estrogen treatment included in the 2019 review. The difference was statistically significant with the addition of the new study (RR: 0.58; 95% CI: 0.39 to 0.87), but the strength of evidence remained low and did not change with the addition of this study.

As part of shared decision-making, the clinician should weigh the risks associated with vaginal estrogen therapy with its benefits in reducing UTIs. Given low systemic absorption, systemic risks association with vaginal estrogen therapy are minimal. Vaginal estrogen therapy has not been shown to increase risk of cancer recurrence in women undergoing treatment for or with a personal history of breast cancer.[273-275] Therefore, vaginal estrogen therapy should be considered in the prevention of UTI in women with a personal history of breast cancer in coordination with the patient's oncologist.

TABLE 5: Commonly used vaginal estrogen therapy		
Formulation	**Composition**	**Strength and Dosage**
Vaginal tablet	Estradiol hemihydrate*	10 mcg per day for 2 weeks, then 10 mcg 2 to 3 times weekly
Vaginal ring	17b-estradiol	2 mg ring released 7.5 mcg per day for 3 months (changed by patient or provider)
Vaginal cream	17b-estradiol	2 g daily for 2 weeks, then 1 g 2 to 3 times per week
	Conjugate equine estrogen	0.5 g daily for 2 weeks, then 0.5 g twice weekly

* Estradiol hemihydrate comes in a 4 mcg tablet; however, this has not been studied for prevention of rUTI.

FUTURE DIRECTIONS

A better understanding of rUTI pathophysiology will greatly aid in our ability to design more effective, mechanistically-based treatments. Critical expansion of our understanding of both host and pathogen factors that result in rUTI is mandated. Additionally, refinement of how UTI is defined must be considered. Indeed, delineating differences between ASB with concomitant non-specific LUTS secondary to storage dysfunction or diverse conditions such as interstitial cyctitis (IC)/bladder pain syndrome (BPS) and OAB versus true rUTI may eventually rely on development of innovative urine or serum biomarkers that can differentiate between these entities.[276] Relying on results from the urinary dipstick test, including leukocyte esterase and nitrate, lacks the necessary level of sensitivity and specificity for diagnostic accuracy. Currently, there are no symptoms that are definitively predictive of acute cystitis, and no good quality evidence that can support a specific pathogen. While some recent studies, such as Niaz et al.,[277] suggest that dysuria may be a predictive factor, other studies like Meckes et al.,[278] contradict this finding. Very low-quality evidence indicates that different symptom clusters could be linked to varying culture results, proposing potential groupings for these clusters and outcomes. However, the data are limited by small sample sizes and is not yet clinically useful.[279] In this context, defining initiatives for algorithms to predict treatment and partnering with our primary care colleagues and patients to provide education regarding rUTI definitions, evaluation, and treatment will provide an impactful narrative for the future.

Urine culture results or any other current methods of bacterial detection do not reflect any aspect of the host response. Investigations of more defined host biomarkers, such as cytokines or serum inflammatory markers, may allow more precise analysis of the host response which reflects a true UTI. Further refinements of bacterial molecular genetic technologies may help point-of-care testing with faster identification of potential uropathogens. By extension, the types and content of bacteria which inhabit the urinary tract as part of the native microbiome will change our understanding of how host-bacterial interactions contribute to development of rUTI.

Advanced molecular technologies give a more complete characterization of genitourinary microbes. PCR and NGS-based diagnostic tests have some enticing preliminary studies suggesting their utilization may facilitate equivalent or even better clinical outcomes than standard urine culture while decreasing the time to diagnosis and reducing empiric antibiotic prescriptions.[280] These early studies, however, are complicated by some methodological limitations and a high risk of bias. More data are needed before identifying what role these tests play in UTI diagnostics. In addition, the increased sensitivity of these methods, with almost all asymptomatic patients exhibiting detectable bacteriuria, prompts significant concern that adoption of this technology in the evaluation of lower urinary tract symptoms may lead to overtreatment with antibiotics. The rapidity of these tests may balance out these risks by encouraging better clinical practice by increasing the number of providers attempting bacterial identification, facilitating the rapid administration of appropriate antibiotics, and reducing empiric antibiotic prescribing. These tests, however, are not well-proven enough to displace standard urine culture as the gold standard for UTI diagnosis.

Advances in the accuracy of these tests may be facilitated by their combinations with evolving technologies and evidence. Combination of bacterial detection with biomarker assessments may provide far greater diagnostic utility than either test alone. Recently, a technique utilizing pre-treating urine samples with propidium monoazide (PMA) dye before PCR amplification significantly reduced the detection of DNA from dead bacteria, improving the selective detection of live urinary bacteria.[280] The concern with molecular diagnostics is its inability to detect active infection from residual dead bacteria in the urine, further development of methods such as Viability-qPCR may dramatically improve the sensitivity of molecular diagnostic testing.

As discussed in the introduction, challenges remain in determining the optimal antibiotic for treatment of acute cystitis episodes. Culture-based phenotypic susceptibility results do not account for the presence of transiently silenced resistance genes that may become activated under antibiotic selection pressure, and can contribute to suboptimal clinical responses.[281, 282] In contrast, genotypic resistance assessments may miss resistance-conferring genes that have not been described or are not targeted for detection.[68-71] Studies comparing phenotypic antibiotic susceptibility test results against genotypic resistance assessments have found a low average concordance between these results (ranging from 20.7% to 84.6%).[283-288] Given the limitations of both genotypic and phenotypic methods individually, novel methods to determine the optimal antibiotic for treatment accurately and rapidly are needed. A composite approach combining genetic and phenotypic methods was recently validated to demonstrate high agreement with standard antibiotic susceptibility testing.[289, 290] While it has yet to be validated in clinical scenarios, such complementary combinations of approaches may help maximize the detection of clinically meaningful antimicrobial resistance while tempering potential overdetection.

Promising new antimicrobials such as the first oral carbapenem, sulopenem, may also provide outpatient therapies for patients with high multidrug resistance or antibiotic-intolerant individuals at high risk of infection progression. While additional research and development towards new antimicrobials could provide additional options for care, these agents should still be reserved for those with high multidrug resistance. It may be reasonable to consider collaboration with infectious disease specialists before prescribing to reduce unnecessary prescribing. While these novel antibiotics represent exciting advancements, we also continue to learn more about the short- and long-term consequences of antibiotic use, particularly intermittent, high-dose use. Preliminary data suggests that continuous prophylaxis with low-dose antibiotics leads to lower rates of antibiotic resistance and fewer adverse events than intermittent treatment of symptomatic episodes, but little is understood about the optimal approach to preventing rUTI long-term and the potential complications and sequelae of each approach.

Emerging data regarding the bacterial communities of the human bladder, bowel, and vagina suggests that depletion or alteration of the normal host microbiome may disrupt host innate barriers to infection and alter innate immune system function, contributing to the development of rUTI. Such disruptions in these communities may be the direct consequence of antibiotic treatments. A better understanding of the relationship between the urinary microbiome and bladder health may fundamentally transform our earlier belief that urine is “sterile”. There is a tremendous interest in novel approaches to UTI treatment and prevention that preserve or reconstitute this microbiome.

A worldwide crisis has emerged due to rapid expansion of MDR bacteria, foreshadowing the devastating implications of the eventual inefficacy of many of our broad-spectrum antimicrobial agents.[237] Coupled with fear about the personal consequences of antibiotic overuse, current concepts of antibiotic stewardship have provoked a further initiative to develop agents outside the traditional pipeline of antibiotics. There is renewed vigor in pursuing non-antibiotic approaches to UTI treatment and prevention which would exhibit significantly less collateral damage. One such promising option is the development of bacteriophage therapies, which shows promising efficacy similar to antibiotics with few side effects.[291]

For prevention, the reconstitution of our native immune system, potentially by changing the microbiome of the gut or genitourinary tract may be a pathway to resolution of rUTI for select patients.[292] Prebiotics and probiotics also have been suggested as alternative approaches. While oral probiotics have yet to demonstrate an improvement in clinical outcomes for adult women with rUTI,[293] vaginal *Lactobacillus* probiotics decreased rUTI recurrence in phase I/II clinical trials.[263, 265] Data from those trials, however, suggests that the specific probiotic strain was important for UTI prevention. A recent double-blinded, placebo-controlled trial also supports this conclusion, demonstrating significant reductions in UTI recurrences in women with rUTI using proprietary *Lactobacillus* vaginal probiotic available in India. While currently there is insufficient quality data to recommend a probiotic that is commercially available in the U.S., active research continues to strengthen the case demonstrating the utility of this approach. Additional research attempting to identify and characterize probiotic strains and formulations capable of both *E. coli* inhibition and vaginal colonization is ongoing. Fecal Microbiome Transplant (FMT) has promise for refractory rUTI; in several studies, FMT performed for management of *C. difficile* colitis had the secondary effect of reducing the UTI recurrence in women with comorbid rUTI.[294, 295] The accumulating data suggest that manipulation and preservation of the microbiome may provide effective alternative approaches to UTI treatment and prevention without the side effects of antibiotics.

Modulation of the host response to bacterial infection is a key dynamic for which limited information currently exists but may represent a future direction for prevention strategies.[296] Vaccines for rUTI have demonstrated efficacy in reducing the UTI recurrence and are already in use in many nations.[297] While these are not yet available in the U.S., numerous high quality, prospective trials demonstrate their utility in women with rUTI;[298, 299] significant uncertainty remains, however, about when or if these will become available for use in the U.S. in the future. Use of mannosides as therapeutic entities to prevent bacterial adhesion to the urothelium may represent a narrow-spectrum treatment strategy associated with few systemic manifestations, although clinical outcomes using such approaches have been equivocal.[300] Modulation of host responses, such as the use of non-steroidal anti-inflammatory agents, have been suggested as a useful adjunct in both preclinical and clinical studies.[301, 302]

We must also expand our perspective on rUTI to include prevention. There currently exists an NIH-funded research consortium addressing this mission- the Prevention of Lower Urinary Tract Symptoms (PLUS) Research Consortium.[303] The PLUS consortium is dedicated to promoting prevention of LUTS (including UTIs) across the woman's life spectrum, utilizing a socioecologic construct.[304] Critical to these investigative efforts is the discovery of methods to suppress symptoms without use of antibiotics and direct studies that support a broader view of rUTI from the host-pathogen perspective. The PLUS consortium also seeks to identify modifiable risk factors for acute cystitis which can be tested in a prospective prevention trial. Through multiple efforts, which include identifying modifiable socioecological risk factors, understanding host responses involved in UTI and understanding pathogen virulence factors, we will discover new methods in diagnosis and treatment of rUTI.

ABBREVIATIONS

AHRQ	Agency for Healthcare Research and Quality
AMR	Antimicrobial resistance
ASB	Asymptomatic bacteriuria
ASM	American Society for Microbiology
AST	Antimicrobial susceptibility testing
AUA	American Urological Association
CFU	Colony-forming unit
CUA	Canadian Urological Association
DNA	Deoxyribonucleic acid
EPC	Evidence-based Practice Center
EQUC	Expanded quantitative urine culture
ESBL	Extended-spectrum β-lactamase
FDA	U.S. Food and Drug Administration
FMT	Fecal microbiome transplant
hpf	High power field
IDSA	Infectious Diseases Society of America
IVU	Intravenous urography
LUTS	Lower urinary tract symptoms
MDR	Multidrug-resistant
NGS	Next generation sequencing
OAB	Overactive bladder
PAC	Proanthocyanidin
PCR	Polymerase chain reaction
PGC	Practice Guidelines Committee
PLUS	Prevention of Lower Urinary Tract Symptoms
PMA	Propidium monoazide

PPV	Positive predictive value
PUF	Pelvic Pain and Urgency/Frequency
RCT	Randomized controlled trial
rUTI	Recurrent urinary tract infection
SQC	Science & Quality Council
SUFU	Society of Urodynamics, Female Pelvic Medicine & Urogenital Reconstruction
TMP-SMX	Trimethoprim-sulfamethoxazole
UTI	Urinary tract infection

RECURRENT URINARY TRACT INFECTION PANEL, CONSULTANTS, AND STAFF

Panel 2019
Jennifer Anger, MD, MPH (Chair)
Cedars-Sinai Medical Center

Toby C. Chai, MD (Vice Chair)
Yale School of Medicine

Melissa R. Kaufman, MD, PhD (PGC Rep)
Vanderbilt University Medical Center

Mary Ann Rondanina (Pt. Advocate)

A. Lenore Ackerman, MD, PhD
Cedars-Sinai Medical Center

Bilal Chughtai, MD
Weill Cornell Medicine

J. Quentin Clemens, MD
University of Michigan

Duane Hickling, MD, MSCI
University of Ottawa

Anil Kapoor, MD
McMaster University

Kimberly S. Kenton, MD, MS
Northwestern Medicine

Una Lee, MD
Virginia Mason

Ann Stapleton, MD
University of Washington

Lynn Stothers, MD
The University of British Columbia

Panel 2022 (Update)
Jennifer Anger, MD, MPH (Chair)
US San Diego Health

Una Lee, MD
Virginia Mason

Yahir A. Santiago-Lastra, MD
UC San Diego Health

Amendment Panel 2025
A. Lenore Ackerman, MD, PhD (Chair)
David Geffen School of Medicine at UCLA

Duane Hickling, MD, MSCI
University of Ottawa

Megan Bradley, MD
University of Pittsburgh Medical Center – Magee Womens Hospital

Consultants 2019
Roger Chou, MD
Jessica Griffin, MS

Consultants 2022
Shelley S. Selph, MD, MPH
Rebecca S. Holmes, MD, MS

Consultants 2025
Kristen E. D'Anci, PhD

Staff 2022
Erin Kirkby, MS
Leila Rahimi, MHS
Brooke Bixler, MPH
Sennett Kim

Staff 2025
Erin Kirkby, MS
Sennett Kim
Leila Rahimi, MHS
Lauren J. Pak, MHS, MS

CONFLICT OF INTEREST DISCLOSURES 2019

All panel members completed COI disclosures. Disclosures listed include both topic– and non-topic-related relationships.

Consultant/Advisor: Toby Chai, Avadel; **A. Lenore Ackerman**, Aquinox Pharmaceuticals; **Bilal Chughtai**, Boston Scientific; **J. Quentin Clemens**, Aquinox, Medtronic; **Duane Hickling**, Astellas, Pfizer, Allergan; **Anil Kapoor**, Pfizer, Bayer Oncology, Novartis Oncology; **Melissa Kaufman**, Boston Scientific; **Kimberly Kenton**, Boston Scientific; **Ann Stapleton**, Paratek
Meeting Participant or Lecturer: Bilal Chughtai, Allergan; **J. Quentin Clemens**, Allergan; **Duane**

Hickling, Astellas, Pfizer, Allergan; **Anil Kapoor**, Pfizer, Bayer Oncology, Novartis Oncology; **Una Lee**, Medtronic
Scientific Study or Trial: Jennifer Anger, Boston Scientific, AMS; **Bilal Chughtai**, American Urological Association, Boston Scientific; **Duane Hickling**, Astellas; **Anil Kapoor**, Pfizer, Novartis Oncology; **Kimberly Kenton**, Boston Scientific; **Lynn Stothers**, IPSEN
Investment Interest: J. Quentin Clemens, Merck
Health Publishing: J. Quentin Clemens, UpToDate
Other: Jennifer Anger, Boston Scientific; **Melissa Kaufman**, Boston Scientific, Cook Myosite; **Mary Ann Rondanina**, Theravance Biopharma

CONFLICT OF INTEREST DISCLOSURES 2022

All panel members completed COI disclosures. Disclosures listed include both topic– and non-topic-related relationships.

Scientific Study or Trial: Jennifer Anger, Medtronic; **Una Lee**, Patient-Centered Outcomes Research Institute

Meeting Participant or Lecturer: Una Lee, Medtronic, Contura, Axonics

Consultant or Advisor: Una Lee, Laborie

CONFLICT OF INTEREST DISCLOSURES 2025

All panel members completed COI disclosures. Disclosures listed include both topic– and non-topic-related relationships.

Consultant or Advisor: A. Lenore Ackerman, Watershed Medical, Abbvie, Desert Harvest; **Duane Hickling**, Allergan

Peer Reviewers 2019

We are grateful to the persons listed below who contributed to the Guideline by providing comments during the peer review process. Their reviews do not necessarily imply endorsement of the Guideline.

AUA (Board of Directors, Science and Quality Council, Practice Guidelines Committee, Journal of Urology)
Peter C. Albertsen, MD
Linda Baker, MD
Peter E. Clark, MD
Robert C. Flanigan, MD
David Ginsberg, MD
David F. Green MD
Louis R. Kavoussi, MD
Kevin McVary, MD
Roger E. Schultz, MD
Anthony Smith, MD
Thomas F. Stringer, MD
Martha Terris, MD

External Reviewers (Non-AUA Affiliates)
Sarah Adelstein, MD
Rahul Bansal, MD
Brook Brown, MD
Linda Brubaker, MD
Benjamin Brucker, MD
Anne Cameron, MD
Laura Chang Kit, MD
Kimberly L. Cooper, MD
Elodi Dielubanza, MD
Roger Dmochowski, MD
Karyn Eilber, MD
Ekene Enemchukwu, MD
Howard Goldman, MD
Alexander Gomelsky, MD
Gary Gray, MD
Michael Kennelly, MD
Kathleen Kobashi, MD
Shahid Lambe, MD
Gary Lemack, MD
Sara Lenherr, MD
Rena Malik, MD
Dena Moskowitz, MD
Laura Nguyen, MD
J. Curtis Nickel, MD
Lee Richter, MD
Eric Rovner, MD
Matthew Rutman, MD
Anthony Schaeffer, MD
Angela Schang, MD
Anne Suskind, MD
Suzette Sutherland, MD
Jannah Thompson, MD
Christian Twiss, MD
Sandip Vasavada, MD
Blayne Welk, MD
Chris Wu, MD

Public Commenters (Via public notice on AUA website)
Kirill Shiranov, MD
Jordan Dimitrakoff, MD

Peer Reviewers 2025

We are grateful to the persons listed below who contributed to the Guideline by providing comments during the peer review process. Their reviews do not necessarily imply endorsement of the Guideline.

AUA (Board of Directors, Science and Quality Council, Practice Guidelines Committee, Journal of Urology)
Erin Travis Bird, MD
Stephen A. Boorjian, MD
Arthur Louis Burnett, II, MD
David Alan Ginsberg, MD
Andrew Harris, MD
Kathleen Kobashi, MD
Tobias Steen Kohler, MD
Stephen Kraus, MD
Janet Baack Kukreja, MD
Gary Evan Lemack, MD
Jay Alan Motola, MD
Stephen Y. Nakada, MD
Matthew Edward Nielsen, MD
Lane Stuart Palmer, MD
Vernon M. Pais, Jr., MD
Phillip M. Pierorazio, MD
Sima Porten, MD
Christopher Porter, MD
Sarah Psutka, MD
Hassan Razvi, MD
Angela B. Smith, MD
Thomas F. Stringer, MD
Anne Suskind, MD
Arthur Edgar Tarantino, MD
Mary Westerman, MD

External Reviewers (Non-AUA Affiliates)
Lysanne Campeau, MD
Kevin Carlson, MD
Bilal Chughtai, MD
Jacques Corcos, MD
Chris Doiron, MD
Ann Gormley, MD
Melissa Kaufman, MD
Keith Kaye, MD
Casey Kowalik, MD
Una Lee, MD
Jennifer Locke, MD
Jerry Lowder, MD
Rena Malik, MD
Genevieve Nadeau, MD
Curtis Nickel, MD
Lee Ann Richter, MD
Anthony Schaeffer, MD
Nazema Siddiqui, MD
Lynn Stothers, MD
Blayne Welk, MD

Public Commenters (Via public notice on AUA website)
Gina Badalato, MD
Indraneel Banerjee, MD
Katie Canalichio, MD
Maude Carmel, MD
Kelly Casperson, MD
Lauren Corona, MD
Amy Dobberfuhl, MD
Mia Duncan, MD
Khadijah Eid, MBBS
Debra Fromer, MD
Lamia Gabal, MD
Howard Goldman, MD
Alex Gomelsky, MD
Forrest Jellison, MD
Jason Joseph, MD
Saad Juma, MD
Anjali Kapur, MD
Kinnari Kennedy, MD
Howard Korman, MD
Tamra Lewis, MD
Frank Lin, MD
Ayman Mahdy, MD
Matan Mekayten, MD
Peyman Mokhtarzadehazar, MD
Courtenay Moore, MD
Priya Padmanabhan, MD
Steven Penman, MD
Jennifer Peters, MD
Allison Polland, MD
Pablo Santamaria, MD
Scott Shie, MD
Kirill Shiranov, MD
Gillian Stearns, MD
Patrick Sweeney, MD
Elizabeth Takacs, MD
Justina Tam, MD
Aravind Viswanathan, MD
Glen Werneburg, MD
Guobing Xiong, MD
Yang Zhang, MD

DISCLAIMER

This document was written by the Recurrent Urinary Tract Infection Guideline Panel of the American Urological Association Education and Research, Inc., which was created in 2017. The Practice Guidelines Committee (PGC) of the AUA selected the committee chair. Panel members were selected by the chair in coordination with the Canadian Urological Association (CUA) and the Society of Urodynamics, Female Pelvic Medicine & Urogenital Reconstruction (SUFU). Membership of the panel included specialists with specific expertise on this disorder. The mission of the panel was to develop recommendations that are analysis-based or consensus-based, depending on panel processes and available data, for optimal clinical practices in the diagnosis and treatment of recurrent urinary tract infection.

Funding of the panel was provided by the AUA with contributions from CUA and SUFU. Panel members received no remuneration for their work. Each member of the panel provides an ongoing conflict of interest disclosure to the AUA.

While these guidelines do not necessarily establish the standard of care, AUA seeks to recommend and to encourage compliance by practitioners with current best practices related to the condition being treated. As medical knowledge expands and technology advances, the guidelines will change. Today these evidence-based guidelines statements represent not absolute mandates but provisional proposals for treatment under the specific conditions described in each document. For all these reasons, the guidelines do not pre-empt physician judgment in individual cases.

Treating physicians must take into account variations in resources, and patient tolerances, needs, and preferences. Conformance with any clinical guideline does not guarantee a successful outcome. The guideline text may include information or recommendations about certain drug uses (‘off label‘) that are not approved by the Food and Drug Administration (FDA), or about medications or substances not subject to the FDA approval process. AUA urges strict compliance with all government regulations and protocols for prescription and use of these substances. The physician is encouraged to carefully follow all available prescribing information about indications, contraindications, precautions and warnings. These guidelines and best practice statements are not intended to provide legal advice about use and misuse of these substances.

Although guidelines are intended to encourage best practices and potentially encompass available technologies with sufficient data as of close of the literature review, they are necessarily time-limited. Guidelines cannot include evaluation of all data on emerging technologies or management, including those that are FDA-approved, which may immediately come to represent accepted clinical practices.

For this reason, the AUA does not regard technologies or management which are too new to be addressed by this guideline as necessarily experimental or investigational.

REFERENCES

1. Paterson DL: "Collateral damage" from cephalosporin or quinolone antibiotic therapy. Clin Infect Dis 2004; **38 Suppl 4:** S341
2. Wagenlehner F, Wullt B, Ballarini S et al: Social and economic burden of recurrent urinary tract infections and quality of life: A patient web-based study (gesprit). Expert Rev Pharmacoecon Outcomes Res 2018; **18:** 107
3. Foxman B: Urinary tract infection syndromes: Occurrence, recurrence, bacteriology, risk factors, and disease burden. Infect Dis Clin North Am 2014; **28:** 1
4. Geerlings SE: Clinical presentations and epidemiology of urinary tract infections. Microbiol Spectr 2016; **4**
5. Gupta K and Trautner BW: Diagnosis and management of recurrent urinary tract infections in non-pregnant women. Bmj 2013; **346:** f3140
6. Foxman B: Epidemiology of urinary tract infections: Incidence, morbidity, and economic costs. Am J Med 2002; **113 Suppl 1A:** 5s

7. Dason S, Dason JT and Kapoor A: Guidelines for the diagnosis and management of recurrent urinary tract infection in women. Can Urol Assoc J 2011; **5:** 316
8. Finucane TE: "Urinary tract infection"-requiem for a heavyweight. J Am Geriatr Soc 2017; **65:** 1650
9. Malik RD, Wu YR and Zimmern PE: Definition of recurrent urinary tract infections in women: Which one to adopt? Female Pelvic Med Reconstr Surg 2018; **24:** 424
10. Nicolle LE, Bradley S, Colgan R et al: Infectious diseases society of america guidelines for the diagnosis and treatment of asymptomatic bacteriuria in adults. Clin Infect Dis 2005; **40:** 643
11. Stamm WE: Measurement of pyuria and its relation to bacteriuria. Am J Med 1983; **75:** 53
12. Bilsen MP, Conroy SP, Schneeberger C et al: A reference standard for urinary tract infection research: A multidisciplinary delphi consensus study. Lancet Infect Dis 2024; **24:** e513
13. Hooton TM: Clinical practice. Uncomplicated urinary tract infection. N Engl J Med 2012; **366:** 1028
14. Bent S, Nallamothu BK, Simel DL et al: Does this woman have an acute uncomplicated urinary tract infection? Jama 2002; **287:** 2701
15. Juthani-Mehta M, Quagliarello V, Perrelli E et al: Clinical features to identify urinary tract infection in nursing home residents: A cohort study. J Am Geriatr Soc 2009; **57:** 963
16. Mody L and Juthani-Mehta M: Urinary tract infections in older women: A clinical review. Jama 2014; **311:** 844
17. Medina-Bombardó D, Seguí-Díaz M, Roca-Fusalba C and Llobera J: What is the predictive value of urinary symptoms for diagnosing urinary tract infection in women? Fam Pract 2003; **20:** 103
18. Boscia JA, Kobasa WD, Abrutyn E et al: Lack of association between bacteriuria and symptoms in the elderly. Am J Med 1986; **81:** 979
19. Stone ND, Ashraf MS, Calder J et al: Surveillance definitions of infections in long-term care facilities: Revisiting the mcgeer criteria. Infect Control Hosp Epidemiol 2012; **33:** 965
20. High KP, Bradley SF, Gravenstein S et al: Clinical practice guideline for the evaluation of fever and infection in older adult residents of long-term care facilities: 2008 update by the infectious diseases society of america. J Am Geriatr Soc 2009; **57:** 375
21. Loeb M, Bentley DW, Bradley S et al: Development of minimum criteria for the initiation of antibiotics in residents of long-term-care facilities: Results of a consensus conference. Infect Control Hosp Epidemiol 2001; **22:** 120
22. American geriatrics society identifies another five things that healthcare providers and patients should question. J Am Geriatr Soc 2014; **62:** 950
23. Behzadi P, Behzadi E, Yazdanbod H et al: A survey on urinary tract infections associated with the three most common uropathogenic bacteria. Maedica (Bucur) 2010; **5:** 111
24. Colgan R and Williams M: Diagnosis and treatment of acute uncomplicated cystitis. Am Fam Physician 2011; **84:** 771
25. Hilt EE, McKinley K, Pearce MM et al: Urine is not sterile: Use of enhanced urine culture techniques to detect resident bacterial flora in the adult female bladder. J Clin Microbiol 2014; **52:** 871
26. Xu R, Deebel N, Casals R et al: A new gold rush: A review of current and developing diagnostic tools for urinary tract infections. Diagnostics (Basel) 2021; **11**
27. Kass EH: Asymptomatic infections of the urinary tract. Trans Assoc Am Physicians 1956; **69:** 56
28. Platt R: Quantitative definition of bacteriuria. Am J Med 1983; **75:** 44
29. Pollock HM: Laboratory techniques for detection of urinary tract infection and assessment of value. Am J Med 1983; **75:** 79
30. Sanford JP, Favour CB, Mao FH and Harrison JH: Evaluation of the positive urine culture; an approach to the differentiation of significant bacteria from contaminants. Am J Med 1956; **20:** 88
31. Tapsall JW, Taylor PC, Bell SM and Smith DD: Relevance of "significant bacteriuria" to aetiology and diagnosis of urinary-tract infection. Lancet 1975; **2:** 637
32. Brubaker L, Chai T, Horsley H et al: Tarnished gold—the “standard” urine culture: Reassessing the characteristics of a criterion standard for detecting urinary microbes. Frontiers in Urology 2023; **3**
33. Kunin CM, White LV and Hua TH: A reassessment of the importance of "low-count" bacteriuria in young women with acute urinary symptoms. Ann Intern Med 1993; **119:** 454
34. Komaroff AL: Acute dysuria in women. N Engl J Med 1984; **310:** 368
35. Stamm WE, Counts GW, Running KR et al: Diagnosis of coliform infection in acutely dysuric women. N Engl J Med 1982; **307:** 463
36. Naber KG, Schito G, Botto H et al: Surveillance study in europe and brazil on clinical aspects and antimicrobial resistance epidemiology in females with cystitis (aresc): Implications for empiric therapy. Eur Urol 2008; **54:** 1164

37. De Backer D, Christiaens T, Heytens S et al: Evolution of bacterial susceptibility pattern of escherichia coli in uncomplicated urinary tract infections in a country with high antibiotic consumption: A comparison of two surveys with a 10 year interval. J Antimicrob Chemother 2008; **62:** 364
38. Heytens S, Boelens J, Claeys G et al: Uropathogen distribution and antimicrobial susceptibility in uncomplicated cystitis in belgium, a high antibiotics prescribing country: 20-year surveillance. Eur J Clin Microbiol Infect Dis 2017; **36:** 105
39. Hooton TM, Roberts PL, Cox ME and Stapleton AE: Voided midstream urine culture and acute cystitis in premenopausal women. N Engl J Med 2013; **369:** 1883
40. Stamm WE, Wagner KF, Amsel R et al: Causes of the acute urethral syndrome in women. N Engl J Med 1980; **303:** 409
41. Giesen LG, Cousins G, Dimitrov BD et al: Predicting acute uncomplicated urinary tract infection in women: A systematic review of the diagnostic accuracy of symptoms and signs. BMC Fam Pract 2010; **11:** 78
42. Kunin C: Urinary tract infections, in detection, prevention and management. Lea & Febiger: Philadelphia 1997;
43. Bekeris LG, Jones BA, Walsh MK and Wagar EA: Urine culture contamination: A college of american pathologists q-probes study of 127 laboratories. Arch Pathol Lab Med 2008; **132:** 913
44. LaRocco MT, Franek J, Leibach EK et al: Effectiveness of preanalytic practices on contamination and diagnostic accuracy of urine cultures: A laboratory medicine best practices systematic review and meta-analysis. Clin Microbiol Rev 2016; **29:** 105
45. Silver SA, Baillie L and Simor AE: Positive urine cultures: A major cause of inappropriate antimicrobial use in hospitals? Can J Infect Dis Med Microbiol 2009; **20:** 107
46. Costello EK, Stagaman K, Dethlefsen L et al: The application of ecological theory toward an understanding of the human microbiome. Science 2012; **336:** 1255
47. Whiteside SA, Razvi H, Dave S et al: The microbiome of the urinary tract--a role beyond infection. Nat Rev Urol 2015; **12:** 81
48. Ackerman AL and Underhill DM: The mycobiome of the human urinary tract: Potential roles for fungi in urology. Ann Transl Med 2017; **5:** 31
49. Barber AE, Norton JP, Spivak AM and Mulvey MA: Urinary tract infections: Current and emerging management strategies. Clin Infect Dis 2013; **57:** 719
50. Cai T, Mazzoli S, Mondaini N et al: The role of asymptomatic bacteriuria in young women with recurrent urinary tract infections: To treat or not to treat? Clin Infect Dis 2012; **55:** 771
51. Darouiche RO, Green BG, Donovan WH et al: Multicenter randomized controlled trial of bacterial interference for prevention of urinary tract infection in patients with neurogenic bladder. Urology 2011; **78:** 341
52. Coyne KS, Sexton CC, Thompson CL et al: The prevalence of lower urinary tract symptoms (luts) in the USA, the uk and sweden: Results from the epidemiology of luts (epiluts) study. BJU Int 2009; **104:** 352
53. Mandell G, Bennett J and Dolin R: Mandell, douglas, and bennett's principles and practice of infectious diseases. Lancet Infect Dis 2010; **10:** 303
54. Scott VCS, Thum LW, Sadun T et al: Fear and frustration among women with recurrent urinary tract infections: Findings from patient focus groups. J Urol 2021; **206:** 688
55. Hatfield KM, Kabbani S, See I et al: Use of multiplex molecular panels to diagnose urinary tract infection in older adults. JAMA Netw Open 2024; **7:** e2446842
56. Thapaliya J, Khadka P, Thapa S and Gongal C: Enhanced quantitative urine culture technique, a slight modification, in detecting under-diagnosed pediatric urinary tract infection. BMC Res Notes 2020; **13:** 5
57. Barnes HC, Wolff B, Abdul-Rahim O et al: A randomized clinical trial of standard versus expanded cultures to diagnose urinary tract infections in women. J Urol 2021; **206:** 1212
58. Hochstedler BR, Burnett L, Price TK et al: Urinary microbiota of women with recurrent urinary tract infection: Collection and culture methods. Int Urogynecol J 2022; **33:** 563
59. Perez-Carrasco V, Soriano-Lerma A, Soriano M et al: Urinary microbiome: Yin and yang of the urinary tract. Front Cell Infect Microbiol 2021; **11:** 617002
60. Moustafa A, Li W, Singh H et al: Microbial metagenome of urinary tract infection. Sci Rep 2018; **8:** 4333
61. Dixon M, Sha S, Stefil M and McDonald M: Is it time to say goodbye to culture and sensitivity? The case for culture-independent urology. Urology 2020; **136:** 112
62. McDonald M, Kameh D, Johnson ME et al: A head-to-head comparative phase ii study of standard urine culture and sensitivity versus DNA next-generation sequencing testing for urinary tract infections. Rev Urol 2017; **19:** 213
63. Daly A, Baunoch D, Rehling K et al: Utilization of m-pcr and p-ast for diagnosis and management of urinary tract infections in home-based primary care. JOJ Urology & Nephrology 2020; **7:** 2

64. Haley E, Luke N, Korman H et al: Improving patient outcomes while reducing empirical treatment with multiplex-polymerase-chain-reaction/pooled-antibiotic-susceptibility-testing assay for complicated and recurrent urinary tract infections. Diagnostics (Basel) 2023; **13**
65. Spangler FL, Williams C, Aberger ME et al: Clinical utility of pcr compared to conventional culture and sensitivity testing for the management of complicated urinary tract infections in adults: Part i. Assessment of clinical outcomes, investigator satisfaction scores, and turnaround times. Diagn Microbiol Infect Dis 2025; **111:** 116601
66. Spangler FL, Williams C, Aberger ME et al: Clinical utility of pcr compared to conventional culture and sensitivity testing for management of complicated urinary tract infections in adults: Part ii.Evaluation of diagnostic concordance, discordant results, and antimicrobial selection efficacy. Diagn Microbiol Infect Dis 2025; **111:** 116646
67. Harris M, Fasolino T, Ivankovic D et al: Genetic factors that contribute to antibiotic resistance through intrinsic and acquired bacterial genes in urinary tract infections. Microorganisms 2023; **11**
68. Nicoloff H, Hjort K, Levin BR and Andersson DI: The high prevalence of antibiotic heteroresistance in pathogenic bacteria is mainly caused by gene amplification. Nat Microbiol 2019; **4:** 504
69. Pereira C, Larsson J, Hjort K et al: The highly dynamic nature of bacterial heteroresistance impairs its clinical detection. Commun Biol 2021; **4:** 521
70. Levin BR, Berryhill BA, Gil-Gil T et al: Theoretical considerations and empirical predictions of the pharmaco- and population dynamics of heteroresistance. Proc Natl Acad Sci U S A 2024; **121:** e2318600121
71. Roch M, Sierra R and Andrey DO: Antibiotic heteroresistance in eskape pathogens, from bench to bedside. Clin Microbiol Infect 2023; **29:** 320
72. Korman HJ, Mathur M, Luke N et al: Multiplex polymerase chain reaction/pooled antibiotic susceptibility testing was not associated with increased antibiotic resistance in management of complicated urinary tract infections. Infect Drug Resist 2023; **16:** 2841
73. Haley E, Luke N, Korman H et al: Comparing prescribing behaviors and clinician experiences between multiplex pcr/pooled antibiotic susceptibility testing and standard urine culture in complicated uti cases. J Clin Med 2024; **13**
74. Lewis DA, Brown R, Williams J et al: The human urinary microbiome; bacterial DNA in voided urine of asymptomatic adults. Front Cell Infect Microbiol 2013; **3:** 41
75. Tchesnokova V, Avagyan H, Rechkina E et al: Bacterial clonal diagnostics as a tool for evidence-based empiric antibiotic selection. PLoS One 2017; **12:** e0174132
76. Schito GC, Naber KG, Botto H et al: The aresc study: An international survey on the antimicrobial resistance of pathogens involved in uncomplicated urinary tract infections. Int J Antimicrob Agents 2009; **34:** 407
77. Ho PL, Yip KS, Chow KH et al: Antimicrobial resistance among uropathogens that cause acute uncomplicated cystitis in women in hong kong: A prospective multicenter study in 2006 to 2008. Diagn Microbiol Infect Dis 2010; **66:** 87
78. Linder JA, Huang ES, Steinman MA et al: Fluoroquinolone prescribing in the united states: 1995 to 2002. Am J Med 2005; **118:** 259
79. Zatorski C, Zocchi M, Cosgrove SE et al: A single center observational study on emergency department clinician non-adherence to clinical practice guidelines for treatment of uncomplicated urinary tract infections. BMC Infect Dis 2016; **16:** 638
80. Chappidi MR, Kates M, Stimson CJ et al: Causes, timing, hospital costs and perioperative outcomes of index vs nonindex hospital readmissions after radical cystectomy: Implications for regionalization of care. J Urol 2017; **197:** 296
81. Barlam TF, Cosgrove SE, Abbo LM et al: Implementing an antibiotic stewardship program: Guidelines by the infectious diseases society of america and the society for healthcare epidemiology of america. Clin Infect Dis 2016; **62:** e51
82. Caterino JM, Ting SA, Sisbarro SG et al: Age, nursing home residence, and presentation of urinary tract infection in u.S. Emergency departments, 2001-2008. Acad Emerg Med 2012; **19:** 1173
83. Suskind AM, Saigal CS, Hanley JM et al: Incidence and management of uncomplicated recurrent urinary tract infections in a national sample of women in the united states. Urology 2016; **90:** 50
84. Kahlmeter G: An international survey of the antimicrobial susceptibility of pathogens from uncomplicated urinary tract infections: The eco.Sens project. J Antimicrob Chemother 2003; **51:** 69
85. Graninger W: Pivmecillinam--therapy of choice for lower urinary tract infection. Int J Antimicrob Agents 2003; **22 Suppl 2:** 73
86. Knothe H, Schäfer V, Sammann A and Shah PM: Influence of fosfomycin on the intestinal and pharyngeal flora of man. Infection 1991; **19:** 18
87. Mavromanolakis E, Maraki S, Samonis G et al: Effect of norfloxacin, trimethoprim-sulfamethoxazole and nitrofurantoin on fecal flora of women with recurrent urinary tract infections. J Chemother 1997; **9:** 203

88. Sullivan A, Edlund C and Nord CE: Effect of antimicrobial agents on the ecological balance of human microflora. Lancet Infect Dis 2001; **1:** 101
89. Köves B, Cai T, Veeratterapillay R et al: Benefits and harms of treatment of asymptomatic bacteriuria: A systematic review and meta-analysis by the european association of urology urological infection guidelines panel. Eur Urol 2017; **72:** 865
90. Cai T, Nesi G, Mazzoli S et al: Asymptomatic bacteriuria treatment is associated with a higher prevalence of antibiotic resistant strains in women with urinary tract infections. Clin Infect Dis 2015; **61:** 1655
91. Woodford HJ, Graham C, Meda M and Miciuleviciene J: Bacteremic urinary tract infection in hospitalized older patients-are any currently available diagnostic criteria sensitive enough? J Am Geriatr Soc 2011; **59:** 567
92. Ferry SA, Holm SE, Stenlund H et al: Clinical and bacteriological outcome of different doses and duration of pivmecillinam compared with placebo therapy of uncomplicated lower urinary tract infection in women: The lutiw project. Scand J Prim Health Care 2007; **25:** 49
93. Christiaens TC, De Meyere M, Verschraegen G et al: Randomised controlled trial of nitrofurantoin versus placebo in the treatment of uncomplicated urinary tract infection in adult women. Br J Gen Pract 2002; **52:** 729
94. Falagas ME, Kotsantis IK, Vouloumanou EK and Rafailidis PI: Antibiotics versus placebo in the treatment of women with uncomplicated cystitis: A meta-analysis of randomized controlled trials. J Infect 2009; **58:** 91
95. Foxman B: The epidemiology of urinary tract infection. Nat Rev Urol 2010; **7:** 653
96. Gágyor I, Hummers-Pradier E, Kochen MM et al: Immediate versus conditional treatment of uncomplicated urinary tract infection - a randomized-controlled comparative effectiveness study in general practices. BMC Infect Dis 2012; **12:** 146
97. Ferry SA, Holm SE, Stenlund H et al: The natural course of uncomplicated lower urinary tract infection in women illustrated by a randomized placebo controlled study. Scand J Infect Dis 2004; **36:** 296
98. Finucane TE: 'Urinary tract infection' and the microbiome. Am J Med 2017; **130:** e97
99. Scholes D, Hooton TM, Roberts PL et al: Risk factors for recurrent urinary tract infection in young women. J Infect Dis 2000; **182:** 1177
100. Abbo LM and Hooton TM: Antimicrobial stewardship and urinary tract infections. Antibiotics (Basel) 2014; **3:** 174
101. Hooton TM, Vecchio M, Iroz A et al: Effect of increased daily water intake in premenopausal women with recurrent urinary tract infections: A randomized clinical trial. JAMA Intern Med 2018; **178:** 1509
102. Scholes D, Hawn TR, Roberts PL et al: Family history and risk of recurrent cystitis and pyelonephritis in women. J Urol 2010; **184:** 564
103. Methods guide for effectiveness and comparative effectiveness reviews. Rockville, MD: Agency for Healthcare Research and Quality: AHRQ Publication No. 10(14)-EHC063-EF, January 2014
104. Koradia P, Kapadia S, Trivedi Y et al: Probiotic and cranberry supplementation for preventing recurrent uncomplicated urinary tract infections in premenopausal women: A controlled pilot study. Expert Rev Anti Infect Ther 2019; **17:** 733
105. Murina F, Vicariotto F and Lubrano C: Efficacy of an orally administered combination of lactobacillus paracasei lc11, cranberry and d-mannose for the prevention of uncomplicated, recurrent urinary tract infections in women. Urologia 2021; **88:** 64
106. Babar A, Moore L, Leblanc V et al: High dose versus low dose standardized cranberry proanthocyanidin extract for the prevention of recurrent urinary tract infection in healthy women: A double-blind randomized controlled trial. BMC Urol 2021; **21:** 44
107. Bruyère F, Azzouzi AR, Lavigne JP et al: A multicenter, randomized, placebo-controlled study evaluating the efficacy of a combination of propolis and cranberry (vaccinium macrocarpon) (duab®) in preventing low urinary tract infection recurrence in women complaining of recurrent cystitis. Urol Int 2019; **103:** 41
108. Iroz A, Dornic Q, Seksek I and Vecchio M: Fp144increased water intake for the prevention of recurrent urinary tract infections does not affect quality of life in women. Nephrology Dialysis Transplantation 2018; **33:** i25
109. Lenger SM CC, Chetti C et al: D-mannose for recurrent urinary tract infection prevention in post-menopausal women using vaginal estrogen: A randomized controlled trial feasibility and interim analysis. Female pelvic med 2020; **26:** S104
110. Lenger SM, Bradley MS, Thomas DA et al: D-mannose vs other agents for recurrent urinary tract infection prevention in adult women: A systematic review and meta-analysis. Am J Obstet Gynecol 2020; **223:** 265.e1
111. Botros C, Lozo S, Iyer S et al: Methenamine hippurate compared with trimethoprim for the prevention of recurrent urinary tract infections: A randomized clinical trial. Int Urogynecol J 2022; **33:** 571
112. Bakhit M, Krzyzaniak N, Hilder J et al: Use of methenamine hippurate to prevent urinary tract infections in community adult women: A systematic review and meta-analysis. Br J Gen Pract 2021; **71:** e528

113. Chen YY, Su TH and Lau HH: Estrogen for the prevention of recurrent urinary tract infections in postmenopausal women: A meta-analysis of randomized controlled trials. Int Urogynecol J 2021; **32:** 17
114. Ferrante KL, Wasenda EJ, Jung CE et al: Vaginal estrogen for the prevention of recurrent urinary tract infection in postmenopausal women: A randomized clinical trial. Female Pelvic Med Reconstr Surg 2021; **27:** 112
115. Harris RP, Helfand M, Woolf SH et al: Current methods of the us preventive services task force: A review of the process. Am J Prev Med 2001; **20:** 21
116. Shea BJ, Reeves BC, Wells G et al: Amstar 2: A critical appraisal tool for systematic reviews that include randomised or non-randomised studies of healthcare interventions, or both. Bmj 2017; **358:** j4008
117. The Nordic Cochrane Centre: Review manager (revman), Version 5.3 ed. Copenhagen: The Chochrane Collaboration 2014
118. Guyatt G, Oxman AD, Akl EA et al: Grade guidelines: 1. Introduction-grade evidence profiles and summary of findings tables. J Clin Epidemiol 2011; **64:** 383
119. Balshem H, Helfand M, Schünemann HJ et al: Grade guidelines: 3. Rating the quality of evidence. J Clin Epidemiol 2011; **64:** 401
120. Faraday M, Hubbard H, Kosiak B and Dmochowski R: Staying at the cutting edge: A review and analysis of evidence reporting and grading; the recommendations of the american urological association. BJU Int 2009; **104:** 294
121. Hsu C-C and Sandford B: The delphi technique: Making sense of consensus. Practical Assessment, Research and Evaluation 2007; **12**
122. Weiss JM: Pelvic floor myofascial trigger points: Manual therapy for interstitial cystitis and the urgency-frequency syndrome. J Urol 2001; **166:** 2226
123. Wolff BJ, Joyce CJ, Brincat CA et al: Pelvic floor myofascial pain in patients with symptoms of urinary tract infection. Int J Gynaecol Obstet 2019; **145:** 205
124. Latour K, De Lepeleire J, Catry B and Buntinx F: Nursing home residents with suspected urinary tract infections: A diagnostic accuracy study. BMC Geriatr 2022; **22:** 187
125. Bruxvoort KJ, Bider-Canfield Z, Casey JA et al: Outpatient urinary tract infections in an era of virtual healthcare: Trends from 2008 to 2017. Clin Infect Dis 2020; **71:** 100
126. Werneburg GT, Lewis KC, Vasavada SP et al: Urinalysis exhibits excellent predictive capacity for the absence of urinary tract infection. Urology 2023; **175:** 101
127. Advani SD, Turner NA, North R et al: Proposing the "continuum of uti" for a nuanced approach to diagnosis and management of urinary tract infections. J Urol 2024; **211:** 690
128. Ourani M, Honda NS, MacDonald W and Roberts J: Evaluation of evidence-based urinalysis reflex to culture criteria: Impact on reducing antimicrobial usage. Int J Infect Dis 2021; **102:** 40
129. Bradbury SM: Collection of urine specimens in general practice: To clean or not to clean? J R Coll Gen Pract 1988; **38:** 363
130. Miller JM, Binnicker MJ, Campbell S et al: A guide to utilization of the microbiology laboratory for diagnosis of infectious diseases: 2018 update by the infectious diseases society of america and the american society for microbiology. Clin Infect Dis 2018; **67:** 813
131. Johnson JD, O'Mara HM, Durtschi HF and Kopjar B: Do urine cultures for urinary tract infections decrease follow-up visits? J Am Board Fam Med 2011; **24:** 647
132. Frawley H, Shelly B, Morin M et al: An international continence society (ics) report on the terminology for pelvic floor muscle assessment. Neurourol Urodyn 2021; **40:** 1217
133. McCarter YS BE, Hall GS, Zervos M, Sharp SE: Laboratory diagnosis of urinary tract infections: ASM Press, 2009
134. Blake DR and Doherty LF: Effect of perineal cleansing on contamination rate of mid-stream urine culture. J Pediatr Adolesc Gynecol 2006; **19:** 31
135. Holliday G, Strike PW and Masterton RG: Perineal cleansing and midstream urine specimens in ambulatory women. J Hosp Infect 1991; **18:** 71
136. Schlager TA, Smith DE and Donowitz LG: Perineal cleansing does not reduce contamination of urine samples from pregnant adolescents. Pediatr Infect Dis J 1995; **14:** 909
137. Schneeberger C, van den Heuvel ER, Erwich J et al: Contamination rates of three urine-sampling methods to assess bacteriuria in pregnant women. Obstet Gynecol 2013; **121:** 299
138. Lifshitz E and Kramer L: Outpatient urine culture: Does collection technique matter? Arch Intern Med 2000; **160:** 2537
139. Beeson PB: The case against the catheter. Am J Med 1958; **24:** 1
140. Boshell BR and Sanford JP: A screening method for the evaluation of urinary tract infections in female patients without catheterization. Ann Intern Med 1958; **48:** 1040

141. Hooton TM, Scholes D, Hughes JP et al: A prospective study of risk factors for symptomatic urinary tract infection in young women. N Engl J Med 1996; **335:** 468
142. Walter FG and Knopp RK: Urine sampling in ambulatory women: Midstream clean-catch versus catheterization. Ann Emerg Med 1989; **18:** 166
143. Immergut MA, Gilbert EC, Frensilli FJ and Goble M: The myth of the clean catch urine specimen. Urology 1981; **17:** 339
144. Lemieux G and St-Martin M: Reliability of clean-voided mid-stream urine specimens for the diagnosis of significant bacteriuria in the female patient. Can Med Assoc J 1968; **98:** 241
145. Valenstein P and Meier F: Urine culture contamination: A college of american pathologists q-probes study of contaminated urine cultures in 906 institutions. Arch Pathol Lab Med 1998; **122:** 123
146. Gupta K, Hooton TM, Naber KG et al: International clinical practice guidelines for the treatment of acute uncomplicated cystitis and pyelonephritis in women: A 2010 update by the infectious diseases society of america and the european society for microbiology and infectious diseases. Clin Infect Dis 2011; **52:** e103
147. Porter IA and Brodie J: Boric acid preservation of urine samples. Br Med J 1969; **2:** 353
148. Hindman R, Tronic B and Bartlett R: Effect of delay on culture of urine. J Clin Microbiol 1976; **4:** 102
149. Lum KT and Meers PD: Boric acid converts urine into an effective bacteriostatic transport medium. J Infect 1989; **18:** 51
150. Wright DN, Boshard R, Ahlin P et al: Effect of urine preservation on urine screening and organism identification. Arch Pathol Lab Med 1985; **109:** 819
151. Hubbard WA, Shalis PJ and McClatchey KD: Comparison of the b-d urine culture kit with a standard culture method and with the ms-2. J Clin Microbiol 1983; **17:** 327
152. Weinstein MP: Evaluation of liquid and lyophilized preservatives for urine culture. J Clin Microbiol 1983; **18:** 912
153. Baerheim A, Digranes A and Hunskaar S: Evaluation of urine sampling technique: Bacterial contamination of samples from women students. Br J Gen Pract 1992; **42:** 241
154. Moore T, Hira NR and Stirland RM: Differential urethrovesical urinary cell-count. A method of accurate diagnosis of lower-urinary-tract infections in women. Lancet 1965; **1:** 626
155. Pagano MJ, Barbalat Y, Theofanides MC et al: Diagnostic yield of cystoscopy in the evaluation of recurrent urinary tract infection in women. Neurourol Urodyn 2017; **36:** 692
156. Santoni N, Ng A, Skews R and Aboumarzouk OM: Recurrent urinary tract infections in women: What is the evidence for investigating with flexible cystoscopy, imaging and urodynamics? Urol Int 2018; **101:** 373
157. Little MA, Stafford Johnson DB, O'Callaghan JP and Walshe JJ: The diagnostic yield of intravenous urography. Nephrol Dial Transplant 2000; **15:** 200
158. Fair WR, McClennan BL and Jost RG: Are excretory urograms necessary in evaluating women with urinary tract infection? J Urol 1979; **121:** 313
159. Nickel JC, Wilson J, Morales A and Heaton J: Value of urologic investigation in a targeted group of women with recurrent urinary tract infections. Can J Surg 1991; **34:** 591
160. Melekos MD, Asbach HW, Gerharz E et al: Post-intercourse versus daily ciprofloxacin prophylaxis for recurrent urinary tract infections in premenopausal women. J Urol 1997; **157:** 935
161. Zhong YH, Fang Y, Zhou JZ et al: Effectiveness and safety of patient initiated single-dose versus continuous low-dose antibiotic prophylaxis for recurrent urinary tract infections in postmenopausal women: A randomized controlled study. J Int Med Res 2011; **39:** 2335
162. Wong ES, McKevitt M, Running K et al: Management of recurrent urinary tract infections with patient-administered single-dose therapy. Ann Intern Med 1985; **102:** 302
163. Nicolle LE, Gupta K, Bradley SF et al: Clinical practice guideline for the management of asymptomatic bacteriuria: 2019 update by the infectious diseases society of america. Clin Infect Dis 2019; **68:** 1611
164. Nicolle LE, Gupta K, Bradley SF et al: Clinical practice guideline for the management of asymptomatic bacteriuria: 2019 update by the infectious diseases society of america. Clin Infect Dis 2019; **68:** e83
165. Blondel-Hill E, Patrick D, Nott C et al: Ammi canada position statement on asymptomatic bacteriuria. Journal of the Association of Medical Microbiology and Infectious Disease Canada 2018; **3:** 4
166. Dull RB, Friedman SK, Risoldi ZM et al: Antimicrobial treatment of asymptomatic bacteriuria in noncatheterized adults: A systematic review. Pharmacotherapy 2014; **34:** 941
167. Coussement J, Scemla A, Abramowicz D et al: Antibiotics for asymptomatic bacteriuria in kidney transplant recipients. Cochrane Database Syst Rev 2018; **2:** Cd011357
168. Advani SD, Ratz D, Horowitz JK et al: Bacteremia from a presumed urinary source in hospitalized adults with asymptomatic bacteriuria. JAMA Netw Open 2024; **7:** e242283
169. Wolf JS BC, Dmochowski RR et al: Urologic surgery antimicrobial prophylaxis. 2012;

170. Zalmanovici Trestioreanu A, Green H, Paul M et al: Antimicrobial agents for treating uncomplicated urinary tract infection in women. Cochrane Database Syst Rev 2010**:** Cd007182
171. Knottnerus BJ, Grigoryan L, Geerlings SE et al: Comparative effectiveness of antibiotics for uncomplicated urinary tract infections: Network meta-analysis of randomized trials. Fam Pract 2012; **29:** 659
172. Huttner A, Kowalczyk A, Turjeman A et al: Effect of 5-day nitrofurantoin vs single-dose fosfomycin on clinical resolution of uncomplicated lower urinary tract infection in women: A randomized clinical trial. Jama 2018; **319:** 1781
173. Faltinsen EG, Storebø OJ, Jakobsen JC et al: Network meta-analysis: The highest level of medical evidence? BMJ Evid Based Med 2018; **23:** 56
174. Kettlewell R, Jones C, Felton TW et al: Insights into durability against resistance from the antibiotic nitrofurantoin. NPJ Antimicrob Resist 2024; **2:** 41
175. Fda updates warnings for oral and injectable fluoroquinolone antibiotics due to disabling side effects
176. Kaye KS, Santerre Henriksen A, Sommer M and Frimodt-Møller N: Safety and tolerability of pivmecillinam during more than four decades of clinical experience: A systematic review. Clin Infect Dis 2025; **80:** 280
177. Wagenlehner F, Perry CR, Hooton TM et al: Oral gepotidacin versus nitrofurantoin in patients with uncomplicated urinary tract infection (eagle-2 and eagle-3): Two randomised, controlled, double-blind, double-dummy, phase 3, non-inferiority trials. Lancet 2024; **403:** 741
178. Wagenlehner F, Perry CR, Hooton TM et al: Plain language summary: Efficacy and safety of gepotidacin, a new oral antibiotic, compared with nitrofurantoin, a commonly used oral antibiotic, for treating uncomplicated urinary tract infection. Future Microbiol 2025; **20:** 265
179. Milo G, Katchman EA, Paul M et al: Duration of antibacterial treatment for uncomplicated urinary tract infection in women. Cochrane Database Syst Rev 2005**:** Cd004682
180. Katchman EA, Milo G, Paul M et al: Three-day vs longer duration of antibiotic treatment for cystitis in women: Systematic review and meta-analysis. Am J Med 2005; **118:** 1196
181. Lutters M and Vogt-Ferrier NB: Antibiotic duration for treating uncomplicated, symptomatic lower urinary tract infections in elderly women. Cochrane Database Syst Rev 2008**:** Cd001535
182. Bailey RR, Roberts AP, Gower PE and De Wardener HE: Prevention of urinary-tract infection with low-dose nitrofurantoin. Lancet 1971; **2:** 1112
183. Beerepoot MA, ter Riet G, Nys S et al: Cranberries vs antibiotics to prevent urinary tract infections: A randomized double-blind noninferiority trial in premenopausal women. Arch Intern Med 2011; **171:** 1270
184. Beerepoot MA, den Heijer CD, Penders J et al: Predictive value of escherichia coli susceptibility in strains causing asymptomatic bacteriuria for women with recurrent symptomatic urinary tract infections receiving prophylaxis. Clin Microbiol Infect 2012; **18:** E84
185. Beerepoot MA, ter Riet G, Nys S et al: Lactobacilli vs antibiotics to prevent urinary tract infections: A randomized, double-blind, noninferiority trial in postmenopausal women. Arch Intern Med 2012; **172:** 704
186. Brumfitt W, Cooper J and Hamilton-Miller JM: Prevention of recurrent urinary infections in women: A comparative trial between nitrofurantoin and methenamine hippurate. J Urol 1981; **126:** 71
187. Brumfitt W and Hamilton-Miller JM: A comparative trial of low dose cefaclor and macrocrystalline nitrofurantoin in the prevention of recurrent urinary tract infection. Infection 1995; **23:** 98
188. Brumfitt W, Hamilton-Miller JM, Gargan RA et al: Long-term prophylaxis of urinary infections in women: Comparative trial of trimethoprim, methenamine hippurate and topical povidone-iodine. J Urol 1983; **130:** 1110
189. Brumfitt W, Hamilton-Miller JM, Smith GW and al-Wali W: Comparative trial of norfloxacin and macrocrystalline nitrofurantoin (macrodantin) in the prophylaxis of recurrent urinary tract infection in women. Q J Med 1991; **81:** 811
190. Brumfitt W, Smith GW, Hamilton-Miller JM and Gargan RA: A clinical comparison between macrodantin and trimethoprim for prophylaxis in women with recurrent urinary infections. J Antimicrob Chemother 1985; **16:** 111
191. Costantini E, Zucchi A, Salvini E et al: Prulifloxacin vs fosfomycin for prophylaxis in female patients with recurrent utis: A non-inferiority trial. Int Urogynecol J 2014; **25:** 1173
192. Gower PE: The use of small doses of cephalexin (125 mg) in the management of recurrent urinary tract infection in women. J Antimicrob Chemother 1975; **1:** 93
193. Guibert J, Humbert G, Meyrier A et al: [antibioprevention of recurrent cystitis. A randomized double-blind comparative trial of 2 dosages of pefloxacin]. Presse Med 1995; **24:** 213
194. Kranjčec B, Papeš D and Altarac S: D-mannose powder for prophylaxis of recurrent urinary tract infections in women: A randomized clinical trial. World J Urol 2014; **32:** 79
195. Martens MG FL: Daily cinoxacin as prophylaxis for urinary tract infections in mature women: A prospective trial. Adv Ther 1995; **12:** 207

196. Martorana G, Giberti C and Damonte P: [preventive treatment of recurrent cystitis in women. Double-blind randomized study using cinoxacin and placebo]. Minerva Urol Nefrol 1984; **36:** 43
197. McMurdo ME, Argo I, Phillips G et al: Cranberry or trimethoprim for the prevention of recurrent urinary tract infections? A randomized controlled trial in older women. J Antimicrob Chemother 2009; **63:** 389
198. Możdżan, Ruxer J, Siejka A et al: The efficacy of chronic therapy of recurrent lower urinary tract infections with fosfomycin and nitrofurantoin in type 2 diabetic patients. Advances in Clinical and Experimental Medicine 2007; **16**
199. Nicolle LE, Harding GK, Thompson M et al: Prospective, randomized, placebo-controlled trial of norfloxacin for the prophylaxis of recurrent urinary tract infection in women. Antimicrob Agents Chemother 1989; **33:** 1032
200. Nunez U SZ: Macrocrystalline nitrofurantoin versus norfloxacin as treatment and prophylaxis in uncomplicated recurrent urinary tract infection. Curr Ther Res Clin Exp 1990; **48:** 234
201. Porru D, Parmigiani A, Tinelli C et al: Oral d-mannose in recurrent urinary tract infections in women: A pilot study. Journal of Clinical Urology 2014; **7:** 208
202. Raz R, Colodner R, Rohana Y et al: Effectiveness of estriol-containing vaginal pessaries and nitrofurantoin macrocrystal therapy in the prevention of recurrent urinary tract infection in postmenopausal women. Clin Infect Dis 2003; **36:** 1362
203. Rugendorff E and Haralambie E: Low-dose norfloxacin versus placebo for long-term prophylaxis of recurrent uncomplicated urinary tract infection. Chemioterapia 1987; **6:** 533
204. Schaeffer AJ, Jones JM and Flynn SS: Prophylactic efficacy of cinoxacin in recurrent urinary tract infection: Biologic effects on the vaginal and fecal flora. J Urol 1982; **127:** 1128
205. Scheckler WE, Burt RA and Paulson DF: Comparison of low-dose cinoxacin therapy and placebo in the prevention of recurrent urinary tract infections. J Fam Pract 1982; **15:** 901
206. Seppänen J: Cinoxacin vs trimethoprim--safety and efficacy in the prophylaxis of uncomplicated urinary tract infections. Drugs Exp Clin Res 1988; **14:** 669
207. Stamm WE, Counts GW, Wagner KF et al: Antimicrobial prophylaxis of recurrent urinary tract infections: A double-blind, placebo-controlled trial. Ann Intern Med 1980; **92:** 770
208. Stapleton A, Latham RH, Johnson C and Stamm WE: Postcoital antimicrobial prophylaxis for recurrent urinary tract infection. A randomized, double-blind, placebo-controlled trial. Jama 1990; **264:** 703
209. Fda drug safety communication. In: U.S. Food and Drug Administration, vol. 2019, 2016
210. Kouyos RD, Abel Zur Wiesch P and Bonhoeffer S: Informed switching strongly decreases the prevalence of antibiotic resistance in hospital wards. PLoS Comput Biol 2011; **7:** e1001094
211. Brown EM and Nathwani D: Antibiotic cycling or rotation: A systematic review of the evidence of efficacy. J Antimicrob Chemother 2005; **55:** 6
212. Holmberg L, Boman G, Böttiger LE et al: Adverse reactions to nitrofurantoin. Analysis of 921 reports. Am J Med 1980; **69:** 733
213. Linnebur SA and Parnes BL: Pulmonary and hepatic toxicity due to nitrofurantoin and fluconazole treatment. Ann Pharmacother 2004; **38:** 612
214. Mulberg AE and Bell LM: Fatal cholestatic hepatitis and multisystem failure associated with nitrofurantoin. J Pediatr Gastroenterol Nutr 1993; **17:** 307
215. Sherigar JM, Fazio R, Zuang M and Arsura E: Autoimmune hepatitis induced by nitrofurantoin. The importance of the autoantibodies for an early diagnosis of immune disease. Clin Pract 2012; **2:** e83
216. D'Arcy PF: Nitrofurantoin. Drug Intell Clin Pharm 1985; **19:** 540
217. Huttner A, Verhaegh EM, Harbarth S et al: Nitrofurantoin revisited: A systematic review and meta-analysis of controlled trials. J Antimicrob Chemother 2015; **70:** 2456
218. Claussen K, Stocks E, Bhat D et al: How common are pulmonary and hepatic adverse effects in older adults prescribed nitrofurantoin? J Am Geriatr Soc 2017; **65:** 1316
219. American geriatrics society 2015 updated beers criteria for potentially inappropriate medication use in older adults. J Am Geriatr Soc 2015; **63:** 2227
220. Holmberg L and Boman G: Pulmonary reactions to nitrofurantoin. 447 cases reported to the swedish adverse drug reaction committee 1966-1976. Eur J Respir Dis 1981; **62:** 180
221. Padley SP, Adler B, Hansell DM and Müller NL: High-resolution computed tomography of drug-induced lung disease. Clin Radiol 1992; **46:** 232
222. Sovijärvi AR, Lemola M, Stenius B and Idänpään-Heikkilä J: Nitrofurantoin-induced acute, subacute and chronic pulmonary reactions. Scand J Respir Dis 1977; **58:** 41
223. Hainer BL and White AA: Nitrofurantoin pulmonary toxicity. J Fam Pract 1981; **13:** 817
224. Goemaere NN, Grijm K, van Hal PT and den Bakker MA: Nitrofurantoin-induced pulmonary fibrosis: A case report. J Med Case Rep 2008; **2:** 169

225. Schattner A, Von der Walde J, Kozak N et al: Nitrofurantoin-induced immune-mediated lung and liver disease. Am J Med Sci 1999; **317:** 336
226. Reynolds TD and Thomas J: Nitrofurantoin related pulmonary disease: A clinical reminder. BMJ Case Rep 2013; **2013**
227. Martin WJ, 2nd: Nitrofurantoin. Potential direct and indirect mechanisms of lung injury. Chest 1983; **83:** 51s
228. Bernstein LS: Adverse reactions to trimethoprim-sulfamethoxazole, with particular reference to long-term therapy. Can Med Assoc J 1975; **112:** 96
229. Iarikov D, Wassel R, Farley J and Nambiar S: Adverse events associated with fosfomycin use: Review of the literature and analyses of the fda adverse event reporting system database. Infect Dis Ther 2015; **4:** 433
230. Gleckman R, Blagg N and Joubert DW: Trimethoprim: Mechanisms of action, antimicrobial activity, bacterial resistance, pharmacokinetics, adverse reactions, and therapeutic indications. Pharmacotherapy 1981; **1:** 14
231. Ho JM and Juurlink DN: Considerations when prescribing trimethoprim-sulfamethoxazole. Cmaj 2011; **183:** 1851
232. Costelloe C, Metcalfe C, Lovering A et al: Effect of antibiotic prescribing in primary care on antimicrobial resistance in individual patients: Systematic review and meta-analysis. Bmj 2010; **340:** c2096
233. Vosti KL: Recurrent urinary tract infections. Prevention by prophylactic antibiotics after sexual intercourse. Jama 1975; **231:** 934
234. Pfau A, Sacks TG and Shapiro M: Prevention of recurrent urinary tract infections in premenopausal women by post-coital administration of cinoxacin. J Urol 1988; **139:** 1250
235. Pfau A and Sacks TG: Effective prophylaxis of recurrent urinary tract infections in premenopausal women by postcoital administration of cephalexin. J Urol 1989; **142:** 1276
236. Pfau A, Sacks T and Engelstein D: Recurrent urinary tract infections in premenopausal women: Prophylaxis based on an understanding of the pathogenesis. J Urol 1983; **129:** 1153
237. World health organization: Antimicrobial resistance, 2018
238. Xiong Z, Gao Y, Yuan C et al: Preventive effect of cranberries with high dose of proanthocyanidins on urinary tract infections: A meta-analysis and systematic review. Front Nutr 2024; **11:** 1422121
239. Kontiokari T, Sundqvist K, Nuutinen M et al: Randomised trial of cranberry-lingonberry juice and lactobacillus gg drink for the prevention of urinary tract infections in women. Bmj 2001; **322:** 1571
240. Maki KC, Kaspar KL, Khoo C et al: Consumption of a cranberry juice beverage lowered the number of clinical urinary tract infection episodes in women with a recent history of urinary tract infection. Am J Clin Nutr 2016; **103:** 1434
241. Stothers L: A randomized trial to evaluate effectiveness and cost effectiveness of naturopathic cranberry products as prophylaxis against urinary tract infection in women. Can J Urol 2002; **9:** 1558
242. Takahashi S, Hamasuna R, Yasuda M et al: A randomized clinical trial to evaluate the preventive effect of cranberry juice (ur65) for patients with recurrent urinary tract infection. J Infect Chemother 2013; **19:** 112
243. Vostalova J, Vidlar A, Simanek V et al: Are high proanthocyanidins key to cranberry efficacy in the prevention of recurrent urinary tract infection? Phytother Res 2015; **29:** 1559
244. Walker EB, Barney DP, Mickelsen JN et al: Cranberry concentrate: Uti prophylaxis. J Fam Pract 1997; **45:** 167
245. Rondanelli M, Mansueto F, Gasparri C et al: Supplementation with highly standardized cranberry extract phytosome achieved the modulation of urinary tract infection episodes in diabetic postmenopausal women taking sglt-2 inhibitors: A rct study. Nutrients 2024; **16**
246. Tsiakoulias E, Gravas S, Hadjichristodoulou C et al: Randomized, placebo-controlled, double-blinded study of prophylactic cranberries use in women with recurrent uncomplicated cystitis. World J Urol 2024; **42:** 27
247. Anger J, Lee U, Ackerman AL et al: Recurrent uncomplicated urinary tract infections in women: Aua/cua/sufu guideline. J Urol 2019; **202:** 282
248. Moro C, Phelps C, Veer V et al: Cranberry juice, cranberry tablets, or liquid therapies for urinary tract infection: A systematic review and network meta-analysis. Eur Urol Focus 2024; **10:** 947
249. Anger JT, Bixler BR, Holmes RS et al: Updates to recurrent uncomplicated urinary tract infections in women: Aua/cua/sufu guideline. J Urol 2022; **208:** 536
250. Hayward G, Mort S, Hay AD et al: D-mannose for prevention of recurrent urinary tract infection among women: A randomized clinical trial. JAMA Intern Med 2024; **184:** 619
251. Lenger SM, Chu CM, Ghetti C et al: D-mannose for recurrent urinary tract infection prevention in postmenopausal women using vaginal estrogen: A randomized controlled trial. Urogynecology (Phila) 2023; **29:** 367
252. Lee BS, Bhuta T, Simpson JM and Craig JC: Methenamine hippurate for preventing urinary tract infections. Cochrane Database Syst Rev 2012; **10:** Cd003265
253. Harding C, Mossop H, Homer T et al: Alternative to prophylactic antibiotics for the treatment of recurrent urinary tract infections in women: Multicentre, open label, randomised, non-inferiority trial. Bmj 2022; **376:** e068229

254. Nahata MC, Cummins BA, McLeod DC et al: Effect of urinary acidifiers on formaldehyde concentration and efficacy with methenamine therapy. Eur J Clin Pharmacol 1982; **22:** 281
255. Thomlinson J, Williams JD and Cope E: Persistence of bacteriuria following gynaecological surgery: A trial of methenamine hippurate. Br J Urol 1968; **40:** 479
256. Albrecht U, Goos KH and Schneider B: A randomised, double-blind, placebo-controlled trial of a herbal medicinal product containing tropaeoli majoris herba (nasturtium) and armoraciae rusticanae radix (horseradish) for the prophylactic treatment of patients with chronically recurrent lower urinary tract infections. Curr Med Res Opin 2007; **23:** 2415
257. Genovese C, Davinelli S, Mangano K et al: Effects of a new combination of plant extracts plus d-mannose for the management of uncomplicated recurrent urinary tract infections. J Chemother 2018; **30:** 107
258. Damiano R, Quarto G, Bava I et al: Prevention of recurrent urinary tract infections by intravesical administration of hyaluronic acid and chondroitin sulphate: A placebo-controlled randomised trial. Eur Urol 2011; **59:** 645
259. De Vita D and Giordano S: Effectiveness of intravesical hyaluronic acid/chondroitin sulfate in recurrent bacterial cystitis: A randomized study. Int Urogynecol J 2012; **23:** 1707
260. Wagenlehner FM, Ballarini S, Pilatz A et al: A randomized, double-blind, parallel-group, multicenter clinical study of escherichia coli-lyophilized lysate for the prophylaxis of recurrent uncomplicated urinary tract infections. Urol Int 2015; **95:** 167
261. Minardi D, d'Anzeo G, Parri G et al: The role of uroflowmetry biofeedback and biofeedback training of the pelvic floor muscles in the treatment of recurrent urinary tract infections in women with dysfunctional voiding: A randomized controlled prospective study. Urology 2010; **75:** 1299
262. Baerheim A, Larsen E and Digranes A: Vaginal application of lactobacilli in the prophylaxis of recurrent lower urinary tract infection in women. Scand J Prim Health Care 1994; **12:** 239
263. Czaja CA, Stapleton AE, Yarova-Yarovaya Y and Stamm WE: Phase i trial of a lactobacillus crispatus vaginal suppository for prevention of recurrent urinary tract infection in women. Infect Dis Obstet Gynecol 2007; **2007:** 35387
264. Reid G BA, Taylor M: Instillation of lactobacillus and stimulation of indigenous organisms to prevent recurrence of urinary tract infections. Microecol Ther 1995; **32**
265. Stapleton AE, Au-Yeung M, Hooton TM et al: Randomized, placebo-controlled phase 2 trial of a lactobacillus crispatus probiotic given intravaginally for prevention of recurrent urinary tract infection. Clin Infect Dis 2011; **52:** 1212
266. Gupta V, Mastromarino P and Garg R: Effectiveness of prophylactic oral and/or vaginal probiotic supplementation in the prevention of recurrent urinary tract infections: A randomized, double-blind, placebo-controlled trial. Clin Infect Dis 2024; **78:** 1154
267. Rahn DD, Carberry C, Sanses TV et al: Vaginal estrogen for genitourinary syndrome of menopause: A systematic review. Obstet Gynecol 2014; **124:** 1147
268. Perrotta C, Aznar M, Mejia R et al: Oestrogens for preventing recurrent urinary tract infection in postmenopausal women. Cochrane Database Syst Rev 2008**:** Cd005131
269. Cardozo L, Benness C and Abbott D: Low dose oestrogen prophylaxis for recurrent urinary tract infections in elderly women. Br J Obstet Gynaecol 1998; **105:** 403
270. Kirkengen AL, Andersen P, Gjersøe E et al: Oestriol in the prophylactic treatment of recurrent urinary tract infections in postmenopausal women. Scand J Prim Health Care 1992; **10:** 139
271. Eriksen B: A randomized, open, parallel-group study on the preventive effect of an estradiol-releasing vaginal ring (estring) on recurrent urinary tract infections in postmenopausal women. Am J Obstet Gynecol 1999; **180:** 1072
272. Raz R and Stamm WE: A controlled trial of intravaginal estriol in postmenopausal women with recurrent urinary tract infections. N Engl J Med 1993; **329:** 753
273. Ponzone R, Biglia N, Jacomuzzi ME et al: Vaginal oestrogen therapy after breast cancer: Is it safe? Eur J Cancer 2005; **41:** 2673
274. Le Ray I, Dell'Aniello S, Bonnetain F et al: Local estrogen therapy and risk of breast cancer recurrence among hormone-treated patients: A nested case-control study. Breast Cancer Res Treat 2012; **135:** 603
275. O'Meara ES, Rossing MA, Daling JR et al: Hormone replacement therapy after a diagnosis of breast cancer in relation to recurrence and mortality. J Natl Cancer Inst 2001; **93:** 754
276. Averbeck MA, Rantell A, Ford A et al: Current controversies in urinary tract infections: Ici-rs 2017. Neurourol Urodyn 2018; **37:** S86
277. Niaz WA, Yasmeen H and Shamsher S: Evaluation of the diagnostic accuracy of lower urinary tract symptoms (luts) in uncomplicated uti in pakistani women. J Pak Med Assoc 2024; **74:** 1811

278. Meckes NA, Melnyk AI, Guirguis M et al: Factors associated with a positive urine culture in women seeking urogynecologic care for urinary tract infection symptoms. Female Pelvic Med Reconstr Surg 2022; **28:** 408
279. Burnett LA, Hochstedler BR, Weldon K et al: Recurrent urinary tract infection: Association of clinical profiles with urobiome composition in women. Neurourol Urodyn 2021; **40:** 1479
280. Lee AS, Lamanna OK, Ishida K et al: A novel propidium monoazide-based pcr assay can measure viable uropathogenic e. Coli in vitro and in vivo. Front Cell Infect Microbiol 2022; **12:** 794323
281. Kime L, Randall CP, Banda FI et al: Transient silencing of antibiotic resistance by mutation represents a significant potential source of unanticipated therapeutic failure. mBio 2019; **10**
282. Wagner TM, Howden BP, Sundsfjord A and Hegstad K: Transiently silent acquired antimicrobial resistance: An emerging challenge in susceptibility testing. J Antimicrob Chemother 2023; **78:** 586
283. Szlachta-McGinn A, Douglass KM, Chung UYR et al: Molecular diagnostic methods versus conventional urine culture for diagnosis and treatment of urinary tract infection: A systematic review and meta-analysis. Eur Urol Open Sci 2022; **44:** 113
284. Barraud O, Ravry C, François B et al: Shotgun metagenomics for microbiome and resistome detection in septic patients with urinary tract infection. Int J Antimicrob Agents 2019; **54:** 803
285. Hasman H, Saputra D, Sicheritz-Ponten T et al: Rapid whole-genome sequencing for detection and characterization of microorganisms directly from clinical samples. J Clin Microbiol 2014; **52:** 139
286. Sun Z, Liu W, Zhang J et al: The direct semi-quantitative detection of 18 pathogens and simultaneous screening for nine resistance genes in clinical urine samples by a high-throughput multiplex genetic detection system. Front Cell Infect Microbiol 2021; **11:** 660461
287. Baunoch D, Luke N, Wang D et al: Concordance between antibiotic resistance genes and susceptibility in symptomatic urinary tract infections. Infect Drug Resist 2021; **14:** 3275
288. Zhang L, Huang W, Zhang S et al: Rapid detection of bacterial pathogens and antimicrobial resistance genes in clinical urine samples with urinary tract infection by metagenomic nanopore sequencing. Front Microbiol 2022; **13:** 858777
289. Festa RA, Cockerill FR, Pesano RL et al: Pooled antibiotic susceptibility testing for polymicrobial uti performs within clsi validation standards. Antibiotics (Basel) 2025; **14**
290. Haley E, Cockerill FR, Pesano RL et al: Pooled antibiotic susceptibility testing performs within clsi standards for validation when measured against broth microdilution and disk diffusion antibiotic susceptibility testing of cultured isolates. Antibiotics (Basel) 2024; **13**
291. Leitner L, Ujmajuridze A, Chanishvili N et al: Intravesical bacteriophages for treating urinary tract infections in patients undergoing transurethral resection of the prostate: A randomised, placebo-controlled, double-blind clinical trial. Lancet Infect Dis 2021; **21:** 427
292. Dutta R, Stothers L and Ackerman AL: Manipulating the gut microbiome in urinary tract infection-prone patients. Urol Clin North Am 2024; **51:** 525
293. Schwenger EM, Tejani AM and Loewen PS: Probiotics for preventing urinary tract infections in adults and children. Cochrane Database Syst Rev 2015; **2015:** Cd008772
294. Tariq R, Pardi DS, Tosh PK et al: Fecal microbiota transplantation for recurrent clostridium difficile infection reduces recurrent urinary tract infection frequency. Clin Infect Dis 2017; **65:** 1745
295. Tariq R, Tosh PK, Pardi DS and Khanna S: Reduction in urinary tract infections in patients treated with fecal microbiota transplantation for recurrent clostridioides difficile infection. Eur J Clin Microbiol Infect Dis 2023; **42:** 1037
296. Timm MR, Russell SK and Hultgren SJ: Urinary tract infections: Pathogenesis, host susceptibility and emerging therapeutics. Nat Rev Microbiol 2025; **23:** 72
297. Mak Q, Greig J, Dasgupta P et al: Bacterial vaccines for the management of recurrent urinary tract infections: A systematic review and meta-analysis. Eur Urol Focus 2024; **10:** 761
298. Nickel JC, Kelly KL, Griffin A et al: Mv140 sublingual vaccine reduces recurrent urinary tract infection in women results from the first north american clinical experience study. Can Urol Assoc J 2024; **18:** 25
299. Lorenzo-Gómez MF, Foley S, Nickel JC et al: Sublingual mv140 for prevention of recurrent urinary tract infections. NEJM Evid 2022; **1:** EVIDoa2100018
300. Spaulding CN, Klein RD, Schreiber HLt et al: Precision antimicrobial therapeutics: The path of least resistance? NPJ Biofilms Microbiomes 2018; **4:** 4
301. Gágyor I, Bleidorn J, Kochen MM et al: Ibuprofen versus fosfomycin for uncomplicated urinary tract infection in women: Randomised controlled trial. Bmj 2015; **351:** h6544
302. Hannan TJ, Roberts PL, Riehl TE et al: Inhibition of cyclooxygenase-2 prevents chronic and recurrent cystitis. EBioMedicine 2014; **1:** 46

303. Harlow BL, Bavendam TG, Palmer MH et al: The prevention of lower urinary tract symptoms (plus) research consortium: A transdisciplinary approach toward promoting bladder health and preventing lower urinary tract symptoms in women across the life course. J Womens Health (Larchmt) 2018; **27:** 283
304. Brady SS, Bavendam TG, Berry A et al: The prevention of lower urinary tract symptoms (plus) in girls and women: Developing a conceptual framework for a prevention research agenda. Neurourol Urodyn 2018; **37:** 2951